2) The glucose molecules is converted to another six carbon sugar (**Fructose 1, 6 - biphosphate**).

3) The Fructose 1, 6 – biphosphate split in to two molecules of three – carbon-sugar **glyceraldehydes – 3 – phosphate** (GP).

The above three reactions can be shortly summarized as:

$$\text{Glucose} + 2ATP \longrightarrow 2GP - 2ADP - 2pi$$

4) Each molecule of **GP** is converted in to **pyruvate** (3 carbon compound). During these reactions two hydrogen atoms are removed and enough chemical energy is released to produce four ATP molecules.

$$2GP - 4\,ADP - 4Pi \longrightarrow 2\ \text{pyruvate} - 4H - 4ATP$$

However, since two ATP molecuels are consumed earlier to split and initiate the oxidation of glucose molecule, the process of glycolysis yields a profit of two ATP molecules.

The two hydrogen atoms released from **GP** combine with **NAD**, which is a hydrogen carrier molecule.

$$NAD^{+} + 2H \longrightarrow NAD - H + H^{+}$$

The overall stages of glycolysis can be summarized as:

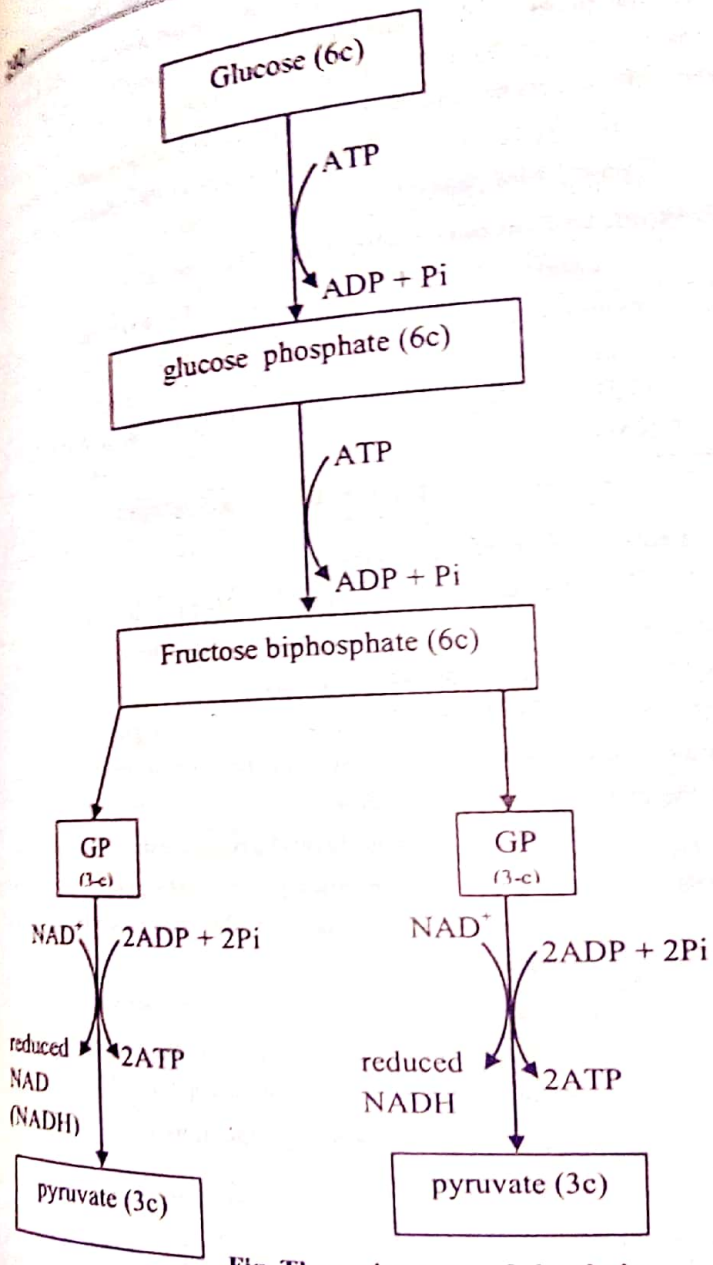

Fig. The main stages of glycolysis

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- The overall reactions of gllycolysis can be summarized as:

$C_6H_{12}O_6 + 2ATP + 4ADP + 2Pi + 2NAD \longrightarrow 2C_3H_6O_3 + 2ADp + 2NAD$ reduced $+ 2H^+ + 2H_2O$.

Note: The necessary ingredients of cellular respiration such as ADP, NAD and phosphate float freely in the cytoplasm of a cell and are used when needed.

The inputs and out puts of glycolysis

Total input	Total out put
✓ 1 molecule of glycose ($C_6H_{12}O_6$) ✓ 2 ATP ✓ 2 NAD^+	✓ 2 molecules of pyruvate (3c) ✓ 4 ATP ✓ 2NADH or reduced NAD
The overall profit Net 2ATP and 2NADH	

Table 5.1. Budjet for glycolysis

2. The link reaction (2^{nd} stage of aerobic respiration)

- The link reaction acts as a transition or linkage between glycolysis (the first stage) and krebs cycle (the third stage).
- In the link reaction, pyruvate the end product of glycolysis which contains most of the energy present in the glucose molecule enters into the matrix of the mitochondria.
- The pyruvate finally changes into **acetyl group** during the conversion of pyruvate to acetyl group the following events take place:
  - ➢ The carbondioxide is lost by the decarboxylation reaction which diffuses out of the cell.

Note: **Decarboxylation** – is the loss of carbon in reaction

  - ➢ The hydrogens are removed by the process called **dehydrogenation** and they are accepted by NAD^+

Note: **Dehydrogenation** is the removal of hydrogen during the reaction.

- ➢ Finally, the oxidized fragment of acetyl group is attached to **coenzyme A** that contains sulfur atom which helps to attach it to the acetyl group by an unstable bond.

Note: • The coenzyme – A is manufactured in the cell from the vitamin – B groups called **panthothenic acid**.
• Fat and anminoacids are also degraded into acetyl coA and enter the respiratory pathway at this point.

The overall reaction of the link reaction is:

2 pyruvate (3c) + $2NAD^+$ + 2 coenzyme A(2CoA)
2 acetyl CoA + 2NADH + 2H + $2CO_2$

Remember: When the acetyl group reacts with coenzyme A (COA) it forms **acetyl COA.**

The formation of **acetyl COA in the link reaction** can also be discussed as:

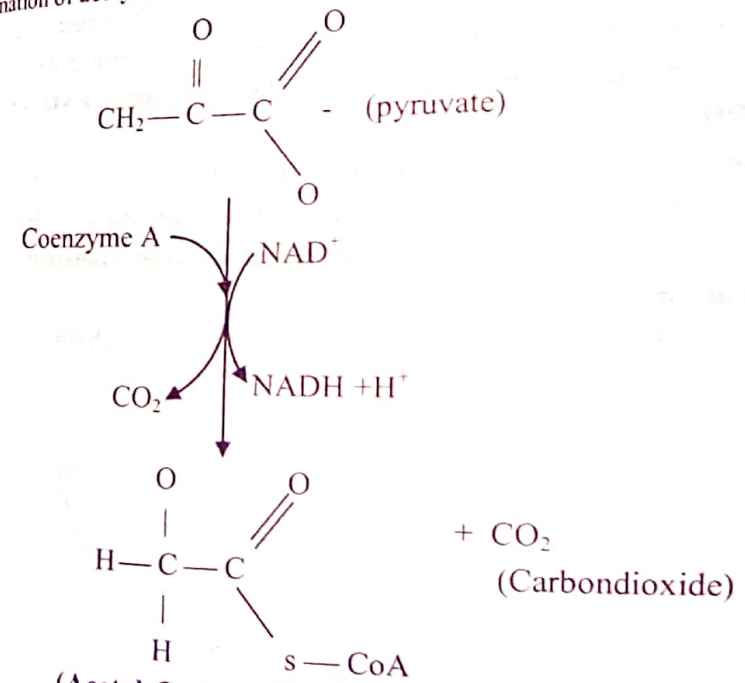

Fig. 5.7. The link reaction

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Summary of the link reaction:

So far, we have seen that:

- It takes place in the matrix of mitochondria,
- The original glucose molecule has been oxidized to two acetyl groups and $2Co_2$ molecules.
- The hydrogens removed are reduced NAD+ to NADH.
- Four NADH molecules have been formed. Two during glycolysis and two during oxidation of private.

3. The Krebs (citric acid) cycle

This cycle is also called **tricarboxylic acid (TCA) cycle**. It is named krebs cycle by the name of sir. Hans kreb, who worked o it 1930's. Sir. Hans kreb was the class mate of an Ethiopian biologist Dr. Getachew Bolodia.

The krebs cycle takes place in the matrix of Mitochondria where it works in the presence of oxygen.

Just like the other stages of cellular respiration the hydrogen ions removed are transferred to NAD in order to produce reduced NAD ($NADH^+$). Moreover, during krebs cycle another electron acceptor **FAD** is reduced to **$FADH_2$.**

Note: The reduced NAD ($NADH+ H^+$) and the reduced FAD ($FADH_2$) release their hydrogen atoms as protons (hydrogen ions) and electrons in the final stage of aerobic respiration called **electron transport chain** and **chemiosmosis.**

- The protons are used in the chemiosmotic synthesis.

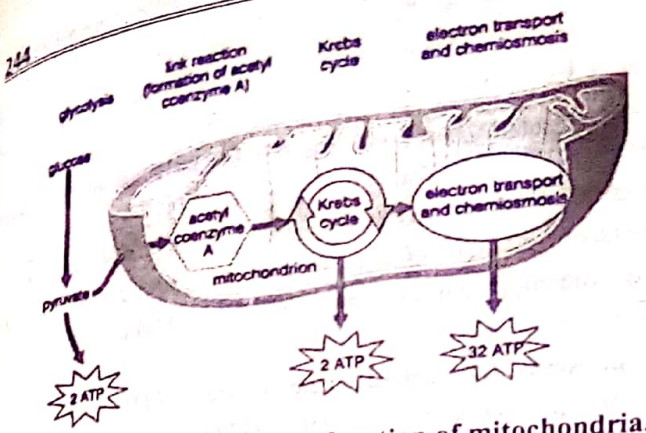

Figure: The structure and function of mitochondria.

What happens in the krebs cycle?

- The two carbon group **Acetyl CoA** reacts with the four carbon compound called **oxaloacetate**. This results in the formation of a six-carbon compound called **citrate.**
- The four carbon compound (oxaloacetate) is found originally in the fluid matrix of mitochondoria. It is used to start the krebs cycle.

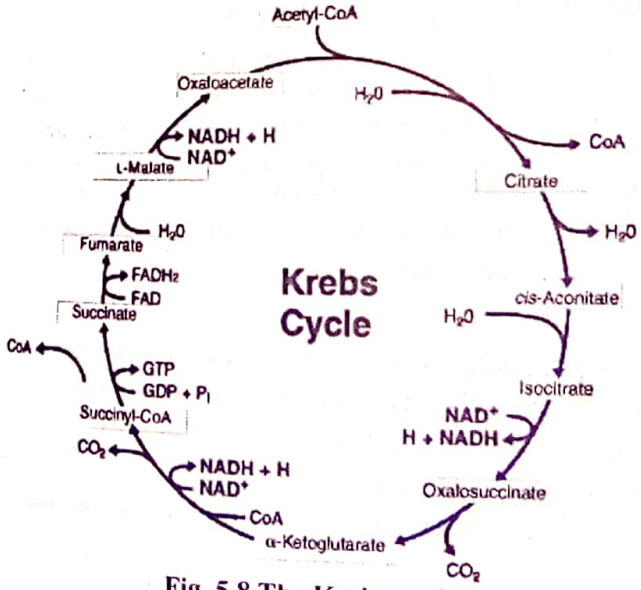

Fig. 5.8 The Krebs cycle

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The key points from krebs cycle

- Acetyl CoA (2c) molecule enter in to the krebs cycle and reacts with four carbon compound called **citrate**.
- In Krebs cycle, one cycle produces **two carbondioxides, three reduced NAD, one reduced FAD** and **one ATP.**
- Two pyravates enter the krebs cycle from one glucose molecule. Thus, two citric acid cycles occur for one glucose molecule. Thus totally, the following are produced during krebs cycle from one glucose molecule.
  - 6NAD is reduced to NADH (reduced NAD)
  - 2FAD is reduced to $2FADH_2$ (reduced FAD)
  - Four carbon dioxide molecules are released.
  - Two ATPs are produced.

Table 5.2: The inputs and out puts of krebs cycle per glucose molecule are as follows:

Input	Out put
2Acetyl groups	$6CO_2$
2ADP + 2Pi	2ATP
6 NAD	6NADH
2FAD	$2FADH_2$

Note: The link reaction, krebs cycle and electron transport chain are called **the aerobic stages of cellular respiration**. While, glycolysis is called the **anaerobic stage.**

4. The electron transport chain and chemiosmosis

These are the stages at which the hydrogen ions that are passed to NAD^+ and FAD during the stages that we have seen so far are used.

The electron transport chain and chemiosmosis together make up the process of **oxidative phosphorylation**. This is because the phosphorylation

of ADP to from ATP is coupled with the flow of electrons they are transferred along the acceptors in the electron transport chain.

- Both processes take place in the inner mitochondorial membrane.

Electron Transport Chain

The process of electron transport chain takes place in the inner folds of the mitochondrial membrane called **cristae** through the following events:

- The hydrogen atoms which are carried by reduced NAD (NADH) and reduced FAD ($FADH_2$) enter electron transport chain and split in to **protons** and **electrons**. When the hydrogen protons (H^+) separate from their electrons, they are released in to the surrounding medium.
- The electrons that have relatively high energy content are passed along a series of electron carriers that form the transport chain lose energy as they pass from one carrier to the next.
- The electron carriers that are used at this stage are proton pumps that move protons from the matrix of mitochondorion to the inner membrane space. The molecules that act as electron carriers are:
  - Reduced NAD dehydrogenase
  - Ubiquinone
  - Cytochromes (proteins that contain iron)
- As the electrons are transferred through these three proton pumps, they lose energy which pumps the protons in to the inner membrane space.
- The electrons from NAD powers all the three pumps to move the protons in to the inter membrane space.
- Oxygen is the **terminal (final) electron acceptor**. As electrons passed along from one cytochrome to the next in the chain, loosing energy as they go, the last cytochrome (cytochrome a_3) Passes the two electrons on to molecular oxygens and the chemical union of hydrogen and oxygen produces water. When no oxygen is available to accept the hydrogen, the last cytochrome in the chain

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(cytochromme a_3) gets stuck. This, may cause the entire system to get blocked and prevents further production of ATP in the electron transport chain.

Note: Cytochromes are molecules that have iron as a central atom. The electron from the hydrogen atom combines with this iron.

- Each pair of electrons that passes from one carrier to the next carrier of electron transport chain releases enough energy to allow the production of ATP.
- Protons accumulate in the inter membrane space creating a higher concentration than in the matrix.

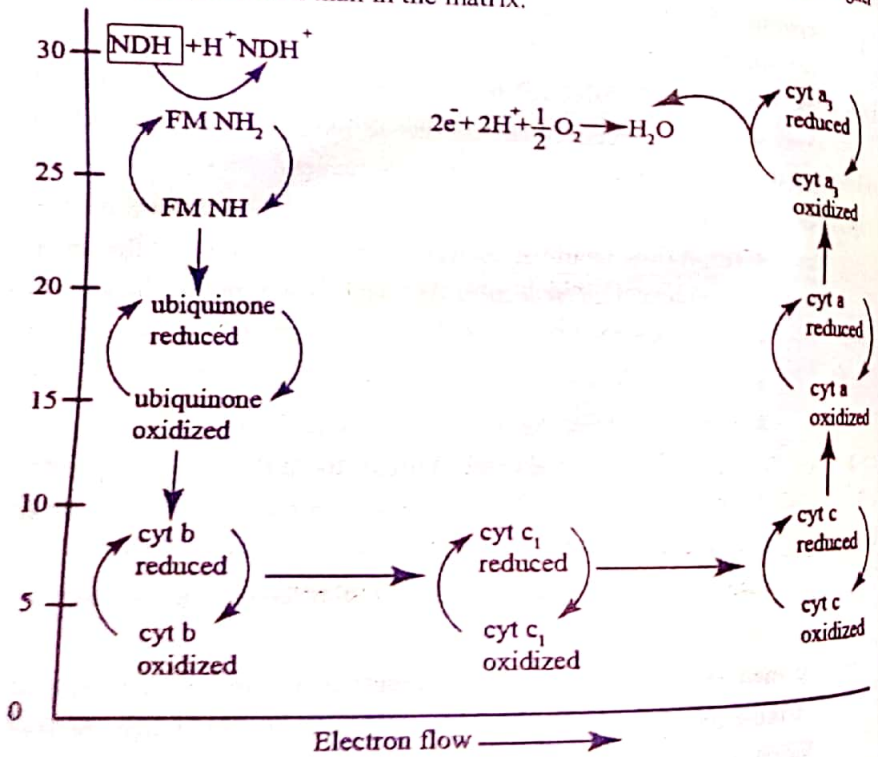

Fig. 5.9. The electron transport chain

Chemiosmosis

The proton gradient concentration results in protons diffusing through the ATP synthase rotor spin and produce ATP from ADP and Pi. The diffusion of Hydrogen ions through the ATP synthase is called **chemiosmosis.**

Therefore;

1. The oxidation of one molecule of reduced NAD (NADH) results in six protons passing through ATP synthase and three molecules of ATP are synthesized.
2. The oxidation of one molecule of reduced FAD results in four protons passing through ATP synthase and two ATP molecules are synthesized.

Utilization of other organic substances as energy source

As seen earlier glucose is the major substance that is used for respiration and fermentation. However, other organic substances such as lipids and proteins can also be used as respiratory subtrates. These substances can enter the aerobic respiration pathway.

The major stages through which proteins and lipids enter the aerobic pathway will be discussed as:

Stage I – The proteins, polysaccharides and lipids are hydrolyzed into their component building blocks.

Stage II - The building blocks are degraded in to a product that can enter the aerobic pathway.

Stage III – The degraded substances enter the krebs cycle and forms and products such as **water, CO2** and **ammonia**.

Note: The metabolism of proteins, lipids and carbohydrates converges on the krebs cycle.

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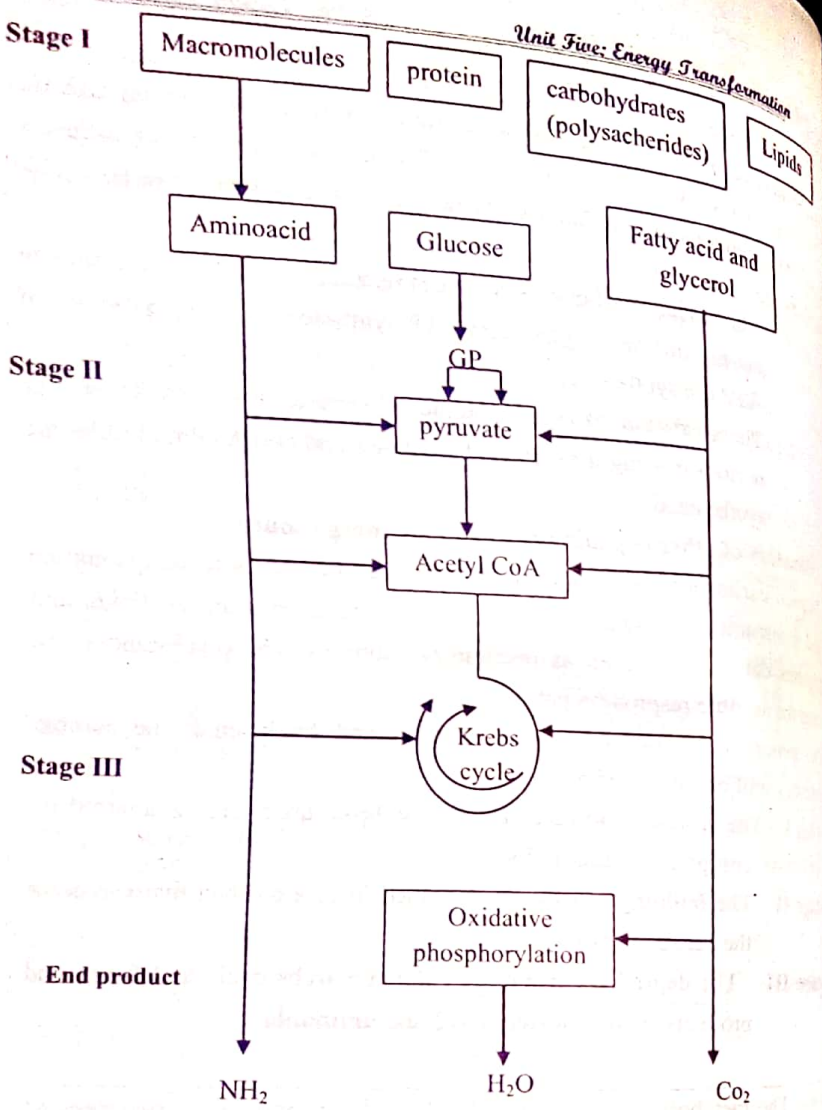

Fig. 5.15 The metabolic pathway of carbohydrate, lipids and proteins and their respired end products.

Illustrative Examples

1. Which of the following steps in cellular respiration can take place in the absence of oxygen?
   A. Electron transport chain
   B. Glycolysis
   C. Krebs cycle
   D. Acetyl CoA formation

Explanation: The krebs cycle, electron – transport chain and the link reaction takes place in the presence of oxygen in the mitochondoria of the cell. However, glycosis the anaerobic stage occurs in the absence of oxygen.

Answer: B

2. Which of the following molecules is known as the energy currency of the cell?
   A. Glucose
   B. Starch
   C. Fat
   D. ATP

Explanation: ATP (Adenosine triphosphate) is the direct source of energy which is also known as:

- ✓ Energy currency of the cell.
- ✓ Cash source of energy.

Answer: D

3. Which of the following processes releases oxygen to the atmosphere?
   A. Respiration
   B. Photosynthesis
   C. Transpiration
   D. Burning of fossil fuels

Explanation: Photosynthesis and respiration are two important life processes in which:

- ✓ Photosynthesis releases oxygen in to the atmosphere.
- ✓ Respiration utilized oxygen from the atmosphere.

Answer: B

4. Choose the final electron acceptor in the electron transport chain during aerobic respiration of eukaryotic cells?
   A. H_2O
   B. O_2
   C. Co_2
   D. NADP

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Explanation: The final electron acceptor in the electron in the electron transport chain is oxygen (O_2).

Answer: B

5. Where in the mitochondria does the krebs cycle take place?
   A. On the cristae.
   B. In the matrix.
   C. Between the outer and inner membrane.
   D. On the inner surface of the outer membrane.

Explanation: The electron transport chain takes place in the cristae of mitochondria, while krebs cycle takes place in fluid filled matrix of mitochondria.

Answer: B

6. Which of the following defines ATP?
   A. Adenine + Adenosine + phosphate
   B. Adenosine + Ribose sugar + double phosphate
   C. Adenine + ribose sugar + three phosphate
   D. Nitrogen + sugar + monophosphate

Explanation:

- ATP the immediate source of energy is composed:
  - Adenosine
  - Ribose sugar
  - Three phosphate

Answer: C

7. Which process of respiration helps to release most of the energy stored in glucose?
   A. Oxidative phosphorylation
   B. Glycolysis
   C. Fermentation reaction
   D. Anaerobic respiration

Explanation: Oxidative phosphorylation takes place in the mitochondria in the presence of oxygen by using an enzyme ATP synthase for the formation of ATP.

Answer: A

8. If an animal inhales a radioactive form of oxygen, in which of the following products of the cellular respiration would the radioactivity be detected?
   A. water
   B. carbondioxide
   C. ATP
   D. NADH

Explanation: Radioactivity is mostly observed on carbon element. The product of cellular respiration which is used to detect the radioactivity are CO_2 and water. CO_2 is the end product of cellular respiration and exhaled during cellular respiration is used to detect radioactivity.

Answer: B

9. What term is used to describe the chemical process of living organisms that involves both the joining up of monomers to produce polymers and the splitting of polymers to monomers?
   A. Anabolism
   B. Catabolism
   C. Metabolism
   D. Polymorphism

Explanation:

- ✓ **Metabolism** is the sum of all the chemical reactions that occur inside the body. It involves anabolism and catabolism,
- ✓ **Anabolism** involves the joining up of monomer units to form polymers.
- ✓ **Catabolism** is splitting of polymers to monomers.

Answer: C

10. In which process of life do we group organisms as authotrophic and others as hetrotrophic?
    A. Transport system.
    B. Aspect of nutrition
    C. Anaerobic respiration
    D. Coordination system

Explanation: Organisms are grouped based on their mode of nutrition as:

- ✓ **Autotrophs** – organisms that can synthesize their own food.

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✓ **Heterotrophs** – organisms that cannot synthesize their own food.

11. During aerobic respiration of cells, in which cellular part does the krebs cycle takes place?

A. Chloroplast
B. Mitochondria
C. Nuclei
D. Lysosomes

Answer: B

Explanation: The most stages of cellular respiration such as **link reaction, krebs cycle** and **electron transport cahin** occurs in the mitochondria of the cell except **glycolysis** which takes place in the **cytoplasm** of the cell.

12. Which of the following molecule contains phosphate group.

A. Nucleotide
B. Protein
C. Cellulose
D. Simple sugar

Answer: B

Explanation: Nucleotide consists of:

✓ Five carbon sugar.
✓ Phosphate group.
✓ Nitrogenous bases (Adenine, thymine, guanine, cytosine and uracil).

13. Which of the following pairs of organelles are concerned with energy transformation?

A. Nucleus and nucleolus
B. Mitochondoria and nucleus
C. Chloroplasts and vacuole
D. Chloroplasts and mitochondria

Explanation:

- **Chloroplast** is the site of photosynthesis where transformation of light energy in to chemical energy takes place.
- **Mitochondria** is an organelle of a cell which is concerned with the release of energy.

Answer: D

14. Which one of the following is not recycled between an organism and the environment in the ecosystem?

A. Energy
B. Carbon
C. Nitrogen
D. phosphorous

Explanation: In an ecosystem energy always flows in one direction. Thus it cannot be recycled in an ecosystem. However, other substances such as water, carbon, nitrogen, phosphorous and other are recycled between an organism and its environment.

Answer: A

15. Which means of transport requires input of energy by the cells?

A. Simple diffusion
B. Facilitated diffusion
C. Osmosis
D. Active transport

Explanation: Active transport occurs against the concentration gradient. Thus, it requires expenditure of energy. However, activities such as simple diffusion, facilitated diffusion and osmosis are passive transport systems that do not require energy.

Answer: D

16. Which of the following is NOT one of the stages in cellular respiration?

A. Calvin cycle
B. Glycolysis
C. Electron transport
D. Krebs cycle

Explanation: Cellular respiration has four major stages. They are:

✓ Glycolysis
✓ Link reaction
✓ Krebs cycle
✓ Electron transport chain

Answer: C

17. Among the molecules found in cells, which of the following contains less energy?

A. Glucose molecule
B. Amino acid
C. triglyceride
D. Water molecule

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Explanation: Organic molecules such as carbohydrates (glucose), protein (amino acid) and lipids (triglyceride) contain high energy content than inorganic molecules such as water and minerals.

Answer: D

18. What is the correct equation for cellular respiration

A. $6CO_2 + 6H_2O + \text{energy} \longrightarrow 6O_2 + C_6H_{12}O_6$

B. $6O_2 + C_6H_{12}O_6 \longrightarrow 6H_2O + 6CO_2 + \text{energy}$

C. $6O_2 + C_6H_{12}O_6 + \text{energy} \xrightarrow[\text{chlo}]{\text{light}} 6CO_2 + 6H_2O$

D. $6CO_2 + 6H_2O \qquad 6O_2 + C_6H_{12}O_6$

Answer: B

19. The amount of net gain of ATP that glycolysis provides to a cell is

A. 2ATP molecules C. 18ATP molecules

B. 4ATP molecules D. 36ATP molecules

Explanation: During glycolysis the total of **4ATP** molecules are produced. However, since the cell expends **2ATPS** to break down glucose molecule. The Net of 2ATP molecules are produced.

Answer: A

20. Which of the following requires expenditure of ATP?

A. Osmosis C. simple diffusion

B. Facilitated diffusion D. Endocytosis

Explanation: The process of active transport which requires energy involves the processes:

- ✓ Phagocytosis
- ✓ Endocytosis
- ✓ Exocytosis

➢ See unit four for more details

Answer: D

21. How many molecules of ATP will be generated as a result of the oxidation of one molecules of $FADH_2$ in actively respiring mitochondria?

A. 10 B. 3 C. 2 D. 6

Explanation: In electron transport chain the

- ✓ Oxidation of one molecule of NADH produces 3ATPs;
- ✓ Oxidation of one molecule of $FADH_2$ produces 2ATPs;

Answer: C

22. Where is no free oxygen to breathe, which one of the following steps of the respiration process can operate in our body?

A. Glycolysis

B. Krebs cycle

C. Electron transport chain

D. Reaction that links glycolysis and krebs cycle.

Explanation: Glycolysis is the anaerobic stage of cellular respiration that takes place in the absence of oxygen.

Answer: A

23. Which energy rich organic compound contains adenine in its molecule?

A. Lipid C. Glucose

B. Carbohydrate D. ATP

Explanation: ATP molecule which is the energy currency of the cell is composed of:

- ✓ Adenine molecule
- ✓ Phosphate unit.
- ✓ Nitrogenous bases

Answer: D

24. During chemiosmosis, what substance diffuses from one side to the other side of the membrane?

A. Water molecule C. Electrons

B. Protons D. ATP molecules

Explanation: When the concentration of proton gradient increases in the outer mitochondorial membrane, it diffuses through the ATP synthase to the inner membrane of mitochondoria.

Answer: B

25. Which of the following paired organelles are membrane bounded?

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A. Ribosomes and peroxisomes
B. Chloroplasts and ribosomes
C. Mitochondria and ribosomes
D. Chloroplast and mitochondria

Explanation: Cellular structures such as **nucleus, vacuole, endoplasmic reticulum, mitochondria** and **chloroplast** are membrane bounded organelles with the cell. **Ribosomes** and **proxisomes** are not bounded by membrane.

Answer: D

26. The net product of the krebs cycle from one molecule of acetyl coenzyme are:
A. ATP, 4NADH, FADH and $2CO_2$
B. 2ATP, 3NADH, 2FADH and CO_2.
C. ATP, 3NADH, FADH and $2CO_2$.
D. 2ATP, NADH, 3FADH and $2CO_2$.

Explanation: During krebs cycle from one molecule of acetyl COA the products are:

✓ 1ATP, 3NADH, 1FADH and $2Co_2$.

Answer: C

27. Which of the following reaction take place when the cell needs energy?
A. $ATP \longrightarrow ADP + pi$
B. $ADP + pi \longrightarrow ATP$
C. $AMP + pi \longrightarrow ADP$
D. $Co_2 + H_2O \longrightarrow C_6H_{12}O_6 + O_2$

Explanation: When a cell requires energy **ATP** molecule breaks in to ADP (Adenisine diphosphate), **phosphate** and **energy**.

Answer: A

28. If a cell contains 20 NADH and 10 FADH molecules, how many ATP molecuels are produced within the cell?
A. 60 ATP
B. 80ATP
C. 30 ATP
D. 70 ATP

Explanation: During electron transport chain from 1 NADH, 3ATP molecules and from 1 $FADH_2$, 2ATP moleculess are produced. Thus;

$\Rightarrow$ $20NADH \times 3ATP = 60ATP$
$\Rightarrow$ $10\ FADH \times 3ATP = 20ATP$
Total $= 80\ ATP$

Answer: B

29. Which one of the following molecule is NOT the product of krebs cycle in cellular respiration?
A. CO_2
B. NADH
C. ATP
D. Pyruvate

Explanation: **Pyruvate** is three carbon molecule that is produced during glycolysis from the breakdown of glucose molecule.

- NADH, $FADH_2$, ATP and CO_2 produced during krebs cycle.

Answer: D

30. Which of the following statements is NOT true?
A. It takes place in mitochondria.
B. It releases small amount of energy.
C. It does not require oxygen.
D. It produces ATP.

Explanation: Glycolysis which is the anaerobic stage of cellular respiration:

✓ Takes place in the cytosol of the cell.
✓ Releases small amount of energy which is only 2 net ATP molecules from one glucose molecule.
✓ Does not require oxygen.

Answer: A

31. When the muscle cells are in short supply of oxygen, which of the following compounds would be accumulated in them?
A. Ethanol
B. Acetic acid
C. Lactic acid
D. carbondioxide

Explanation: During the time when we perform activities like heavy physical exercises, our muscle cells are in short supply of oxygen. In such a condition glucose molecule breaks in to

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lactic acid molecules which accumulates in our muscles to cause muscle scramp.

Answer: C

32. Which of the following chemical substance absorbs carbondioxide from organisms surrounding?
A. Sodium chloride
B. Potassium hydroxide
C. sodium hydroxide
D. potassium chloride

Explanation: **Potassium hydroxide** is a chemical substance which absorbs CO_2 from organisms surrounding. **Sodium chloride** releases CO_2 in to its surrounding.

Answer: B

33. Which of the following gives the products of fermentation in yeast cells?
A. Lactate, CO_2 and 2ATP
B. CO_2, H_2O and 36ATP
C. Alcohol, CO_2 and 2ATP
D. Alcohol, CO_2 and 36 ATP

Explanation: During anaerobic respiration by the yeast:

$$\underset{(C_6H_{12}O_6}{\text{Glucose}} \longrightarrow \underset{(2C_2H50H)}{\text{Ethylalcohol}} + \underset{(CO_2)}{\text{carbondioxide}} + 2ATP$$

Answer: C

34. Which of the following is the product of both aerobic and anaerobic respiration in plant cell?
A. $2C_2H_5OH$
B. Co_2
C. Lactic acid
D. Water

Explanation: During both aerobic and anaerobic respiration carbondioxide is produced.

i) Cellular respiration (aerobic)

$$C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6CO_2 + \text{energy}$$

ii) Anaerobic respiration by yeast

$$C_6H_{12}O_6 \longrightarrow 2C_2H_5OH + 2CO_2 + \text{energy(2ATP)}$$

Answer: B

35. The removal of amino group from amino acid in the liver is
A. Deamination
B. Hydrolysis
C. Decarboxylation
D. Dehydrogenation

Explanation: **Deamination** the process by which the amino group of the amino acids is removed in the liver. This occurs, because our body cannot store excess amino acids unlike fat and carbohydrates.

Answer: A

36. Lactic acid is converted in to glucose in the:
A. Liver
B. Muscle cell
C. pancreas
D. kidney

Explanation: In animal muscle cells glucose is fermented in to lactic acid and 2AT molecules. the ATP is used immediately by out body as a source of energy. But the lactic acid cause muscle fatigue and removed in the liver by cari cycle.

$$\text{Lactase} \longrightarrow \text{2 pryuvate} \xrightarrow{6ATP \quad 6\,ADP + 6pi} \text{Glucose}$$

Answer: A

37. How many molecule of ATP is gained by a cell during the process of glycolysis from 10 molecules of glucose?
A. 40 B. 20 C. 380 D. 120

Explanation: Two molecules of ATP (2ATPs) are gained from the glycolysis of one glucose molecule. Thus, from 10 molecules glucose **20ATP** molecules are gained.

Answer: B

38. Which of the following steps in cellular respiration takes place in the absence of oxygen?
A. Link reaction
B. Krebs cycle
C. Glycolysis
D. electron transport chain

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Explanation: Glycolysis is the first stage of cellular respiration that takes place in the absence of oxygen. Thus, it is called the **anaerobic stage** of cellular respiration.

39. Substrate level phosphorylation takes place during which step of cellular respiration?

Answer: C

A. Glycolysis and krebs cycle.
B. Link reaction and electron transport chain.
C. Glycolysis and electron transport chain.
D. Link reaction and krebs cycle.

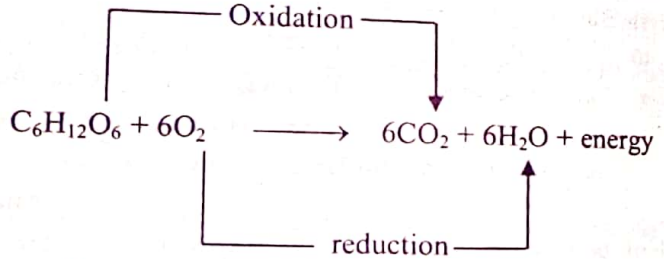
Explanation: Substrate level phosphorylation is the production of ATP which occurs without involvement an enzyme ATP **synthase**. It occurs during glycolysis and krebs cycle.

Answer: A

Conceptual Summary Example

1. Write a summary of reaction for aerobic respiration, showing which reactant become oxidized and which becomes reduced.

Explanation:



2. Indicate where each stage of aerobic respiration takes place in eukaryotic cell.

Explanation: Glycolysis occurs in the cytosol, and the remaining stages of aerobic respiration takes place in the mitochondoria.

3. Add up the energy captured (as ATP, $NADH_1$ and $FADH_2$) in each stage of aerobic respiration.

Explanation:

- In glycolysis ⇒ 2NADH and 2 net ATP molecules are produced.
- In link reaction ⇒ 2NADH are formed
- In the citric acid cycle ⇒ 2 acetyl CoA molecoles are metabolized to form 6NADH, $2FADHA_2$ and 2ATP molecules
- As a result 4ATP, 10 NADH and $2FADH_2$ molecules are formed from the above reactions. When 10 NADH and $2FADH_2$ pass through the electron transport chain, 32 to 34ATP molecules are produced by **chemiosmosis**. Accordingly, each glucose molecule produces 36 to 38 ATP molecules.

4. Describe the process by which the proton gradient drives ATP synthesis in Chemiosmosis.

Explanation: The diffusion of protons through channels formed by the enzyme **ATP synthase**, through the inner mitochondrial membrane from the inter membrane space to the mitochondrial matrix. Provides the energy to synthesize ATP.

5. Compare and contrast anaerobic respiration and fermentation;

Explanation:

- In **anaerobic respiration** electrons are transferred from fuel molecule to an electron transport chain, 2ATPs are produced per glucose molecule.
- **Fermentation** is anaerobic process that do not use an electron transport chain 2ATPs are produced per glucose molecule.

Post test I

1. The process of splitting larger molecules in to smaller ones is an aspect of metabolism called:

A. anabolism
B. fermentation
C. catabolism
D. oxidative phasphorylation
E. chemiosmosis

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2. The synthetic aspect of metabolism is called:
   A. Anabolism
   B. Fermentation
   C. catabolism
   D. oxidative phosphorylation
   E. Chemiosmosis

3. A chemical process during which a substance gains electrons is called:
   A. Oxidation
   B. Oxidative phosphorylation
   C. Deamination
   D. Reduction
   E. Dehydrogenation

4. The pathway through which glucose is degraded to pyruvate is called:
   A. Aerobic respiration
   B. The citric acid cycle
   C. The oxidation of pyruvate
   D. Alcohol fermentation
   E. Glycolysis

5. The reactions of _____ takes place within the cytosol of eukaryotic cells.
   A. Glycosis
   B. Oxidation of pyruvate
   C. Citric acid cycle
   D. chemiosmosis
   E. Electron transport chain

6. Before pyruvate enters the citric acid cycle, it is decarboxylated, oxidized and combined with coenzyme A, forming acety CoA, carbondioxide, and one molecule of.
   A. NADH
   B. $FADH_2$
   C. ATP
   D. ADP
   E. $C_6 H_{12} O_6$

7. In the first step of the citric acid cycle, acety CoA reacts with oxaloacetate to form:
   A. Pyrovate
   B. Citrate
   C. NADH
   D. ATP
   E. Co_2

8. Dehydrogenase enzymes remove hydrogen atoms from fuel molecules and transfer them to acceptors such as:
   A. O_2 and H_2O
   D. CO_2 and H_2O

   E. COA and pyruvate
   B. ATP and FAD
   C. NAD^+ and FAD

9. Which of the following is a major source of electrons for the electron transport chain?
   A. H_2O
   B. ATP
   C. NADH
   D. ATP synthese
   C. Coenzyme A

10. In the process of _____, electron transport and ATP synthesis are coupled by a proton gradient across the inner mitochondrial membrane.
   A. Chemiosmosis
   B. Deamination
   C. Anaerobic respiration
   D. Glycolysis
   E. Decarboxylation

11. Which of the following a common energy flow sequence in aerobic respiration, starting with the energy stored in glucose?
   A. Glucose → NADH → pyrovate → ATP
   B. Glucose → ATP → NADH → Electron transport chain
   C. Glucose → NADH → Electron transport cahin → ATP
   D. Glucose → Oxygen → NADH → water
   E. Glucose → $FADH_2$ → NADH → Coenzyme A

12. When deprived of oxygen, yeast cell obtain energy by fermentation, producing carbondioxide, ATP, and _____.
   A. Acetyl CoA
   B. Ethyl alcohol
   C. Lactate
   D. pyruvate
   E. Cirate

13. During sternous muscle activity, the pyruvate in muscle cells may accept hydrogen from NADH to become _____.
   A. Acetyl CoA
   B. Ethyl alcohol
   C. Lactate
   D. Pyruvate

Answers for Post test 1

1.C	2.A	3.D	4.E	5.A	6.A	7.B	8.C	9.C

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10.A 11.B 12.C

5.2. Photosynthesis

Photosynthesis is the process by which light energy is converted in to chemical energy of organic molecules. So, photosynthesis involves transduction of **light energy** in to **chemical energy.**

Photosynthetic organisms are capable of converting in organic compounds **carbondioxide (CO_2)** and **water (H_2O)** in to organic compounds by using light energy. Light energy is absorbed by a pigment known as **chlorophyll**. The whole process of photosynthesis takes place in a cell organelle called **chloroplast.**

The process of photosynthesis is carried out by groups of organisms called **photoautothrops**. These organisms are plants, algaes and photosynthetic bacterias.

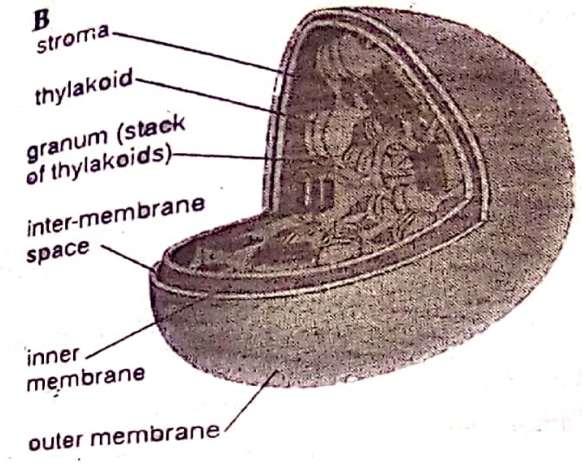
As mentioned above the process of photosynthesis involves the conversion of one form of energy in to another. This process is called energy **transduction**.

5.2.1. Structure and Function of Chloroplast

Chloroplast is an organelle which is enclosed by double membrane. The major parts of chloroplast are:

- ✓ **Grana** – is the sack of thylakoids where the light dependent reaction of photosynthesis takes place.
- ✓ **Thylakoids** – are layers of the inner membranes of chloroplast.
- ✓ **Stroma** – is a part where the light independent reaction (calvin cycle) of photosynthesis takes place.

B



5.2.2. The nature of light

Light is composed of small particles or packets of energy called **photons.** This forms a continous range of radication which is called **electromagnetic** spectrum. Among these, the visible spectrum of light only represents a small portion. Visible spectrum of flight contains all the colours of rainbow. All the radiation in the visible spectrum travels in waves and the distance from one peak to the next is called **wave length**. It is measured in fractions of nanometer.

The visible spectrum of light consists of spectra of the light that human beings can see.

The visible slight spectrum ranges within a wave length of 380 to 760nm. The colours of light in a visible spectrum of light according to increasing level of their wave length can be arranged as:

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V ⇒ violet
I ⇒ Indigo
B ⇒ Blue
G ⇒ Green
Y ⇒ Yellow
O ⇒ Orange
R ⇒ Red

Accordingly, violet has the shortest wavelength and red has the long wave length.

5.2.3. The Process of Photosynthesis

The process of photosynthesis involves two major stages:

- ✓ Light dependent stage (Light reaction)
- ✓ Light independent stage (Dark reaction)

Light dependent stage is called **light reaction** because it is the stage in which light energy is directly required. While, the light independent stage is called **dark reaction** because at this stage light energy is not directly required.

Note: Both light dependent and light independent stages of photosynthesis occur at the day time when there is light.

i) Light dependent stage of photosynthesis

During the light dependent reactions, the radiant energy from the sun is used to:

- ✓ Phosphorylates ADP in order to produce ATP.
- ✓ Reduces $NADP^+$ in order to form NADPH.

Therefore, the light energy that is captured by chlorophyll is temporarily stored in ATP and NADPH. Thus:

- ✓ **ATP** – is used to provide energy for the light independent reactions of photosynthesis.

- ✓ **Reduced NADP (NADPH)** – provides hydrogen ions for reduction reaction in which CO_2 gets reduced to form glucose ($C_6H_{12}O_6$).

Note: NADP and **NAD** are very similar in function because they are used in transporting (carrying) hydrogen ions.

The light dependent reactions of photosynthesis begins when chlorophyll and other pigments absorb light. The chlorophyll molecules, accessory pigments and associated electron acceptors are organized in to units called **photosystems.**

Photosystems are biochemical substances by which chlorophyll absorbs light energy. There are two different photosystems each containing light harvesting complex pigment. This cluster of pigment molecules are called **antenna complex.**

Different pigment molecules in the antenna complex can absorb different wave lengths of light. These pigments include:

- Chlorophyll – a
- Chlorophyll – b
- Carotenoids

Before going in to the detail discussion about photosystems in a light dependent stage of photosynthesis, let us discuss some about the pigment molecules that are absorbed by antenna complex.

- ✓ **Chlorophyll – a** is the reaction center molecule which is very important in initiating the light dependent reactions of photosynthesis.
  - It appears bright green.
- ✓ **Chlorophyll – b** is accessory pigment. It differs from chlorophyll – a by its functional group in the prophyrin ring.

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- The methyl group (CH_3) in chlorophyll – a is replaced in chlorophyll – b by a terminal carbonyl group (– CHO).
- Chlorophyll – b appears yellow green.

✓ **Carotenoids** are the accessory pigments used in photosynthesis. They appear yellow and green.

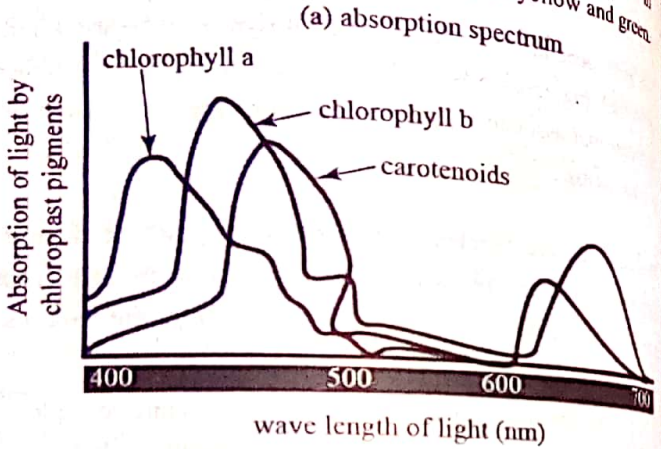

Fig. The structure of chloroplyll

Fig. The absorption spectrum of Chlorophylla, chlorophyll – b and carotenoids.

The two photosystems that are involved in the light dependent reactions of photosynthesis are **photosystem – I** and **photosystem – II**. They contain about 200 to 300 pigment molecules which are highly associated with specific enzymes and other proteins.

Photosystem I contains special form of chlorophyll – a which has an absorption peak at 700nm. Thus, it is referred to as **p – 700.**

Photosystem II contains another form of chlorophyll – a with an absorption peak of 680nm. Thus, it is referred to as **p – 680.**

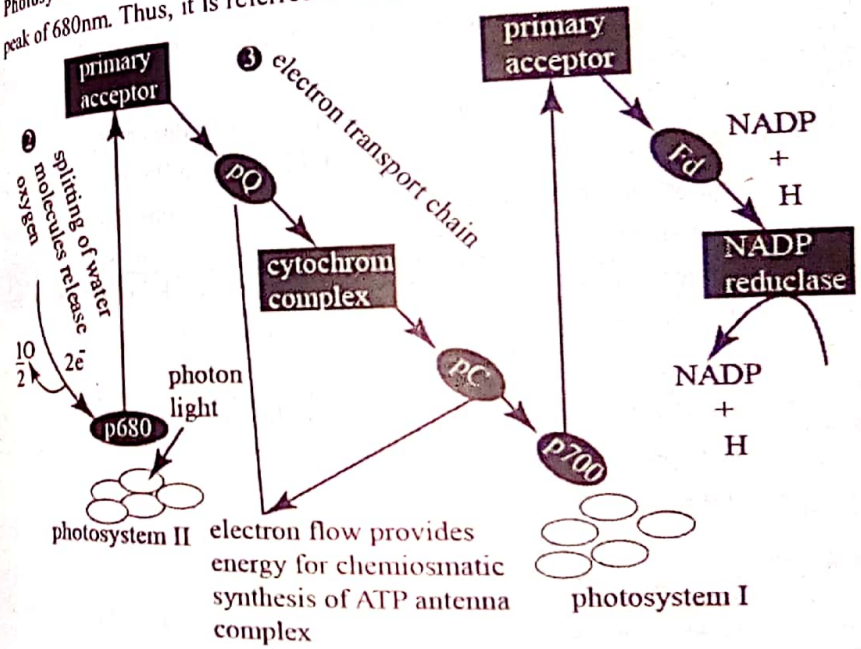

Fig. The Light dependent reactions of photosynthesis

As illustrate in figure above:

1) Electron (e) chlorophyll molecules in photosystem II are excited by energy in photons (sunlight) and they become more energetic. The electrons escape from the chlorophyll and pass into primary electron acceptor.

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2) The light energy splits the water molecule in to hydrogen ions and oxygen.

The splitting of water molecules by light energy in to hydrogen ions and oxygen gas is called **photolysis.**

$$2H_2O \longrightarrow O_2 + 4H^+ + 4e^-$$

The electrons pass to the primary electron acceptor and the process occurs in reaction center which is ps II (P 680)

3) The primary electron acceptor passes the electrons to electron transport chain (ETC). The electrons pass through **plauto quinine (pq), cytochromes,** and **plastocyanine (Pc)** which if the last carrier in the chain.

The electrons lose energy as they pass from one carrier to the next.

4) In the cytochrome complex of ETC a proton (hydrogen ion) pump between the outer membrane to the inner stroma. As the electrons transferred to the ETC the energy they lose powers the pump which moves protons from the stroma to the space inside the thylakoid of chloroplast. At this time, it drives the chemiosmotic synthesis of ATP.

5) Electrons in photosystem I (chlorophyll molecules) are excited when this photosystem absorbs photons of light and escape from the molecule to the primary acceptor.

6) The electrons then pass along a second ETC involving ferredoxin (Fd) and NADP reductase. Finally this electron transport chain reacts with protons and NADP in the stroma to form reduced NADP ($NADPH^+$)

Note: Photosystem II is active up to a maximum of 680nm wavelength of light. While photosystem I is active with in maximum of 700nm wave length of light.

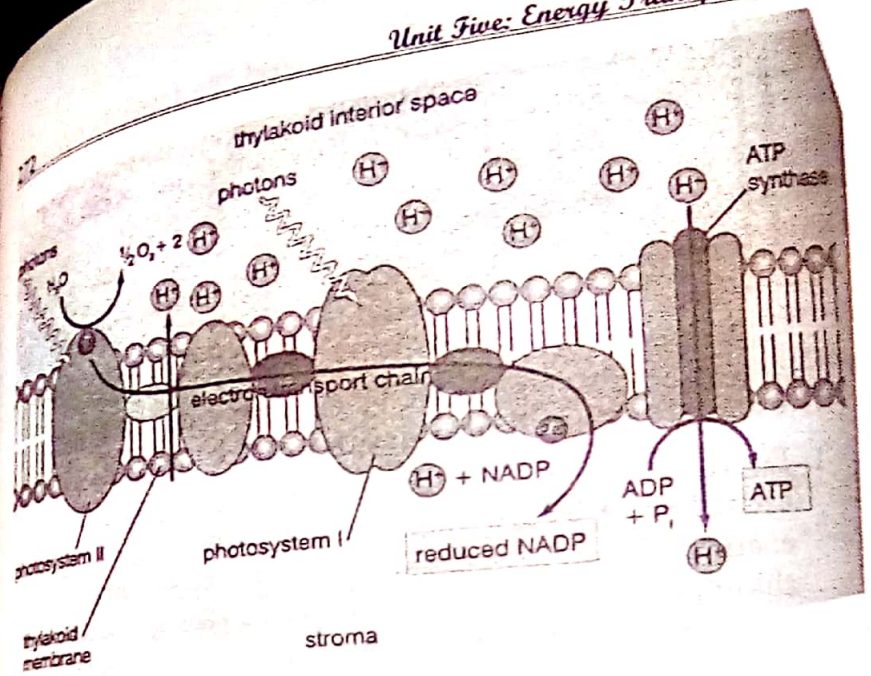

Figure: The arrangement of molecules in a photosynthetic unit

The efficiency of photosynthesis increases as the intensity of light increases up to a limiting value. At the saturation intensity, photosynthesis occurs at its maximum rate. If the light intensity is increased beyond the saturation point, the rate of photosynthesis cannot be increased.

Photosynthic unit is the arrangement of molecules capable of carrying out all the reactions in the light – dependent stage of photosynthesis

Photophosphorylation

Photophosphorylation is the process by which electrons energized by photons (light particles) contribute their energy to the phosphorylation of **ADP** in synthesizing ATP. There are two forms of phosphorylation that are used in the light dependent reaction. They are **cyclic photophosphorylation** and **non-cyclic photophorylation**

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1) Cyclic photophorylation

- Cyclic photophorylation is the simplest route of light dependent reaction in which only photosystemm I is involved. The pathway is cyclic because exicited electrons originate from p 700 at the reaction center and eventually return to p700.
- During the cyclic photophorylation:
  - ✓ no oxygen and no reduced NADP are formed.
  - ✓ Energized electrons (e^-) leave the PSI reaction center chlorophyll-a and are taken up by an electron acceptor.
  - ✓ The whole photosynthetic units are not involved
- Cyclic photophosphorylation is known as back to back phosphorylation from PSI to PSI for the production of ATP. In this process no NADPH is produced. Accordingly, the process of cyclic photophorylation is not a basis for photosynthesis, because NADPH is necessary for CO_2 to be reduced to carbohydrate.
- It is believed by most biologists that, cyclic photophorylation may occur in plants when there is too little $NADP^+$ to accept the electrons from ferredoxin and it is used by ancient bacterias to produce ATP from light energy.

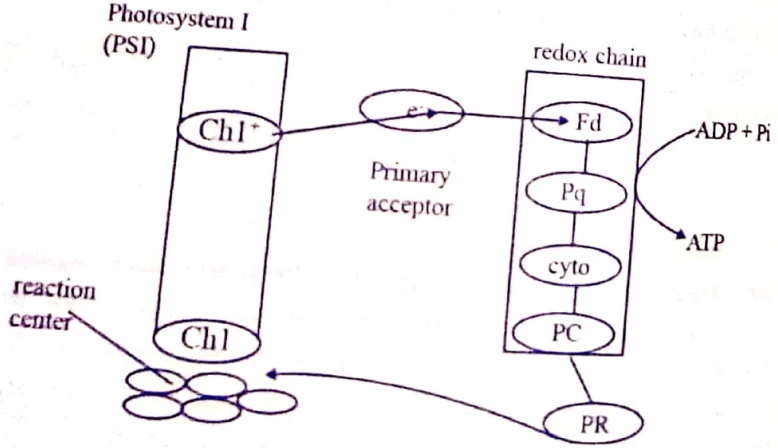

Fig. cyclic photophosphorylation

2) Non-cyclic photophorylation

Non-cyclic photophosphorylation is the most common type of photophosphorylation. During this process.

- ✓ Both photosystems (p-690 and p-700) are used.
- ✓ There is one way of flow of electrons from water to $NADP^+$
- ✓ For every electrons that enter the path way there is an energy yield of two ATP molecules and one NADPH molecule.

Non cyclic photophosphorylation is the more common pathway in which both **photosystem – I** and **photosystem – II** are used. It involves one way flow of electrons from water to $NADP^+$. For every two electrons that enter this pathway, there is an energy yield of two ATP molecules and one NADPH molecule.

The non-cyclic photophorylation occurs through the following pathway of events:

1. Electrons (e^-) in chlorophyll molecules in photosystem II are excited by the energy in photons of light and become energetic. The extra energy escape from the chlorophyll (PS – II) and pass to the **primary electron acceptor**.

   The energy created in ps II causes a water molecule to split in to hydrogen ion and oxygen.

   $$2H_2O \longrightarrow O_2 + 4H^+ + 4e^-$$

   The process by which the light energy splits water molecule is called **photolysis**.
2. The electrons pass from the primary electron acceptor to the **plastoquinine** (pq) in an electron transport chain.

   The electrons again pass to the cyctochrome complex and then to the **plastocyanin** (Pc) which is the last electron carrier in the chain. There is loss of energy of the electrons as they pass from one carrier to the next.

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The electrons that flow (are lost) provide energy for chemiosmotic synthesis of ATP.

3. The PSI absorbs photons of light which makes the electrons in the PS – I to get exited. After getting excited the electrons start to escape and pass along the electron transport chain through **ferredoxin** (Fd) and **NADP** reductase.

Note: NADP reductase are enzymes that make the NADP to react with hydrogen ions in order to form a reduced NADP (NADPH).

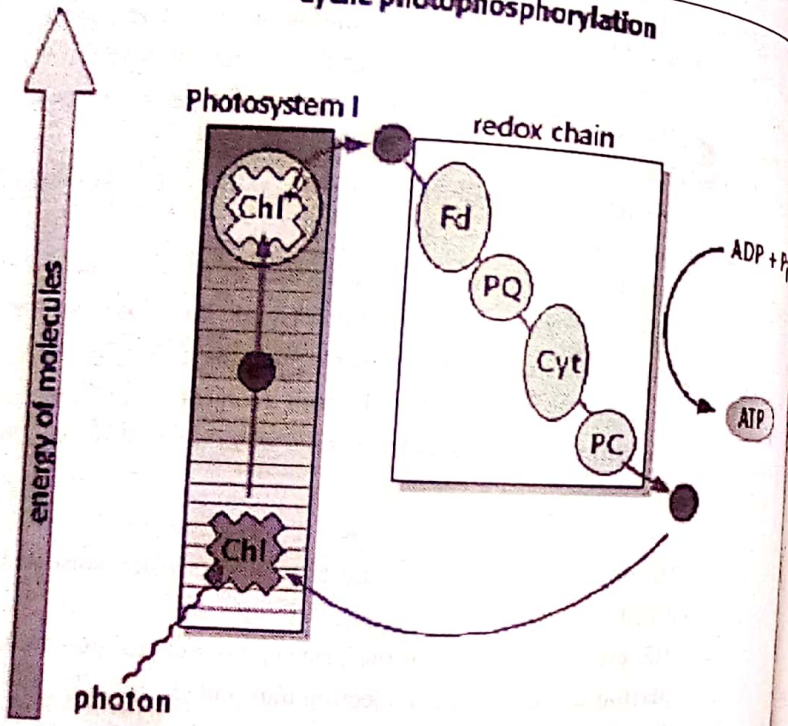

Fig. 5.21 The light dependent reactions of photosynthesis in non-cyclic photophosphorylation

The arrangement of molecules in the photosynthesis unit is shown in the following figure.

Table 5:5. The comparision of the cyclic and non-cyclcic photophorylation

	Non-cyclic photophosphorylation	Cyclic photophosphorylation
Pathway of electron	Non – cyclic	Cyclic
First electron donor (source of electron)	Water	Photosystem (PSI) p700
Last electron acceptor	NADP	Photosystem I (p 700)
Products	ATP, NADP H^+ and oxygen	ATP only
Photosystem involves	PSI and PS II	PSI only

A summary of the light dependent reaction

- Light energy is used to excite electrons in the chlorophyll.
- The excited electrons cause protons to pass to inside of the **thylakoid membrane.**
- The passage of electrons along the first electron transport chain leads to synthesis of ATP.
- The excited electrons react with **hydrogen ions** and **NADP** at the end of the second electron transport chain to form a **reduced NADP($NADPH^+$).**
- Finally, the **ATP** and reduced **NADP** ($NADPH^+$) are used to drive the synthesis of carbohydrates in the light in dependent reaction of photosynthesis.

5.2.4. Factors affecting the rate of photosynthesis

The process of photosynthesis can be affected by number of factors. Among these, the major ones are environmental factors. These include:

a. Light intensity

Light intensity is the amount of light in an environment and it varies in different ways.

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✓ **Cold bright day (arctic area)**

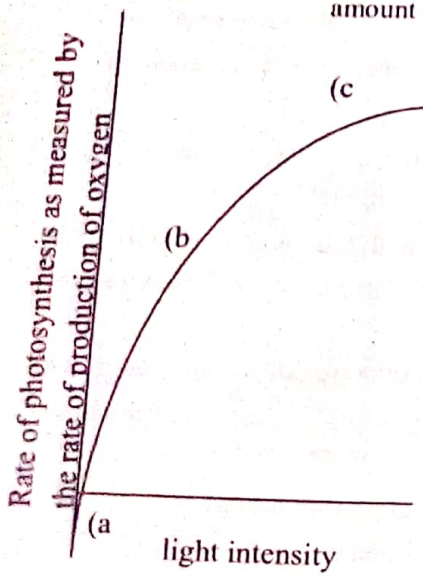
➢ High light intensity but low temperature. Thus, the low temperature affects the enzymatic activity and limit the rate of photosynthesis.

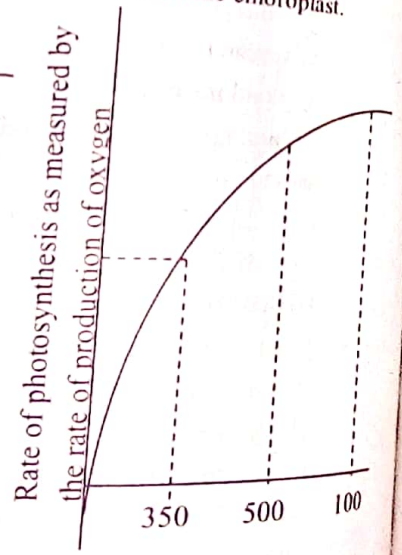
✓ **On warm cloudy day**

➢ This condition shows low supply of light intensity but high temperature. Thus, low light intensity affects the number of electrons excited in the light dependent reaction on antenna complex and reaction center of chlorophyll molecule.

✓ **On warm sunny day**

➢ This is a condition where the temperature and light intensity is high. The rate of photosynthesis is affected since the stomatal pores close and sufficient amount of Co_2 do not enter the chloroplast.





b. **The effect of the concentration of carbondioxide on the rate of photosynthesis**

At very low concentration of CO_2:

✓ The rate of photosynthesis declines although the process of respiration takes place by using oxygen. Therefore, the CO_2 concentration increases in the atmosphere.

c. **The effect of temperature on the rate of photosynthesis**

Many reactions of photosynthesis in both light dependent reactions and light independent reactions are regulated by enzymes. Thus, the change in temperature affects enzymatic activity that regulate the rate of photosynthesis.

Other factors that affect the rate of photosynthesis

d. **The wave length of light**

- Photosynthesis takes place efficiently in red and blue wave length than other wavelengths. This is because red and blue wavelength absorb more light than others.

e. **The amount of Chlorophyll Concentration**

- Low chlorophyll concentration can result in decreased rate of photosynthesis.

f. **Water**

- Water is the raw material in photosynthesis. The lack of water affect the rate of photosynthesis. It may even result in wilting and drying of the plant.
- When the amount of water is very high from the normal condition, it affects the rate of the soil. This decreases the rate of photosynthesis.

Summary of factors that affect the rate of photosynthesis

- The rate of photosynthesis is influenced by:
  - ✓ Light intensity
  - ✓ Carbondioxide concentration

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✓ Temperature

- Low light intensity affects the rate of photosynthesis. It also affects the amount of temperature and carbondioxide concentration.

Illustrative Examples

40. Where in plant cell does the krebs cycle take place?
A. Nucleus C. Mitochondoria
B. Cytoplasm D. Chloroplast

Explanation: Krebs cycle (citric acid) cycle takes place in the mitochondria of the cell.

Answer: C

41. An enzyme known as catalase converts hydrogen peroxide in to water and oxygen. In this reaction, which one is the substrate?
A. Catalase C. Water and oxygen
B. Hydrogen peroxide D. water only

Explanation: **Subtrate** is a **substance** on which an enzyme acts. Thus, catalase is an enzyme and hydrogen peroxide is a substrate.

Answer: B

42. Which of the following is true about the first stage of photosynthesis?
A. Light dependent C. ATP driven
B. Temperature dependent D. Glucose driven

Explanation: The first stage of photosynthesis is the light dependent stage in which the light energy splits water molecule in to hydrogen ion and oxygen.

$$12H_2O \xrightarrow[Chlorophyll]{Light} 24H^+ + 6O_2$$

Answer: A

43. The carbon dioxide produced during the respiratory cycle and other processes is largely absorbed by which of the following?
A. Ozone layer C. Plants

B. Animals D. Sky

Explanation: Plants require carbondioxie to synthesize glucose during the process of photosynthesis. This process occurs during the light independent stage of photosynthesis.

$$24H^+ + 6Co_2 \longrightarrow C_6H_{12}O_6 + + 6H_2O$$

Answer: C

44. In cyclic photosphosphorylation, what is the source of the recycled electron?
A. Reduced NADP.
B. Chlorophyll molecule.
C. Adenosine triphosphate.
D. Photolysis of water molecule.

Explanation: In cyclic photophosphorylation only ATP is regenerated in photosystem I, no oxygen and no NADPH and formed. It is the cyclic event between PSI and ETC to regenerate ATP.

Answer: C

45. What is the source of oxygen,that is produced during the process of photosynthesis by higher plants?
A. Co_2 B. H_2O C. ATP D. Chlorophyll

Explanation: The source of oxygen during the process of photosynthesis is water molecule. For more explanation see example number 43.

Answer: B

46. Which of the following is NOT true about photosystem II?
A. It is reaction center molecule is P680.
B. It passes its excited electron to photosystem I.
C. The energy lost from its exited electrons reduces NADP
D. It replenishes its lost electrons from photolysis of water.

Explanation: In photosystem II energy is not lost but it is gained and used to excite electrons to reduce NADP.

Answer: C

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47. What is the importance of chemiosmosis in photosynthesis and cellular respiration?
   A. Splitting water molecule.
   B. Operating the proton pump.
   C. Combining hydrogen and carbon.
   D. Synthesizing ATP.

Explanation: **Chemiosmosis** is the diffusion of hydrogen ions or proton pumps through ATP synthase to produce ATP.

Answer: B

48. Where does the light dependent reaction of photosynthesis occur in the chloroplast?
   A. In the thylakoid membrane
   B. In the fluid of the stroma.
   C. In all parts of the chloroplast.
   D. In the stomatal opening.

Explanation: The light dependent reaction of photosynthesis is the first stage of photosynthesis. It takes place in the staks of granna in the thylakoid membrane.

49. Light independent reaction of photosynthesis is called dark reaction because:
   A. It takes place in the stroma.
   B. It doesn't require light.
   C. It's end products are glucose and oxygen.
   D. It takes place only at the night time.

Explanation: The light independent stage of photosynthesis is called dark **reaction** because it do not require light. However, this does not mean it takes place at a dark time.

Answer: B

50. The chlorophyll pigment that can transform light energy directly in to chemical energy is:
   A. Chlorophyll "b"
   C. chlorophyll 'a'

   B. Xanthophylls
   D. carotene

Explanation: Chlorophyll "b" and other accessory pigments are found in the antenna complex, but chlorophyll 'a' is found in the reaction center (p680 and P700).

Answer: A

51. What is the name of an enzyme that catalyze the combination of RuBP with CO_2 molecule in the calvin cycle of photosynthesis?
   A. RuBP carboxylase.
   B. PEP carbozxylase.
   C. Reduced NAD dehydrogenase.
   D. ATP synthase.

Explanation: In light dependent reaction (calvincycle) Ribulose which is found in the stroma bind with carbondioxide by an enzyme **RuBP carboxylase** (Rubisco).

Answer: A

52. Based on the given chemical equation.

$12H_2O + 6Co_2 \longrightarrow C_6H_{12}O_6 + 6O_2 + 6H_2O$

The metabolic process in directed by the above reaction is:
   A. Photosynthesis
   B. Respiration
   C. Transpiration
   D. Osmosis

Explanation: The metabolic process that is indicated above is **photosynthesis**.

Answer: A

53. Which of the following process occurs during the light independent reactions of photosynthesis?
   A. ATP formation
   B. Splitting of water
   C. Excitation of electrons
   D. fixation of carbondioxide

Explanation: Splitting of water molecules, exciting of electrons and ATP formation takes place in thylakoid membrane during **light dependent reaction**. The fixation of CO2 takes place in the

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fluid stroma of the chloroplast during light independent reaction.

54. The reactive chlorophyll molecule of photosystem I is: **Answer: B**

A. P600 C. P680
B. P700 D. P 550

Explanation: The reaction center the chlorophylia molecule of night spectrum in the photosystem is P700.

55. Plants obtain nitrogen for protein synthesis from the soil in the form: **Answer: B**

A. Nitrogen gas C. Nitrate
B. Ammonia D. Nitrite

56. Which of the following event does not happen during light phase of photosynthesis?

A. Reduced NADP C. photolysis of water
B. ATP generation D. Reduction of CO_2

Explanation:

- During the light dependent state of photosynthesis
  - ✓ ATP is generated.
  - ✓ Water molecule splits (photolysis).
  - ✓ NADP is reduced and $NADPH^+$ is formed.

Answer: D

57. When a photon strikes photosystem I (PSI). The electron carrier that pass electron to NADP is:

A. Primary electron acceptor C. Ferridoxin
B. Plasto cyanine D. obiquinone

Answer: C

58. Which of the following molecule links light and dark reaction of photosynthesis?

A. Acety – CoA C. Pyruvate
B. ATP D. NADP

Explanation: Acety CoA and pyruvate are the products of respiration.
- The products of light reaction which are transferred to dark reaction are ATP and **NADPH**.

Answer: B

59. The ultimate source of electron for light dependent reaction is:

A. Water C. p700
B. Glucose D. light

Explanation: The electrons and protons are formed from the hydrolysis of water. The electrons pass from p 680 to p 700 and then to NADpH.

Answer: A

60. Which of the following is common to both light and dark reaction of photosynthesis?

A. Photosystem C. Redox reaction
B. Electron transport chain D. ATP synthesis

Explanation: Photosystem, electron transport chain and ATP synthesis are involved in light reaction. However, redox reaction is involved in both light and dark reaction of photosynthesis.

Answer: C

61. Which of the following occurs in both photosynthesis and respiration?

A. Glycolysis C. krebs cycle
B. Chemiosmosis D. calvin cycle

Explanation: Chemiosmosis is the process which is used to produce ATP by diffusion of protons pomp through ATP synthase.

Answer: B

62. All the following are the inputs of calvin cycle except:

A. ATP C. CO_2
B. NADPH D. O_2

Explanation: The raw materials for the light independent reactions are ATP, NADPH, CO_2 and RuBP, but oxygen is a byproduct.

Answer: D

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The light independent Reactions of photosynthesis

The light independent reaction of photosynthesis involves fixation of carbondioxide from the air for synthesis of glucose (sugar).

During the carbon fixation, the energy from ATP and NADPH is used for the formation of organic molecules from Co_2.

The reaction of carbon fixation can be summarized as:

$$12NADPH + 18ATP + 6Co_2 \longrightarrow C_6H_{12}O_6 + 12NADP^+ + 18ADP + 18\ pi + 6H_2O$$

The carbon fixation accors in different cycles according to the difference in the nature of the plants.

They are **C_3 pathway** (calvin cycle) and **C_4 pathway**.

C_3 pathway (clavin cycle)

The C_3 pathway is also known as the **calvin cycle**. This cycle involves a sequence of reactions in which the carbon fixation occurs. It takes place in the **stroma** of the chloroplast.

The calvin cycle is first introduced by **Melvin calvin, Andrew Benson** and other team of researchers from California university. For this contribution they have won a nobel prize of chemistry in 1961.

The product in the initial carbon fixation reaction is a thre carbon compound. Plants that use C_3 pathway are called **C3 plants**.

C_3 plants have the following adaptations for photosynthesis:

- ✓ The have broad leaves to catch more sunlight.
- ✓ Their palaside cells are nearest to the upper surface of the leaf to absorb more light as possible.
- ✓ Their stomata are open for most of the time to allow the entry of Co_2, but closed if water loss is too high on a hot day.

- ✓ Spongy mesophyll has air spaces that allow easy diffusion of carbondioxide and oxygen between the palisade layer and stomata.

The calvin cycle occurs through the following three major stages:

1) **Carbondioxide fixation (CO_2 Up take)**
   i) CO_2 reacts with a phosphorylated five carbon compound called **uribulose biphosphate (RUBP)**. This reaction is catalyzed by an enzyme **rubisco bis-phosphate carboxylase/ oxygenase (rubisco)**. As a result 6 carbon containing unstable molecule is formed.
   ii) The unstable six carbon molecule immediately breaks in to two molecules of **phosphoglycerate (PGA)** which consists of 3 carbons each .
2) **Carbon reduction**
   This step involves energy found in ATP and NADPH (produced in light dependent stage) are used to convert the **PGA molecule** to **glycerol dehyde – 3 – phosphate (G_3P)**.
   The reaction of two molecules of GP leads to the formation of glucose or fructose.
3) **RuBP regeneration**
   - Among 6 molecules of G3P, 1G3p molecules were used to synthesize a six carbon molecule **glucose** or **fructose** and 5G3P molecules remain; this contains 15 carbon atoms at all.

     Through a series of reactions 15 carbons and their associated molecules get arranged in to **3 molecules** of **ribulose biphosphate**. Each molecule of ribulose bi phosphate is phosphorylated to become **RuBP** (a 5 – carbon compound with which the cycle started).

> **Note:** Each of the 3 carbon molecules G3P molecules is essentially half of a hexose (six-carbon sugar) molecule, are joined in pairs to produce glucose or fructose. (In some plants like sugar cane they are joined to produce sucrose).

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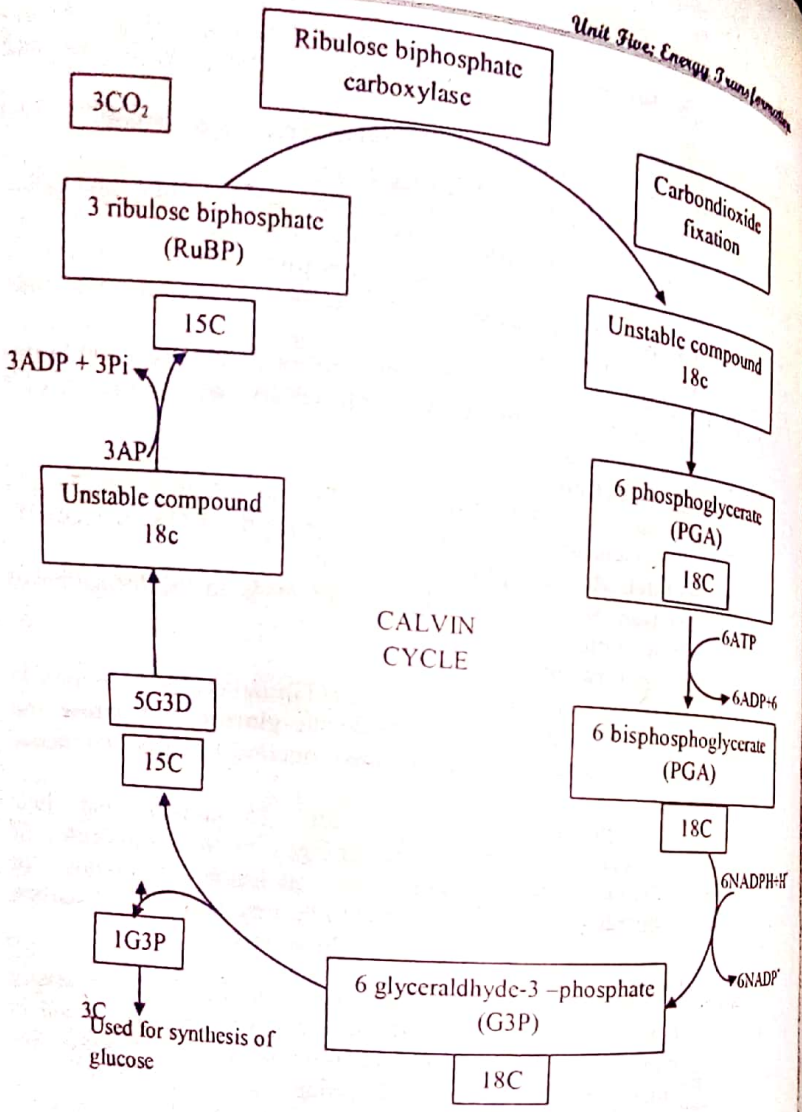

Fig. The calvin cycle

Photorespiration

In some cases, specially during very hot weather most C_3 plants do not yield as much carbondioxide as expected for photosynthesis. This because the stomata closes to prevent loss of water, but this prevents the entrance of CO_2 in to the leaf. Thus, the amount of O_2 becomes higher than CO_2 and O_2 competes with CO_2 for the active site of **RUBP carboxylase.** *RuBP oxidized in to a three carbon molecule* **phosphoglycerote (GP)** *and* **phosphoglycolate** *(a two – carbon molecule).*

Ribulose biphosphate + oxygen ⟶ GP + phospholycolate

Phosphoglycolate *passes through a series of metabolic reactions and forms carbondioxide in mitochondoria.*

Note: The process of photorespiration organelles of the cell; chloroplast, peroxisome and mitochondria.

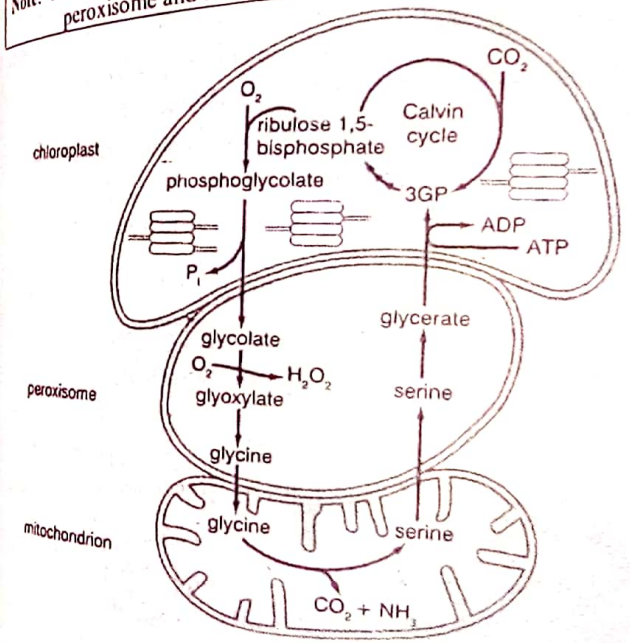

Fig. Reactions of photorespiration

The C_4 pathway

The first compound formed in the light independent reaction is C_4 compound (contains for carbon atoms) unlike C_3 plant in which the first molecule formed ion GP (3 carbon compound).

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There is a difference between the structure of leaf between C_3 and C_4 plant In C_4 plants:

- ✓ The cells of the bundle sheath contain chloroplast.
- ✓ The light independent reactions takes place in mesophyll cells, which have chloroplast with thylakoids.

The plants that use C_4 path way use the following steps instead of using the calvin cycle:

1. Carbondioxide reacts with a three carbon compound called **PEP** to form a fourcarbon compound called **oxaloacetate**. This reaction is catalyzed by an enzyme called **PEP carboxylase.**
2. Oxaloacetate is converted in to another C_4 compound called **malate**, which passes from the Mesophyll in to bundle sheath cell.
3. In the bundle sheath, Malate is converted to pyruvate with the release of a molecule of carbondioxide, which starts the reactions of the calvin cycle by binding with RuBP.
4. The pyruvate is converted back to PEP, this reaction requires ATP.

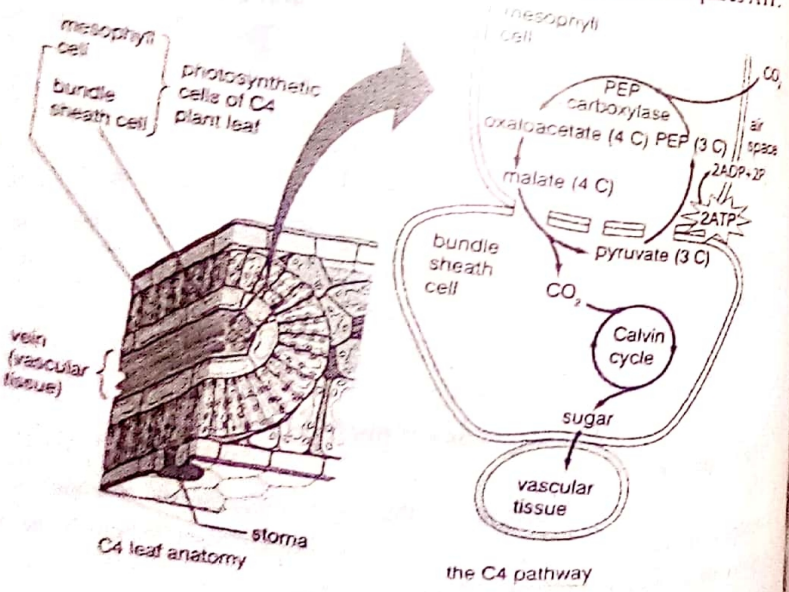

Fig. C_4 pathway

C_4 plants are plants that first appeared in geographical areas with high temperature, high light intensities and limited amount of water. Thus, C_4 plants:

- ✓ Have high optimum temperature.
- ✓ Lose less water by transpiration.
- ✓ Higher rate of photosynthesis and growth than. plants that use only the calvin cycle.

The most essential feature of C_4 plants is, they can concentrate carbondioxide in their cell, (Thus, the process of photosynthesis and growth continue efficiently even under adverse (unsuitable) conditions.

Some examples of C_4 plants are; maize, sorghum and sugar cane.

CAM (Crasslacean acid) Metabolism in plants

CAM uses PEP carboxylase to fix some CO_2 at night time forming C_4 (four carbon) molecules which is stored in large vacuoles of the mosophyll cells. The crassulacean plants are the family of flowering succulent plants that live in the warm and arid regions of the world. CAM plants mostly grow in desert area. These plants include pineapple, cactus, etc.

The CAM photosynthesis occurs in the following sequence:

1) At night plants open their stomata to take in CO_2. The CO_2 then reacts with PEP in mesophyll cells to form oxaloacetate and malate as in C_4 plant pathway.
2) The malate is then stored in the vacuole of mesophyll cells over night.
3) During the day, the light dependent reaction generate ATP and reduced NADP; and the calvin cycle continues.
4) Malate is released from the vacuoles and break down to glycerate, releasing CO_2 fixation at night when the stomata are open. During the day, stored Co_2 is used for the calvin cycle.

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In general, CAM plants use PEP case to carry on CO2 fixation at night, when the stomata are open. During the day to stored CO_2 enter the calvin cycle.

Table: The pathway of CO_2 Fixation

Feature	C_3	C_4	CAM
Leaf structure	Bundle sheath cells lacking chloroplast	Bundle sheath cells having chloroplast and no thylakoids.	Large vacuoles in mesophyll cells.
Enzyme used to fix CO_2	Rubisco	PEP case	PEP case
Optimum temperature	$15 - 25^0$c	$30 - 40^0$c	35^0c
Optimum concentration	700ppm	400ppm	400ppm
Fixation of CO_2	Mesophyll cells	Mesophyll cells	Mesophyll cells
Calvin cycle	Mesophyll cells	Bundle sheath cells	Bundle sheath cells

Summary of the reactions for photosynthesis

- The light dependent reactions:

  $12H_2O + 12NADP^+ + 18ADP + 18Pi \longrightarrow 6O_2 + 12NAPH + 12H^+ + 18ATP$

- The light independent reactions:

  $12NADPH + 12H^+ + 18ATP + 6CO_2 \longrightarrow C6H_{12}O_6 + 12NADP^+ + 18ADP + 18Pi + 6H_2O$

- The overall reaction of photosynthesis:

  $6CO_2 + 12H_2O \longrightarrow C_6H_{12}O_6 + 6O_2 + 6H_2O$

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Summary of principal reactions of photosynthesis

Reaction series	Process	What is needed	End products
1. Light dependent reactions (take place in thylakoid membrane) i) Photochemical reaction	Energy from sunlight used to split water, manufacture ATP, and reduce $NADP^+$ chlorophyll energized; chlorophylla in reaction center gives up energized	Light energy pigments such as chlorophyll	Electrons
ii) Electron transport	electron to electron acceptor Electrons are transported along the chain of electron acceptors in thylakoid membranes; electrons reduced NADP; splitting of water	Electrons, NADP, H_2O , electron acceptors	NADPH + H+, O_2, H^+
iii) Chemiosmosis	provides some of H^+ that accumulates inside thylakoids. H^+ molecules are permitted to move a cross thylakoid membrane down proton gradient; they cross membrane through special channels in ATP	Proton gradiant, ADP + Pi	ATP
Light independent reactions (takes place in stoma)	synthetase; energy released is used to produce ATP Carbon fixation; carbondioxide is	Ribuluse biphosphate CO_2, ATP NADPH+ H^+	Carbohydrate ADP + Pi, $NADP^+$

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Table: Comparision of light dependent and light in dependent reactions of photosynthesis

Feature	Light dependent stage	Light independent stage
Takes place	In the tylakoid membrane of grannel stok structure of chloroplast	In the stroma of chloroplast
Light requirement	Requires light	Do not require light, but occurs at day time
Reaction process	• Light energy excite from the reaction center and primary acceptor to ETC. • Photolysis of water molecule. • Cyclic and non-cyclic photo – phosphorylation occurs.	• CO_2 fixation with ribulose by RuBP enzyme (Rubisco) • Reduction of ATP and NADPH. • Regeneration of RuBP
Raw materials	Light energy, water ADP and NADP	• CO_2, RuBP, ATP and NADPH
End product	18ATP, 12NADPH, 6 molecule of oxygen	• 18 ADP+ 18pi, 12NADP, $C_6H_{12}O_6$ and RuBP
Overall chemical equation	$12H_2O \xrightarrow{Light} 24H^+ + 6O_2$	• $3CO_2 + 24H \xrightarrow[C_6H_{12}O_66 + H_2O]{18ADP + 18Pi}$

The overall reaction of photosynthesis is:

$$6H_2O + 6CO_2 \xrightarrow[chlorophyl]{sun\ light} C_6H_{12}O_66 + 6CO_2$$

Illustrative Example

63. In which of the following activity of C_4 plants are more efficient than C_4 plants?
A. light absorption
B. photolysis
C. chemiosmosis
D. Fixation of CO_2

Explanation: C_4 plants are more efficient in absorption of CO_2, because they produce a compound called **phosphoenol pyruvate** which has high affinity of CO_2.

Answer: D

64. CAM photosynthesis is:
A. The same as C_4 photosynthesis.
B. An adaptation to cool environment.
C. Prevalent in desert plants.
D. Occurs in plants that live in empty areas.

Explanation: CAM photosynthesis is mostly observed in desert plants.

Answer: C

65. Which one of the following should not be associated with electron transport chain (ETC)?
A. Cytochromes.
B. Movement of H^+ to the thylakoid space.
C. Formation of ATP.
D. Absorption of solar energy.

66. A photosystem consists of:
A. Pigments, reaction center and electron acceptor.
B. ADP, pi and hydrogen ions (H^+)
C. Protons, photons and pigments.
D. Electron transport carriers.

Explanation: A photosystem consists of:
- ✓ Chlorophyll molecules.
- ✓ Accessory pigments.
- ✓ Associated electron acceptors.

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✓ Reaction center molecules (P680 and P700).

67. PEP case has an advantage over Rubisco in that: **Answer: A**

A. PEP case is present in both mesophyll and bundle sheath cells, but Rubisco is not.
B. Rubisco fixes CO_2 only in C_3 plants, but PEP case does it in both C_3 and C_4 plants.
C. Rubisco is subject to photo respiration, but PEP case is not.
D. PEP case is fixed with CO_2 in C_3 plants.

Explanation: Rubisco enzyme contains both carboxylase and oxygenase which are used during photorespiration.

68. Which of the following is NOT true about fermentation? **Answer: C**

A. Net gain of only two ATP.
B. Occurs in cytosol.
C. NADH donates electrons to electron transport chain.
D. Begins with glucose.

Explanation: During fermentation NADH do not donate electrons to electron transport chain, because fermentation occurs by glucolysis in the cytosol of the cell.

Answer: C

69. The link reaction is:

A. Connects glycolysis to the krebs cycle.
B. Gives off CO_2.
C. Utilizes NAD.
D. All of these are correct.

Explanation: The link reaction which connects glycolysis to the krebs cycle, gives off CO_2 and utilizes NAD^+.

Answer: D

70. Which of the following is the greatest contributor of electrons to the electron transport chain?

A. Oxygen
B. Glycolysis
C. Krebs cycle
D. link reaction

Explanation: Large number of electrons are produced during electron transport chain during krebs cycle in which 18 ATPs are formed from 6NADH and 4ATPs are produced from $2FADH_2$.

Answer: C

71. Substrate level phosphorylation takes place in:

A. Glycolysis and krebs cycle.
B. The link reaction and electron transport chain.
C. Glycolysis and electron transport chain.
D. The link reaction and krebs cycle.

Explanation: Substrate level phosphorylation is the production of ATP without utilization of an enzyme ATP synthase. It takes place during glycolysis and krebs cycle.

Answer: A

72. Of the 36 ATP molecules that are produced during the complete breakdown of glucose, most are due to the action of:

A. Chemiosmotic phosphorylation.
B. Electron transport chain.
C. Substrate level phosphorylation.
D. Both A and B are correct.

Explanation: About 90% of ATP molecules which are produced by oxidative phosphorylation are during chemiosmosis and electron transport chain. **Answer: D**

73. Which enzyme is NOT found in C_3 plants?

A. PEP carboxylase.
B. Rubilose biphosphate oxygenase.
C. Rubilose biphosphate carboxylase.
D. ATP synthase.

Explanation: PEP carboxylase is used in C_4 plants to catalyze a reaction between carbondioxide and a three carbon compound in

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order to form a four carbon compound called **oxalo acetate.**

74. Light independent reaction in C_4 plants take place in:

A. Mesophyll cells
B. Cortex
C. Bundle sheath cells
D. stomata

Answer: A

Explanation: The light independent of C_4 plants takes place in bundle **sheath cells**, while the calvin cycle of C_3 plants takes place in **mesophyll cells.**

75. In which of the following organelles phosphoglycolate is converted in to glycolate?

A. Peroxisome
B. Lysosome
C. Mitochondoria
D. Chloroplast

Answer: A

Explanation: During photorespiration a molecule phosphoglycolate is converted in to glycolate in **peroxisome.**

76. In photorespiration carbondioxide is produced in:

A. Chloroplast
B. Mitochondoria
C. Lysosome
D. peroxisome

Answer: B

Explanation: In photorespiration carbondioxide is produced in the mitochondoria when glycine is converted to serine.

77. Which of the following is the main objective of photo respiration:

A. To reduce transpiration.
B. To increase the concentration of CO_2.
C. To increase the efficiency of photosynthesis.
D. To increase the production of oxygen.

Explanation: The main purpose of photorespiration is to increase concentration of Co_2 by producing CO_2 from the oxidation of RuBP in a condition where oxygen is deficient in C_3 plants.

Answer: C

78. How do photorespiration decrease the efficiency of photosynthesis?

A. Carbon is oxidized instead of its reduction to make glucose.
B. It passes a lot of steps to remove phosphoglycolate and to reduce the rate of photosynthesis.
C. Extra ATP is used to resynthesize the RuBP.
D. All are possible answers.

Explanation: All the alternatives listed can be reasons for photorespiration to decrease the efficiency of photosynthesis.

Answer: D

79. Photo respiration involves the oxidation of:

A. PGA
B. RuBP
C. Chlorophyll
D. Chlorophyll B

Explanation: Photorespiration involves oxidation of **RuBP.**

RuBP + oxygen Rubisco GP + phosphoglycolate

80. C_4 plants differ from C_3 plants in all of the following except:

Answer: B

A. Fix atmospheric Co_2 by phosphoenol pyruvate.
B. Produce oxaloacetate as their stable product.
C. Phosphoenol pyruvate is the only Co_2 fixing molecule in them.
D. Fix CO_2 twice in their cycle.

Explanation: PEP is not the only one to fix CO_2, but RuBP can also fix CO_2.

81. Which of the following system increases the production of glucose in hot and dry clamate?

A. GA pathway
B. C_3 pathway
C. photorespiration
D. Krebs cycle

Explanation: C_4 plants have an ability to concentrate CO_2 in their leaves. Thus, they can grow land photosynthesize in adverse conditions.

Answer: A

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82. Photorespiration is:
   A. Uses carbondioxide.
   B. Leads to net gain of carbon.
   C. Increase as temperature increases.
   D. Releases CO_2 in the dark.

Explanation: Normally, photorespiration occurs in bright day when the weather is hot and dry, because plants close their stomata to prevent the loss of water through the leaf.

83. The ATP molecule is sometimes described as: **Answer: C**
   A. Phosphorylated nitrogen base.
   B. Phosphorylated nucleotide.
   C. Glycosated nucleotide.
   D. Glycosated nitrogen base

Explanation: ATP is described as **phosphorylated nucleotide**

84. ATP is formed from: **Answer: B**
   A. AMP and Pi
   B. ADP and AMP
   C. ADP and Pi
   D. AMP and A

Explanation: ATP is formed from ADP and phosphate molecule.

$$ADP + pi + energy \longrightarrow ATP$$ **Answer: C**

85. Examples of processes requiring ATP include:
   A. Simple diffusion and active transport.
   B. Active transport and facilitated diffusion.
   C. Conduction of nerve impulses and osmosis.
   D. Active transport and protein synthesis.

Explanation: Active transport and facilitated diffusion occur with expenditure of energy (ATP) from the cell. **Answer: D**

86. The ATP synthase molecule produces ATP when:
   A. Electrons turn the rotor to activate sites in the catalytic knob.
   B. Hydrogen ions spin the catalytic knob.
   C. Electron spin the catalytic knob.
   D. The hydrogen ions turn the rotor to activate the sites in the catalytic knob.

Explanation: ATP is produced when the hydrogen ions (protons) pump through ATP synthase by activating the sites of the catalytic knob. **Answer: D**

87. ATP is ideal energy molecule in cells because it:
   A. Releases energy in small amount.
   B. Releases energy quickly
   C. Move freely in, but do not escape from the cell
   D. All of the above. **Answer: D**

88. Which of the following does not take place during the krebs cycle?
   A. Oxidative phosphorylation
   B. Substrate level phosphorylation
   C. Electron transport chain
   D. The link reaction **Answer: B**

89. During fermentation:
   A. Oxidative phosphorylation does not take place.
   B. Substrate phosphorylation does not take place.
   C. NAD is reduced in glycolysis.
   D. All of the above

Explanation: During fermentation oxidative phosphorylation does not take place. But, ATP is produced by substrate level phosphorylation and NAD is formed reduced NAD (NADH). **Answer: D**

90. Which of the following statements about mitochondria is not true?
   A. The carrier molecule of the ETC are found on inner mitochondorial membrane.
   B. The reactions of the krebs cycle take place inside the mitochondria.
   C. All of the ATP needed by the cell is made in the mitochondria.

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D. Much of the ATP needed by the cell is made in the mitochondria.

Explanation: Mitochondoria is the major site of cellular respiration. However, the process of glycosis which takes place outside mitochondria produces 2 New ATP molecules.

91. In the electron transport chain, electron are passed: Answer: C

A. From the lumen of the mitochondria to the inter membrane space.

B. From the intermmembrane space to the lumen of the miitoochondoria.

C. Through ATP synthase.

D. Along a series of electron carriers.

Explanation: During electron transport chain electrons pass through electron carrers such as ubiquine, cytochromes, etc.

Answer: D

92. Oxidative phosphorylation include:

A. The electron transport chain and chemiosmosis.

B. The electron transport chain and krebs cycle.

C. The krebs cycle and chemosmosis.

D. None of the above

Explanation: Oxidative phosphorylation includes **electron transport chain** and **chemiosmosis**. Because much of the ATP molecules are produced in these stages.

Answer: A

93. In the krebs cycle:

A. Some ATP is made by oxidative phosphorylation.

B. The four carbon compound oxaloacetate is regenerated.

C. ATP is used.

D. The six carbon compound citrate is split in to two three carbon compounds.

Explanation: In the krebs cycle the four carbon compound oxaloacetate is regenerated in to citrate (six carbon compound).

Answer: B

94. When compared with aerobic respiration and fermentation of glucose by yeast:

A. Yield less ATP per molecule of glucose

B. Produces lactate.

C. Produces more CO_2.

D. None of the above.

Explanation: Fermentation of glucose by yeast produces only 2ATPs. While, during aerobic respiration 38ATPs are produced from one glucose molecule.

Answer: A

95. Which of the following statement about aerobic respiration is correct?

A. Glycolysis takes place in the matrix of mitochondoria.

B. Carrier molecule of ETC exist on the outer membrane of the mitochondria.

C. A high concentration of hydrogen ions builds up in the matrix of mitochondria.

D. The kreb's cycle takes place in the matrix on mitochondoria.

Explanation: In aerobic respiration, the krebs cycle takes place in the matrix of mitochondria, because the krebs cycle needs oxygen.

Answer: D

96. In a respirometer:

A. The amount of oxygen used up by the organism is replaced with equal amount of CO_2.

B. The CO_2 given off is absorbed by potassium hydroxide

C. The breathing rate of an organism is measured.

D. We measure the uptake of oxygen by an organism.

Explanation: Spirometer is an apparatus that is used to measure the ratio of the uptake of oxygen and CO_2 given off by an organism.

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97. Which of the following occur in both aerobic respiration and fermentation (anaerobic respiration) in mammals?
A. Substrate level phosphorylation.
B. Chemiosmosis
C. Link reaction
D. Decarboxylation

Explanation: In both aerobic and anaerobic respiration, there is substrate level phosphorylation at which ATP is produced.

Answer: A

98. In the light independent reactions of photosynthesis
A. NADP is reduced.
B. ATP is produced
C. ADP is produced
D. Light energy excites chlorophyll electrons.

Explanation: In light independent reaction NADP is reduced to NAD.

Answer: A

99. In the light dependent reaction of photosynthesis
A. ATP is used to convert Gp to Tp.
B. Reduced NADP in used to convert OP in to Tp.
C. ATP is produced.
D. Carbohydrates are produced

Explanation: During light dependent reaction the raw materials such as ATP and reduced NADP needed for the light independent are produced.

Answer: C

100. Cyclic phosphorylation produces:
A. Oxygen and ATP.
B. Reduced NADP and ATP.
C. Oxygen only.
D. ATP only.

Explanation: Cyclic phosphorylation is the pathway between PSI and back to PSI for the production of ATP only.

Answer: D

101. In light independent reaction, reduced NADP is used to:
A. Oxidize GP to TP
B. Oxidize TP to GP
C. Reduce Tp to Gp
D. Reduce GP to TP

Explanation: Reduced NADP which is produced during light dependent reaction is used to reduce GP to TP in light independent reaction.

Answer: D

102. A photosynthetic unit can carryout
A. Photolysis
B. The synthesis of ATP
C. The synthesis of reduced NADP
D. All are possible answers

Explanation: photosynthetic unit is the arrangement of molecules, capable of carrying out all the reactions of the light dependent reaction. Therefore, photosynthetic unit occurs in photolysis, synthesis of ATP and synthesis of reduced NADP.

Answer: D

103. A photosystem Consists of:
A. A reaction center molecule and ETC.
B. A reaction center molecule and antenna complex.
C. An accessory pigment and antenna complex.
D. An accessory pigment and ETC

Explanation: A photosystem consists of a reaction center molecule and antenna complex.

Answer: D

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104. In photorespiration:

A. Low oxygen concentration cause Rubisco to form more GP than usual.

B. Low oxygen concentration cause Rubisco to form less GP than usual.

C. Low CO_2 concentration cause Rubisco to form less GP than usual.

D. Low CO_2 concentration cause Rubisco to form more GP than usual.

Explanation: During photorespiration C_3 plants absorb less CO_2 concentration and oxygen combine with RuBP by Rubisco biphosphate oxygenase and convert it to one molecule of GP and one molecule of phosphoglycolate.

Answer: C

105. In the C_4 pathway

A. PEP carboxylase catalyzes the reaction of CO_2 with Rubp in the bundle sheath.

B. PEP carboxylase catalyzes the reaction of CO_2 with RuBp in the bundle sheath.

C. PEP carboxylase catalyzes the reaction of CO_2 with PEP in the mosophyll cells.

D. PEP carboxylase catalyzes the reaction of $CO_2$with PEP in the bundle sheath cells.

Answer: C

106. C_4 photosynthesis is more efficient than C_3 photosynthesis in conditions of:

A. High light intensity and high CO_2 concentration

B. Low light intensity and high CO_2 concentration

C. High light intensity and low CO_2 concentration.

D. Low light intensity and low CO_2 concentration.

Explanation: C_4 photosynthesis is more efficient, because;

- It is effective under high light affinity.
- It is efficient under the shortage of Co_2, because C_4 plants can concentrate CO_2.

Answer: C

107. The chloroplasts in the bundle sheath cells of C_4 plants are an adaptation to this pathway because:

A. They contain no calvin cycle enzyme.

B. They do not contain thylakoids.

C. They have large surface area.

D. They produced large amount of oxygen.

Explanation: C_4 plants contain chloroplast in their bundle sheath, but do not contain thylakoids.

Answer: B

108. Which of the following statements about C_4 and CAM photosynthesis is true?

A. In CAM photosynthesis the C_4 stage and the calvin cycle are separated.

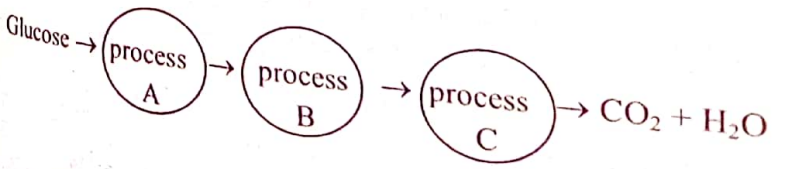
B. In CAM photosynthesis, the C_4 stage and the calvin cycle are separated in space.

C. In C_4 photosynthesis, the C4 stage and calvin cycle are separated in time.

D. In C_4 photosynthesis, the C_4 stage and calvin cycle both occur in the same cell.

Answer: A

109. The three processes represent three major processes in aerobic respiration.



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The process 'A' represents

A. Glycolysis
B. Krebs cycle
C. Electron transport chain
D. Lactic acid fermentation

Explanation: The process is glycolysis which is the first and anaerobic stage of cellular respiration.

Answer: A

110. The process 'A' occurs in

A. Mitochondria
B. Chloroplasts
C. The nucleus
D. The cytoplasm
E. Different organelles in different cells

Explanation: Glycolysis (A) occurs in the cytoplasm of the cell, unlike the other steps of cellular respiration (link reaction, krebs cycle and electron transport chain) which take place in the mitochondria.

Answer: D

111. Process 'C' represents:

A. Glycolysis
B. The krebs cycle
C. The electron transport chain
D. Alcohol fermentation
E. Lactic acid fermentation

Explanation: Process 'C' represents ETC. The first step is glycolysis which involves the breakdown of glucose and production of ATP. The next step is the conversion of pyruvic acid to acetyl CoA, which enters the krebs cycle.

Answer: C

112. Which of the following is correct about these process?

A. Process 'A' requires energy.
B. Process 'B' requires energy.
C. Process 'C' requires energy.
D. Process A depends on a gradient.
E. Process B releases oxygen.

Explanation: The only process which requires ATP is glycolysis (A). During glycolysis 4ATPs are produced from 1 glucose molecule. However, since 2ATP molecules are used up to split glucose molecule, 2 Net ATP molecules are gained.

Answer: C

113. Which is true about process C?

A. It requires water.
B. It is universal
C. It involves the membrane structure ATP synthase.
D. It only occurs in animals, not plants.
E. It results in the by-product pyruvic acid.

Explanation: Process 'C' represents the electron transport chain which uses energy from the flow of electron through cristae membrane to pump protons across the membrane and create a proton gradient. This gradient power phosphorylation of ADP to ATP.

Answer: C

114. Which is true about process B?

A. It produces the most energy of all the processes.
B. It produces energy by chemiosmosis.
C. Its production of energy is explained by the laws of diffusion.
D. The raw material is pyruvic acid.
E. It releases carbondioxide.

Explanation: Process 'B' is krebs cycle. Carbondioxide is the bi-product of the krebs cycle.

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115. Which of the following is probably evolved first? **Answer: E**

A. The krebs cycle
B. Chemiosmosis
C. Glycolysis
C. Electron transport chain
E. the citric acid cycle

Explanation: Glycolysis occurs during the anaerobic phase of respiration and evolved on ancient earth. It was possible the way the first cells produced energy. The krebs cycle and electron transport chain evolved after free oxygen appeared on ancient earth.

116. All of the following are correct about ATP synthase EXCEPT **Answer: C**

A. It is a molecule located within the cristae membrane.
B. It spins like turbine.
C. It binds a phosphate to a molecule of ADP.
D. It is an important component of glycolysis.
E. Its function depends on a proton gradient.

Explanation: ATP synthase is an important component of aerobic respiration. Glycolysis is part of aerobic respiration.

117. The breakdown of glucose in to pyruvic acid occurs during the process of: **Answer: D**

A. Glycolysis
B. Lactic acid formentation
C. Krebs cycle
D. Electron transport chain
E. Alcohol fermentation

Explanation: Glycolysis is the first stage of cellular respiration which involves the breakdown of glucose in to two pyruvic acid molecules.

Answer: A

118. Choose the pair that is correctly matched.

A. Krebs cycle – cytoplasm
B. Electron transport chain – outer compartment of mitochondria
C. Glycolysis – cristae membrane.
D. Glycolysis – mitochondoria
E. Electron transport chain – cristae membrane

Explanation:

Stages of cellular respiration	Takes place in
Glycolysis	Cytoplasm
Krebs cycle	Matrix of mitochondria
Electron transport chain	Inner mitochondorial membrane celled cristae

Answer: E

119. During sternous excercies, skeletal muscles us up _____ and produce large amount of _____, which causes pain and fatigue in the muscle.

A. Pyruvic acid – carbon dioxide
B. Carbondioxide – pyruvic acid
C. Pyruvic acid – lactic acid
D. Oxygen – lactic acid
E. Glycogen – carbon dioxide

Explanation: Lactic acid fermentation occurs during strenuous exercise when the body cannot keep up with the increased demand for oxygen by skeletal muscle. Pyruvic acid converts to lactic acid that builds up in the muscle, causing fatigue and burning.

120. The molecule of which pigment is located at the reaction center of a photosystem? **Answer: C**

A. Chlorphyll a
C. carotenoid

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B. Chlorophyll b D. Accessory pigments

Explanation: Chlorophyll is a light absorbing pigment that is found in the chloroplast of a leaf. The major types of chlorophyll are:

- **Chlorophyll-a** – is bright green pigmented substance and plays central role in initiation of light dependent reaction.
- **Chlorophyll – b** – is an accessory pigment which is yellow green pigmented.
- **Carotenoids** – yellow or orange accessory pigment and are used for broadening the spectrum of light.

Answer: A

121. Which of the following substances is NOT necessary for photosynthesis to take place?

A. Chlorophyll C. Oxygen

B. Carbon dioxide D. water

Explanation: The major requirements for the process of photosynthesis are:

- Chlorophyll
- Light
- Carbondioxide
- Water

Answer: C

122. Which of the following groups of plants carryout light dependent and light independent reactions of photosynthesis in separate cells?

A. C_3 – plants C. CAM plants

B. C_4 – plants D. plants without chlorophyll

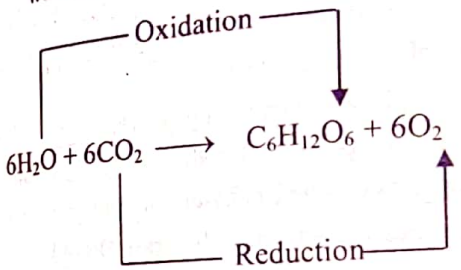
Explanation: C_4 – plants have an ability to concentrate CO_2 in their cells. This is because they are capable of fixing CO_2 in to oxaloacetate compound. They also have chloroplast in their bundle sheath. Thus, the process starts in mesophyll cells and ends in bundle sheath process starts in mesophyll cells and ends in bundle sheath

Answer: B

Conceptual Summary Example

1. Describe photosynthesis as a redox process.

Explanation: During photosynthesis, light energy is captured converted to the chemical energy of carbohydrates, hydrogens from water are used to reduce carbon and oxygen delivered from water becomes oxidized, forming molecular oxygen.



2. Distinguish between the light dependent reactions and carbon fixation reactions of photosynthesis.

Explanation: In the light dependent reaction electrons energized by light are used to generate ATP and NADPH. These, compounds provide energy for the formation of carbohydrate during carbon fixation reaction.

3. Describe the flow of electrons through photosystems I and II in the non cyclic electron transport phathway; contrast this with cyclic electrontransport.

Explanation: Photosystem I and II are the two types of photosynthetic units involved in photosynthesis. Each photosystem includes chlorophyll molecules and accessory pigments organized with pigment binding proteins in to **antenna complex**.

4. Explain how a proton (H^+) gradient is established across the thylakoid membrane and low this gradient functions in ATP synthesis.

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Explanation:

- **Photophosphorylation** is the synthesis of ATP coupled to the transport of electrons energized by photons of light. Some of the energy of the electrons is used to pump protons across the thylakoid membrane, providing the energy to generate ATP by chemiosmosis.
- As protons diffuse through **ATP synthase**, an enzyme complex in the thylakoid membrane, ADP is phosphorylated to form ATP.

5. Summarize the three phases of the calvin cycle, and indicate the roles of ATP and NADpH in the process.

Explanation: The three phases of calvin cycle are:

- ✓ **CO_2 up take phase** – this stage CO_2 is combined with RuBP by an enzyme called **rubisco**, inorder to form a three carbon molecule called **phasphoglycerate (PGA)**
- ✓ **Carbon reduction phase** – the energy and reducing power of ATP and NADPH are used to convert PGA molecules to glycerol dehyde – 3 – phosphate (G-3P). for every $6CO_2$ molecules fixed, 12 molecules of GP are produced, as 2 molecules of GP leave the cycle to produce equivalent of 1 molecule of glucose.
- ✓ **RuBP generation phase** - the remaining GP molecules are modified to regenerate RuBP.

6. Discuss how photorespiration reduces photosynthetic efficiency.

Explanation: **photorespiration** in C_3 plants consume oxygen and generate CO_2 by degrading calvin cycle intermediates but do not produce ATP. Photorespiration is significant on bright, hot, dry days when plants close their stomata in order to conserve water but preventing the passage of CO_2 in to the leaf.

7. Compare the C_4 and CAM pathways.

Explanation:

- In **C_4 pathway**, an enzyme **PEP carboxylase** binds with CO_2 effectively, even when CO_2 is at low concentration. C_4 reactions take place within mesophyll cells. The CO_2 is fixed in **oxalo acetate**, which is then converted in to malate. The malate moves in to a **bundle sheath,** and CO_2 is removed from it. The released CO_2 is removed from it. the released CO_2 then enters the calvin cycle.
- The **crassulacean acid metabolism (CAM)** pathway is similar to the C_4 pathway. PEP carboxylase fixes carbon at night in the mosphyll cells, and the calvin cycle occurs during the day in the same cells.

Post – Test - II

1. Where is the chlorophyll located in the chloroplast?
   A. Thylakoid membranes
   B. Stroma
   C. Matrix
   D. Thylakoid lumen
   E. Between the inner and outer membranes.
2. In photolysis, some of the energy captured by chlorophyll is used to split:
   A. CO_2
   B. ATP
   C. NADPH
   D. H_2O
3. Light is composed of particles of energy called:
   A. Cerotenoids
   B. Reaction centers
   C. Photons
   D. antenna complexes
   E. photosystems
4. The relative effectiveness of different wavelengths of light in photosynthesis is demonstrated by:
   A. An action spectrum
   B. Photolysis
   C. Carbon fixation
   C. photohetrotrophs
   E. an absorption spectrum

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5. In plants, the final electron acceptor in the light dependent reaction is:
   A. $NADP^+$
   B. CO_2
   C. H_2O
   D. O_2
   E. G_3P
6. In addition to chlorophyll, most plants contain accessory photosynthetic pigments such as
   A. PEP
   B. G_3P
   C. carotenoids
   D. PGA
   E. $NADP^+$
7. The part of a photosystem that absorbs light energy is its:
   A. Antenna complexes
   B. Reaction center
   C. Terminal quinine electron
   D. Pigment – building protein
   E. Thylakoid lumen
8. The mechanism by which electron transport is coupled to ATP production by means of a proton gradient is called _____.
   A. Chemiosmosis
   B. Crassulacean acid metabolism
   C. Fluorescence
   D. The C_3 pathway
   E. The C_4 pathway
9. During photosynthesis in eukaryotes, the transfer of electrons through a sequence of electron acceptors provides energy to pump protons across the;
   A. Chloroplast outer membrane.
   B. Chloroplast inner membrane.
   C. Thylakoid membrane.
   D. Inner mitochondorial membrane
   E. Plasma membrane
10. The inputs for _____ are CO_2, $NADPH_2$, and ATP.
   A. Cyclic electron transport.
   B. Carbon fixation reactions.
   C. Non – cyclic electron transport.
   D. Photosystems I and II.

   E. Chemiosmosis.
11. The calvin cycle begins when Co_2 reacts with:
   A. Phosphonol pyruvate
   B. Glyceraidehyde – 3 – phosphate
   C. Ribulose biphosphate
   D. Oxaloacetate
   E. Phosphoglycerate
12. The enzyme that is directly responsible for almost all carbon fixation on earth is:
   A. Rubisco
   B. PEP carboxylase
   C. ATP synthase
   D. phosphofructo kinase
   E. Ligase

Answers to Post test – II

1.A	2. D	3.C	4. A	5.A	6.C
7.A	8.A	9.E	10. B	11.C	12.A

Supplementary Exercise on Unit 5

Fill in the following blank spaces with appropriate answers

1. The region on chloroplast in which the calvin cycle (light independent reaction) of photosynthesis takes place is _____.
2. The conversion of energy from one form to another form is celled _____.
3. The first molecule which accepts the excited electrons displaced from antenna complex is _____
4. The splitting of water molecules during light dependent reaction of photosyntehsis is called _____.
5. Removing of carbon from carbondioxide is called _____.
6. The diffusion of hydrogen ions (proton) through the ATP synthase enzyme is called _____.
7. Chlorophyll 'a' molecules are always found in _____.

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8. The raw materials of photosynthesis are _____, _____ and _____

9. The three steps of light independent reaction or calvin cycle are _____, _____ and _____.

10. The inputs light dependent reactions are _____

Multiple Choice items

11. Which molecule have high energy electron carrier generated during fermentation that allows cells to continue making ATP using photolysis?
    A. FADH
    B. NADH
    C. ADP
    D. AMP

12. Which of the following happens as electrons pass through electron transport chain?
    A. Energy from the moving electron generate ATP used for light independent reaction.
    B. Energy from the moving electron proton from the matrix to inter membrane space.
    C. Energy from the moving electron stored as ATP.
    D. CO_2 is reduced.

13. Which compound produced ATP most when oxidized?
    A. Acetyl – COA
    B. Glyceraldehyde – 3 – phosphate
    C. Glucose
    D. Fructose biphosphate

14. At the end of glycolysis most of the energy remaining from the original glucose molecule is stored in:
    A. NADH
    B. CO_2
    C. Pyruvic acid
    D. ATP

15. In C_4 pathway the light independent reaction occurs in:
    A. Bundle sheath
    B. Mesophyll
    C. Grana
    D. Ribosome

16. Which enzyme has high affinity to produce CO_2?
    A. PEP carboxylase
    B. Ribulose biphosphate
    C. Glyceraldehyde
    D. ATP synthase

17. C_4 photosynthesis:
    A. Lack chloroplast in the bundle sheath.
    B. Advantageous when the weather is hot and dry
    C. Occurs in plants whose bundle sheath contain chloroplast.
    D. B and C are answers.

18. The immediate product of C_3 and C_4 photosynthesis are respectively;
    A. Ribulose 1,5 –biphosphate and malic acid.
    B. Maltase and CO_2
    C. 3 – phosphoglycerate and oxaloacetic acid.
    D. Glyceraldehyde – 3 – phosphate and PEP.

19. Respirometer is used to measure:
    A. The volume of oxygen used in respiration.
    B. The volume of CO_2 used in respiration.
    C. The uptake of oxygen given of CO_2 in respiration
    D. The volume of water vapor.

20. Decarboxylation of pyruvic acid is:
    A. Phosphoenol pyruvate
    B. Glyceraldehyde – 3 – phosphate
    C. Lactic acid
    D. Acetyl

21. How many net ATP molecules are generated through anaerobic respiration when a single glucose molecule is changed to Pyruvate in human body?
    A. Two
    B. three
    C. four
    D. Six

22. During krebs cycle, which of the following molecule temporarily stores most of the energy released from food molecule?

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A. ADP
B. ATP
C. NADH
D. FADH

23. At which stage is most of the ATP generated in aerobic respiration?
A. Glycolysis
B. Link reaction
C. krebs cycle
D. electron transport chain

24. Which of the following substance is NOT necessary for photosynthesis to take place?
A. Chlorophyll
B. Carbondioxide
C. oxygen
D. water

25. The molecule of which pigment is NOT necessary for photosynthesis to take place?
A. Chlorophyll 'a'
B. Chlorophyll 'b'
C. carotenoid
D. xanthophylls

26. What is the ultimate source of energy for all organisms living on the earth?
A. Water
B. Oxygen
C. Nitrogen
D. sun

27. Where exactly in the cell does the krebs cycle takes place?
A. Mitochondorial matrix.
B. Inner mitochondrial membrane.
C. Cytoplasmic fluid.
D. Outer mitochondrial membrane.

28. If present in the fermentation system, which of the following would negatively affect alcohol production by yeast?
A. Water
B. Oxygen
C. Glucose
D. fermentation enzyme

29. Which term describes the process by which water is lost by evaporating through the stomata of leaves?
A. Transport
B. Transpiration
C. Anchorage
D. Adhesion

30. Which of the following is an accurate representation of the relationship between the terms metabolism, anabolism and catabolism?
A. Anabolism = catabolism
B. Metabolism = catabolism
C. Anabolism = catabolism + metabolism
D. Metabolism = catabolism + anabolism

31. In which of the following ways is the carbondioxide of the atmosphere fixed in to the carbon found in organic molecules?
A. In the breathing process of all animals.
B. In the decomposition of organic molecules.
C. In the process of photosynthesis by green plants.
D. In all the cellular respiration processes of organisms.

32. Which of the following groups of plants carry out light dependent and light independent reactions of photosynthesis in separate cells?
A. C3 plants
B. C4 plants
C. CAM plants
D. Plants without chlorophyll

33. From which of the process of cellular respiration is the majority of the ATP is generated?
A. Anaerobic fermentation.
B. Glycolysis and link reaction.
C. Electron transport and chemiosmosis.
D. Krebs cycle and glycolysis.

34. Which one of the following does the rate of algal photosynthesis in the lake depend?
A. The oxygen content of water.
B. The nitrogen content of water.
C. The elevation where the lake is found.
D. The amount of light that penetrates the lake water.

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35. During aerobic respiration, what is the route through which protons return from the mitochondoria inter membrane space back to its matrix?

A. Proton pump
B. ATP sysnthase
C. Ion channel
D. Membrane lipid

36. What does the absence of chloroplasts in a unicellular organism include?

A. Reproduction by sexual means.
B. Autotrophic nutrition.
C. Heterotrophich nutrition.
D. Complete lack of metabolic processes.

37. Which of the following did the early heterotrophs do that probably assisted the origin of autotrophs?

A. They added oxygen and chlorophyll to the environment.
B. They added carbondioxide to the environment.
C. They stored energy in the bonds of inorganic compounds.
D. They manufactured food from carbiondioxide and oxygen.

Answer to Supplementary Exercise on Unit 5

1. Stroma
2. Transduction
3. Primary electron acceptor
4. Photolysis
5. Decarboxylation
6. Chemiosmosis
7. Reaction center of PS II (P680) and PSI (P 700) molecules.
8. H_2O, CO_2, chlorophyll and sunlight
9. CO_2 uptake, carbon reduction, RuBP generation
10. Water and sunlight

11. B	12.B	13.D	14.C15.A	
16. A	17.D	18.C	19.C20.D	
21.A	22.C	23.D	24.C25.A	
26.D	27.A	28.B	29.B30.D	
31.C	32.B	33.C	34.D	35.B
36.C	37.B			

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UNIT SIX

MICRO ORGANISMS

6.1. Types of Micro organisms

Micro organisms - are very small organisms. They are mostly microscopic and unicellular. Unless aided by microscope, human eyes can not see them. They consist of 5 (five) main groups. These are:

- Viruses
- Bacteria
- Protozoa
- Some algae
- Some fungi

Protozoa - are unicellular organisms without cell wall.

Example: Amoeba, Plasmodium, Paramecium, Giardia, Euglena, etc

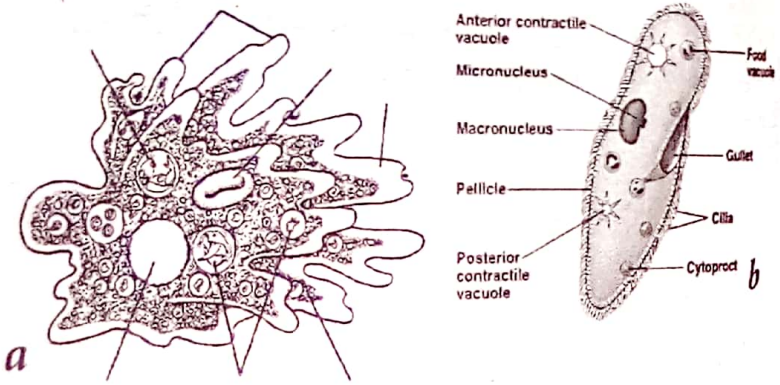

Fig. 6.1. Different protozoans: a=amoeba, b=giardia,

Fungi - are non-photosynthetic.

- Parasitic or saprophytic (live on decaying organic matter).
- Major part of their cell wall is ***Chitin***.
- Store ***oil*** and ***glycogen*** like animals.
- Body of higher fungi is made from thread like structures called **Hyphae** (Hypha, sing.)
- Inter connected mass of Hyphae is called ***Mycellium*** (Mycelia, Pl.)

Example: - Yeasts (used in bakery and brewery)
- Candida (causes mouth thrush)

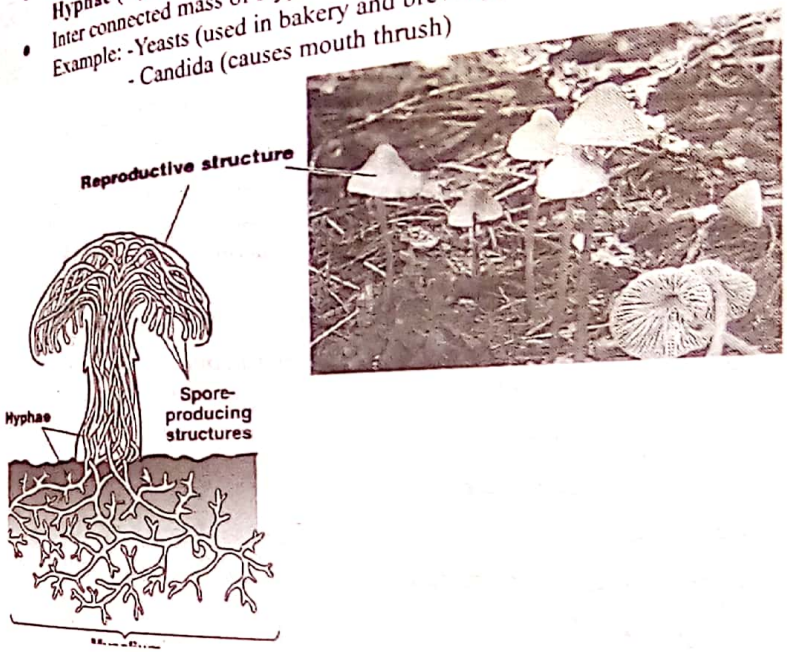

Fig. 6.2 Structure of higher fungi

Algae - includes the green slimes and films on ponds and trees, sea weeds in sea shores and phytoplankton in lakes and seas.

- They are **photosynthetic**, unicellulars or multicellulars.
- Phytoplankton are unicellular , microscopic and act as food for **zooplankton** and smaller animals (**Example:** Fish).
- Highly productive of **Oxygen** and **organic matter** than terrestrial plants.

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Example: Chlamydomonas (dinoflagelates), chlorella, vorticella, etc

Virus - genetic material embedded in **protein coat**, some may have additional membrane

- Use other living cells for reproduction.
- Can not carry out common life-processes independently.
- Therefore, all are parasites.
- Some believe that viruses are border-line between living and non-living things.

Example: HIV, bacteriophages, tobacco mosaic virus (TMV), etc.

Bacteria - are unicellular prokaryotes, non-photosynthetic except cyano-bacteria (blue-green algae)

There are some common structures found in all bacteria. These are:

- Cell wall (made from ***peptidoglycan***)
- Cytoplasm
- Cell membrane
- Ribosome
- DNA

- Some other structures are found in few bacterial types only. These are:
  - Flagella
  - Capsule
  - Pilli (hair-like structures on the surface)
  - Plasmids
- Bacteria are 1 to 10 μm in diameter.
- They lack nucleus, mitochondria and chloroplasts.
- In general, they are devoid of membrane-bound organelles.
- Respiration and photosynthesis take place in the cytoplasm and/or on membranes.

Generalized structure of a prokaryote

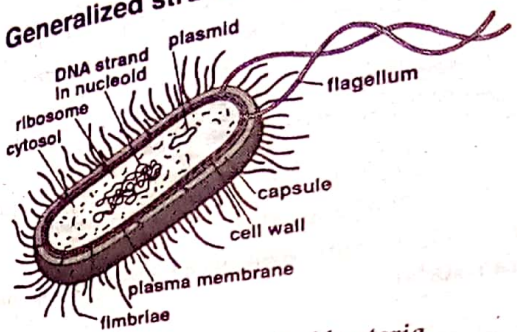


Fig. 6.5 Structure of a typical bacteria

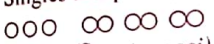
Shapes and arrangements of bacterial cells

- Bacterial cells have different shapes and arrangements. These include:

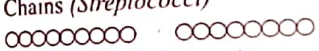
a) ***Coccus (Cocci, pl.)*** - Spherical shaped bacteria

Can be arranged as:

- Singles and pairs *(Diplococci)*



- Chains *(Streptococci)*



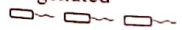
- Clusters *(Staphylococci)*

b) ***Bacillus (Bacilli, pl.)*** - Rod-shaped bacteria

- Singles and pairs

- Chains

- Flagellated



c) ***Spirochactes*** - spiral, cork-screw or comma-shaped. May also be flagellated.

Gram-positive and Gram-negative bacteria

- Bacteria can also be classified as Gram-positive or Gram-negative based on their staining properties.

a. **Gram-positive bacteria** - stained **Purple** by Gram's stain.

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Example: *Clostridium tetani, Bacillus anthracis, Corynebacterium diphtheria, Mycobacterium tuberculosis, Mycobacterium leprea, Staphylococci, Streptococci, Streptomyces,* etc

b. **Gram-negative bacteria** - stained **Pink** by Gram's stain.
Example: *Acetobacter, Enterbacter, Escherichia coli, Helicabacter, Hemophilus influenzae, Klebsiella, Neisseria, Pseudomonas, Ricketsia, Salmonella typhi Treponema Pallidum,* etc.

Gram's stain - is a differential stain since it stains various bacteria differently.

Steps of Gram's staining

- Fixation
- Crystal violet treatment
- Iodine solution addition
- Decolorization by ethanol
- Counter staining by safranin

Fixation - passing microscope slide containing bacterial sample over flame for few times.

Gram's stain reacts with peptidoglycan cell wall of bacteria.

Gram-negative - bacteria have additional membrane outside their cell wall which blocks the entry of **crystal violet and iodine solution**. So, their cell wall fails to interact with the stains. The stains are then, washed away by ethanol. As a result, Gram-negative bacteria retain the last stain (safranin) and stain Pink.

In Gram-positive bacteria, nothing blocks the primary stain (crystal violet) from interacting with the cell wall. Consequently, they retain it and stain purple.

Gram-staining was developed by **Hans Christian Gram** in **1884**.

The outer membrane of Gram-negative bacteria secretes **Endotoxins**.

Generally, diseases caused by Gram-negative bacteria are serious. But, Gram-positive bacteria can also cause serious diseases.

Example: TB.

Types of cells

- Cells are broadly divided into two categories based on their structural complexity.

a. **Prokaryotic cells** - are evolutionarily primitive cells with simple structures.
Example: Kingdom Monera-ordinary bacteria and Cyano bacteria (blue-green algae)

b. **Eukaryotic cells** - advanced cells with complex structures. They have membrane-bound organelles.
Example: Kingdom Protoctists (Protozoa and Algae)
Kingdom Fungi

Kingdom Animalia
Kingdom Plantae

ILLUSTRATIVE EXAMPLES

1. Which of the following is not a fungus?
(A) Yeast (C) Saccharomyces
(B) Candida (D) Plasmodium
Answer: D
Explanation:
- Plasmodium is a protist, not a fungus.

2. Which of the following is a fungus?
(A) Plank ton (C) Plasmodium
(B) Yeast (D) Protozoa
Explanation:
- Yeast is a fungus, plankton is aquatic microscopic algae, plasmodium and protozoa are protists.
Answer: B

3. The structure that is found in all bacteria is
(A) Plasmid (C) DNA
(B) Capsule (D) Flagella
Explanation:
- DNA is common to all bacteria.
Answer: C

4. The structure that is found in some bacteria but not in all is
(A) Cytoplasm (C) Plasmid
(B) Cell wall (D) Ribosome
Explanation.
- Plasmid is peculiar structure in some bacteria only. The others are common to all bacteria.
Answer: C

5. Bacteria cell wall is made from _____.
(A) Peptidoglycan (C) Lipoprotein
(B) Cellulose (D) Phospholipid
Explanation:
- Bacterial cell wall is composed of complex molecules made from **sugars and amino acids**, called **Peptidoglycan**. It forms mesh-like structure which absorbs Gram's stain.
Answer: A

6. It is true to say that all viruses are parasites.
Explanation:
- This is true since viruses depend on other cells for all their life-processes.

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7. In which kingdom of life are the unicellular eukaryotes grouped? **(UEE, 2004 E.C)**
(A) Monera
(B) Protista
(C) Plantae
(D) Animalia

Explanation:
- The unicellular eukaryotes such as Protozoa and Algae are grouped in protista.

Answer: B

8. Which of the following are rod-shaped bacteria? **(UEE, 2003 E.C)**
(A) Cocci
(B) Spirochaetes
(C) Spirilla
(D) Bacilli

Explanation:
- The collective name of rod-shaped bacteria is **Bacilli**.

Answer: D

9. Gram-positive and Gram-negative bacteria are differentially stained by Gram's stain. This is due to the structure difference in their:
(A) Cell membrane
(B) Ribosome
(C) Capsule
(D) Cell wall

Explanation:
- Cell wall: the cell wall of Gram-negative bacteria has additional membrane on its outer surface. This prevents the entry of Gram's stain to react with the cell wall.

Answer: D

10. All serious bacterial diseases are cased by Gram-negative bacteria

Explanation:
- This is false as there are exceptions. Tuberculosis (TB), fore example, is very serious disease caused by a Gram-positive bacteria called *Mycobacterium tuberculosis*.

Answer: False

11. Surface appendages used by bacteria to attach to one another, and to host organisms are:
(A) Mesosomes.
(B) Pilli.
(C) Thylakoids.
(D) Spirilla.

Explanation:
- Pilli: these are hair-like structures on the surface of some bacteria used for attachment.

Answer: B

12. Compared to Gram-negative bacteria, Gram-positive bacteria
(A) Have less complex cell wall.
(B) Are more susceptible to antibiotics.
(C) Retain crystal violet rather than safranin.
(D) All of the above are correct.

Explanation:
- All the statements are right: Gram-negative bacteria have more complex cell wall as there is additional membrane on its outer side. It is easier to treat Gram-positive bacteria than Gram-negative ones. Gram-negative cell wall retain the last stain (safranin), while Gram-positive bacteria retain the primary stain (crystal violet).

Answer: D

13. Grape like aggregate of bacteria is called _____.

Explanation:
- Grape-like bunch (cluster) of bacteria is known as **Staphylococcus.**

14. Paired spherical bacteria is _____, while long chain-like bacterial spheres are called _____.

Explanation:
Diplococcus, Streptococcus.

15. Unlike plant cell walls which are made of cellulose, most bacterial cell walls contain:
(A) Peptidoglycan.
(B) Chitin.
(C) Lipoprotein.
(D) Phospholipid.

Explanation:
- Bacterial cell walls contain **Peptidoglycan**

Answer: A

16. What are the whip-like attachments to bacterial cells? For what purpose are they used?

Explanation:
- Flagella - these are used for locomotion in some bacteria.

17. What is the external enclosing jelly-like layer found in some bacteria?

Explanation:
- Capsule - this makes chemotherapy difficult in bacteria possessing them.

18. Which of he following bacteria are **Gram-negative**?
(A) *Clostridium tetani*
(B) *Mycobacterium leprae*
(C) *Escherichia coli*
(D) *Streptococci*

Explanation:
- Escherichia coli is a Gram-negative bacteria. The rest are Gram-positive bacteria.

Answer: C

19. Which of the following is the **correct order** of using the chemicals in Gram's staining?

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(A) Safranin → Iodine → Ethanol → Crystal violet.
(B) Safranin → Ethanol → Crystal violet → Iodine.
(C) Ethanol → Crystal violet → Iodine → Safranin.
(D) Crystal violet → Iodine → Ethanol → Safranin.

Explanation:

- Staining by crystal Violet and Iodine solution in that order, and decolorization by Ethanol. Finally, the bacterial sample is counter-stained by Safranin.

Answer: D

20. Which of he following nutritions is **not** characteristic of Fungi?
(A) Parasitic.
(B) Saprophytic.
(C) Autotropic.
(D) Hetrotropic.

Explanation:

- Auto tropic - non of the fungi are autotropic. All are hetrotropic (i.e. Parasitic or Saprophytic).

Answer: C

1.2. The Ecology and uses of bacteria

Bacteria are found everywhere. Their study is important because they:

- Cause diseases
- Are used in industry
- Recycle nutrients in nature

Role in Diseases

Germ theory - states that infectious (communicable) diseases are caused by micro-organisms.

Pathogenes - are organisms that cause diseases.
- are also called **Germs**.

Louis Pasteur, Joseph Lister, and Robert Koch involved in the development of germ theory

- Pasteur showed that microbes in air caused wine to go sour (mid 19th century).
- Lister showed that carbolic acid (phenol) acted as disinfectant (1860s)
- Koch identified causes of TB and Cholera (1880).

Koch's postulates

If a micro-organism causes a disease:-

a) It is present in sick organisms and absent from healthy ones.
b) It can be isolated from the infected organism, and be grown in culture.
c) This should cause the disease when artificially inoculated into healthy host.
d) It can be isolated from the newly infected host, and be grown in culture.

How micro-organisms cause diseases

Viruses - incorporate their genetic material into host cell's **DNA**, and disrupt cells metabolism. It is difficult to treat viral diseases by antibiotics.
Example: Influenza, Common cold, Measles, AIDS, etc.

Bacteria - release toxins, and cause physical damage. They are treated by antibiotics.
Example: Pneumonia, TB, Cholera, etc.

Protozoa - Multiply and grow in tissues, consume tissues and cause physical damage. They are treated by anti-protozoans.
Example: Amoebiasis, Giardiasis, Malaria, Sleeping sickness, etc.

Fungi - grow in/on organisms, and cause physical damage. Their hyphae secrete **enzymes and toxins**. They digest and/or cause **allergy**.
Example: "Farmers lung" (Allergy in lung), athlete's foot, mouth thrush, etc. Although they can be treated by anti-fungals, it is difficult to cure them.

Transmission of pathogens

Reservoir of infection - origin/source of infection.
Example: a person, animal, plant, soil, etc.

Disease - is a condition where part or all of a body is made to function abnormally at a lesser efficiency.

Table 6.1 Ways of transmission of pathogens

Disease	Transmission ways	Examples
Respiratory diseases	- Droplet infection (i.e. during sneezing, coughing, talking)	. Influenza . Common cold . Pneumonia . TB
Gut diseases	- Contaminated food and water	. Cholera . Typhoid fever . Salmonelosis . Listeriosis . Botulism
Skin infections	- Direct contact	. Athletes foot . Ring worm

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Sex-organ infections	- Sexual intercourse - Direct body contact - Saliva - Blood contact	· Gonorrhoea · AIDS · Syphilis · Candidiasis · Hepatitis B
Vector borne diseases	- Animal vectors Example: insects such as Mosquitoes, flies, and tsetseflies	· Malaria · Typhoid fever · Sleeping sickness

Table 6.2 Types and causes of diseases

Types	Causes	Examples
Infectious (Communicable) diseases	Micro-organisms (germs, pathogens)	. HIV/AIDS . TB . Malaria . Cholera, etc
Human induced diseases	Person's life style	. Cancer . Obesity . Fibrosis . Diabetes . Hypertnesion, etc
Degenerative disease	Tissue deterioration due to aging	. Deafness . Loss of accurate sight . etc
Genetic diseases	Mutation	. Sickle cell anaemia . Hypertension . Haemophilia
Deficiency disease	Nutrient shortage in diet	. Scurvey . Kwashiorkor . Goiter . Anaemia, etc

Social diseases	Social activities	. Alcoholism . Drug addiction . Isolation . Depression . etc
Functional diseases	- Organ or system's malfunction - No obvious cause	. Large intestine failurity . Chronic fatigue syndrome, etc

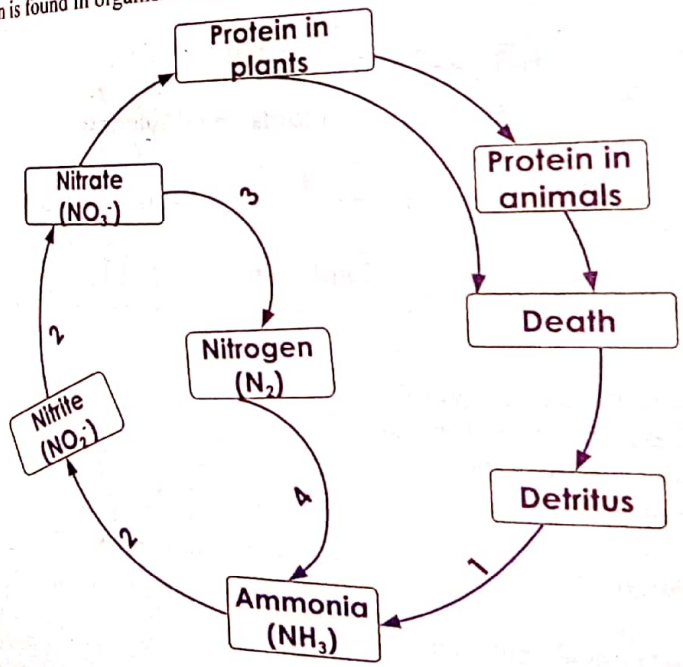
Multi-factorial diseases - are disorders caused by many factors.
Example: Hypertension - may be caused by one's life style or gene.
Atherosclerosis (fat accumulation in arteries can be due to aging, fatty diet or genes.

Role of bacteria in environment

Bacteria decompose dead organic matter, and recycle nutrients such as Carbon, Nitrogen, Sulphur, Phosphorus, etc.

The Nitrogen cycle

Nitrogen is found in organisms in the form of **proteins, DNA, RNA, ATP**, etc.



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Key

Detritus = dead organic matter

1= **Decomposition** by **ammonifying bacteria**

Example: Proteins $\xrightarrow{}$ NH_3

2= **Nitrification** by **nitrifying bacteria**

$NH_3 \xrightarrow{Nitrosomonas} NO_2^-$

$NO_2^- \xrightarrow{Notrobacter} NO_3^-$

3= Denitrification

$NO_3^- \xrightarrow{Pseudomonas} N_2$

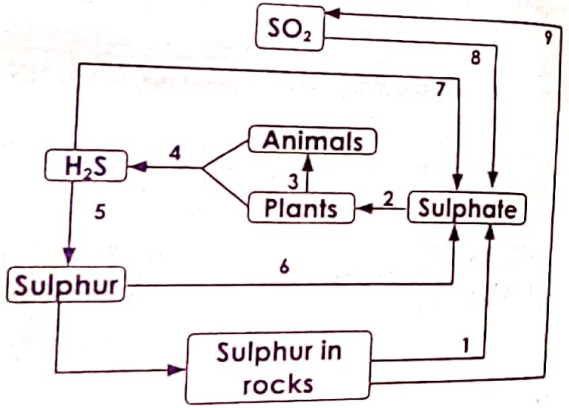
4= Nitrogen fixation

$N_2 \xrightarrow{Rhizobium} NH_3$

Rhizobium lives in root nodules of legumes. *Azotobacter* and Klebsiella live freely in soil, and also involved in **Nitrogen fixation.**

The Sulphur cycle

- Sulphur is found in many proteins



Key

1 = Weathering and oxidation
2 = Sulphate absorption from soil
3 = Feeding and assimilation
4 = Decomposition
5 = Oxidation
6 = Oxidation
7 = Oxidation
8 = Photochemical oxidation
9 = Combustion of fossil fuels

The accumulation of Sulphur and Nitrogen oxides in atmosphere causes pollution, and leads to the formation of **Acid rain**

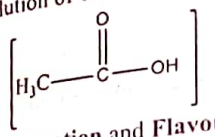
Acid rain contains H_2SO_4 and HNO_3, and acidifies soils and water bodies. It can kill organisms, corrode metals and rocks.

Role of bacteria in industry

Bacteria are used in the making of bread, alcohol, yoghurt and vinegar. They are also used in other process such as sewage treatment and leather processing.

Vinegar production

Vinegar - is dilute solution of ethanoic (acetic) acid.



- Used in food **Preservation** and **Flavouring.**
- **Pickling** - food preservation by using vinegar.
- Produced by oxidation (fermentation) of *ethanol* by *Acetobacter* (a genus of bacteria).

Bacteria are also used in **antibiotic** production.

Antibiotics - drugs used to kill bacteria.

The first antibiotic (i.e Penicillin) was produced by a fungus called *Penicillium*.

Now a days, genetically engineered bacteria are used to produce:-

- Antibiotics
- Insulin
- Human growth hormone
- Enzymes
- Vaccines, etc.

- Sewage (liquid waste) from households, institutions, and factories, is also treated (i.e. made harmless) by bacteria and other micro-organisms before it is released into the field or river. There are two ways of sewage treatment. These are:
  a) The percolating filter method.
  b) The activated sludge method.

Genetic modification of bacteria

Gene - Section of DNA of an organism.
- Codes for a particular protein.

Genetic engineering - is the act of modifying **genome** of an organism, usually by adding a gene from other sources.

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- 3 (three) main requirements for genetic engineering
  a) **Restriction endonucleases** - are enzymes that cut DNA at a specific sequence. They make genes ready for insertion into other DNA molecules.
  b) **DNA ligases** - are enzymes that insert and "tie" genes into DNA molecules.
  c) **Vectors** - are structures used to carry and transfer genes into other cells.

  Example: Viruses, Plasmids (Circular DNAs in bacteria)

Transgenic organisms - are genetically modified organisms. They have foreign genes added to them.

Steps of genetic engineering

The process of genetic engineering has the following steps.

1. Plasmid isolation from bacteria.
2. Plasmid opening by restriction enzymes.
3. Cutting out the gene of interest from donor DNA.
4. Joining together the gene and plasmid to create **recombinant plasmid**.
5. Introducing recombinant plasmid into suitable bacterium.

Important products such as **Insulin** and **Enzymes** are now a days produced by such a bacterium.

Plants are also genetically modified for:-

- Pest and disease resistance.
- Increased yield **quality** and **quantity**.
- Longer **shelf-life**
- Increased drought tolerance, etc.

A bacterium called *Agrobacterium tumefaciens* is used to transfer genes of interest into plant cells. Those plants which can not be genetically engineered by *A. tumefaciens* are modified by **gene gun**. Such plant cells are bombarded by "bullets" of gold pellets covered in DNA (i.e. golden gun).

Example: Maize, tobacco, carrots, soya-beans and apple are modified by gene gun.

Illustrative Examples

21. Where do we find bacteria?
    (A) In soil
    (B) In air
    (C) In water
    (D) All

Explanation:

- Bacteria are ubiquitous (i.e. found everywhere). Soil, air, water and surfaces are habitats for bacteria.

Answer: D

22. Nitrogen fixing bacteria are found in
    (A) Legume root nodules
    (B) The soil
    (C) The plant roots
    (D) A and B are correct

Explanation:

- **Rhizobium** is found in legume root nodules, while free-living Nitrogen fixers such as **Klebsiella** and **Azotobacter** live in the soil.

Answer: D

23. Which of the following scientists did not contribute to the development of germ theory?
    (A) Robert Hooke
    (B) Robert Koch
    (C) Joseph Lister
    (D) Louis Pasteur

Explanation:

- Robert Hooke contributed to cell discovery, not to germ theory. Koch, Lister and Pasteur contributed much to the development of germ theory.

Answer: A

24. In DNA cloning technology, which of the following molecules serves as a vector of gene of interest to be transferred to bacterial host? **(UEE, 2004 E.C)**
    (A) Bacterial DNA
    (B) Plasmid DNA
    (C) Nuclear DNA.
    (D) Mitochondrial DNA

Explanation:

Plasmid DNA.

- Plasmid DNA acts as a vector.

Answer: B

25. Which of the following is an infectious disease?
    (A) Kwashiorkor
    (B) Goitre
    (C) HIV/AIDS
    (D) Obesity

Explanation:

- HIV/AIDS is communicable (infectious) disease since it is caused by a micro-organism (HIV). Kwashiorkor and Goitre are deficiency diseases. Obesity is human induced disorder.

Answer: C

26. Which of the following is a multi-factorial disease?
    (A) Malaria
    (B) Hypertension
    (C) Alcoholism
    (D) Tuberculosis

Explanation:

- Hypertension may be caused due to our life style; or it can also be genetical.

Answer: B

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27. Which of the following is the right vector to transfer alien genes to bacteria through genetic engineering?
(A) Plasmids
(B) Snails
(C) Mosquitoes
(D) Bacteria

Explanation:

- Plasmids are suitable vectors for gene transfer.

Answer: A

28. Which of the following enzymes cut DNA at a specific sequence?
(A) DNA-ligase
(B) DNA-helicase
(C) DNA-polymerase
(D) Restriction endonuclease

Explanation:

- Restriction endonuclease - these search for specific sequences in DNA molecule, and cut it at such points.

Answer: D

29. Why do you think is Nitrogen important for life on earth?

Solution:

- Nitrogen is part of molecules such as **proteins, DNA, RNA and ATP,** and these molecules are essential for life on earth.

30. What is the main cause of acid rain? What are its consequences?

Explanation:

- Atmospheric pollution causes acid rain. **Nitrogen** and **Sulphur** oxides released from different sources accumulate in atmosphere and interact with water vapour to form **Sulphuric acid** and **Nitric acid**. These are washed down to water bodies and terrestrial environments. They corrode rocks and metals. It can also kill organisms.

31. Which of the following is a route of HIV transmission? **(NIMEE, 2004 E.C)**
(A) Insect bites.
(B) Sexual intercourse.
(C) Touching a person infected with HIV.
(D) Breathing air in a room with a person with AIDS.

Explanation:

- **Sexual intercourse** - the major route of transmission of HIV is sexual contact.

Answer: B

32. Which of the following cells can be taken as a good example of a cell that lacks nuclear membrane and mitochondria? **(NIMEE, 2004 E.C)**
(A) Bacterial cell
(B) Alga cell
(C) Fungal cell
(D) Protozoan cell

Explanation:

- Bacteria are in the kingdom Monera, which lacks membrane bound organells.

Answer: A

33. Human system that is highly attacked by HIV is
(A) Circulatory system
(B) Immune system
(C) Skeletal system
(D) Respiratory system

Explanation:

- HIV/AIDS attacks the immune system, and makes the body vulnerable to opportunistic diseases.

Answer: B

34. Which of the following does **not** contain **non-parasitic** members? **(NIMEE, 2004 E.C)**
(A) Bacteria
(B) Fungi
(C) Protozoa
(D) Viruses

Explanation:

- All Viruses are parasitic; bacteria, fungi and protozoa have free-living members.

Answer: D

35. Which of the following human hereditary diseases is caused by mutation in the haemoglobin gene?
(A) Sickle cell anaemia
(B) Phenylketonurea
(C) Color blindness
(D) Haemophilia

Explanation:

- **Sickle cell anaemia** - is caused due to mutation in the gene controlling the formation of haemoglobin.

Answer: A

36. DNA technology that is used to identify individuals is
(A) DNA sequencing
(B) DNA amplification
(C) DNA labeling
(D) DNA finger printing

Explanation:

- **DNA finger printing** - is used to identify individuals participated in crimes, for example. DNA sequencing is used to work out the complete base sequence of a given DNA molecule. DNA amplification is increasing the amount of DNA sample. DNA is labeld, for example, by radioactive elements to trace its position in a genome.

Answer: D

37. Some of the antiretroviral drugs currently in use for HIV treatment inhibit the reverse transcription of the viral genome. Which one of the following processes is inhibited?
(A) Fusion of the viral envelop with host cell membrane.

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(B) Copying of the viral RNA into DNA.
(C) Transcription of viral DNA into RNA.
(D) Assembly of viral components into a whole virus.

Explanation:

- If reverse transcription is inhibited, copying of the viral RNA into DNA is blocked

Answer: B

38. Why are crop plants genetically engineered?
(A) To make them disease resistant.
(B) To render them longer shelf-life.
(C) To improve their yield quality.
(D) All of the above.

Explanation:

- All of the above statements are correct.

Answer: D

39. What is the significance of *Agrobacterium* in agriculture?

Solution:

- *Agrobacterium* is bacterial genus used to transfer genes of interest to crop plant cells for the sake of genetic modification of the crops.

40. Crop plants which can not be genetically engineered can be modified by _____.

Explanation:

- **Gene gun** - genes of interest (DNA parts) are wrapped around gold pelletes. These are used as "bullets" and shot into plant cells. The cells take up the genes and incorporate into their DNA.

41. Which of the following is the mode of genome replication in AIDS virus?
(A) DNA → DNA
(B) RNA → RNA
(C) DNA → RNA
(D) RNA → DNA → RNA

Explanation:

- Viral RNA is reverse transcribed in to DNA, and the DNA is transcribed into viral RNA. Viral proteins are also produced. Then, virus particles are assembled from viral RNA and viral proteins.

Answer: D

42. Which bacterial species is used to transfer recombinant DNA to plants? **(CEE, 1995 E.C)**
(A) *Escherichia coli*
(B) *Agrobacterium*
(C) *Nitrogen fixing bacteria*
(D) *Denitrifying bacteria*

Explanation:

- *Escherichia coli* is a parasitic bacteria in colon; Nitrogen fixing bacteria convert elemental Nitrogen into Ammonia (i.e Rhizobium is

symbiotic in root nodules of legumes, *Azotobacter* and *klebsiella* free-living in soil); denitrifying bacteria convert Nitrate in the soil into elemental Nitrogen and reduce the amount of Nitrogen in the soil.

Answer: B

43. "Farmer's lung" which is allergy in the lung is caused by:
(A) Bacteria
(B) Fungus
(C) Virus
(D) Protozoa

Answer: B

Explanation:

- "Farmer's lung" is a fungal disease.

44. Which of the following is a degenerative disease?
(A) Haemophilia
(B) Scurvy
(C) Goitre
(D) Loss of accurate sight

Explanation:

- As the age of an individual increases, his or her ability to sharply observe things decreases. Haemophilia is a genetic (hereditary) disease; scurvy and goitre are deficiency diseases.

Answer: D

45. Which of the following is the ultimate source of Sulphur for plants?
(A) Atmosphere
(B) Rocks
(C) Water
(D) Animals

Explanation:

- Though atmosphere, water and animals may be sources of Sulphur for plants indirectly, rocks are the original direct source of Sulphur for plants. The **Sulphur** in **rocks** is converted into **Sulphate** by weathering so that plants can absorb and use it.

Answer: B

1.3. Viruses

Virus - means poisonous fluid in Latin.
Viron - Virus particle.
Virus - is genetic material (DNA/RNA) in protein coat (**capsid**). Some may have additional outer membrane made from **lipoprotein**. Other proteins and enzymes are embedded in the capsid. Viruses are not true cells. They are smaller than the smallest bacteria (0.01 - 0.1μm in length or diameter).
They lack organelles, and show no life processes common in other cells. Consequently, all of them are **parasites**. They depend on other living cells for their reproduction. They invade cells and "hijack" its metabolic machinery. Then, the host cells are forced to produce more viruses.

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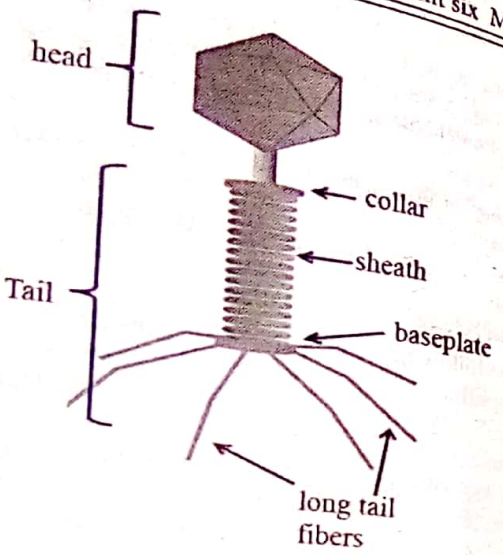

Fig. 6.7 Bacteriophages

Stages of viral infection

Attachment --- Entry --- Uncoating --- Replication ---Assembly --- Budding of or membrane rapture --- release.

Types of viruses

Viruses vary in **shape** and **form**, **way of infection**, and **genetic material**. Based on their host, they can be classified as plant, animal or bacterial infecting viruses. Bacteria infecting viruses are also known as **bacteriophages.**

Based on their genetic material, there are 3 types of viruses. These are:

a) **DNA viruses** - have DNA in their coat.

Example: Herpes simplex (causes cold sores), bacteriophages, adenoviruses, etc

b) **RNA viruses** - have RNA in their coat.

Example: H_1N_1 (Cause swine flu), tobacco mosaic virus, influenza virus, etc.

c) **Retro viruses** - have RNA as their genetic material in their coat, but this is converted into DNA during infection.

Example: HIV - causes AIDS.

HIV RNA $\xrightarrow{\text{Reverse transcriptase from HIV}}$ HIV DNA
(By reverse transcription)

Steps of virus infection

Attachment --- DNA/RNA injection --- Replication --- Assembly --- Release.

Reproductive cycles of viruses

1. **Lytic cycle** - The host cell bursts to release virus particles assembled in it.
2. **Chronic release cycle** - Few viruses are released at a time by **exocytosis** without the host cell bursting.
3. **Lysogenic cycle** - Viral DNA is incorporated into host cell's DNA, and replicated as host cell divides. Later, triggered by environmental stimulus, viral DNA starts producing **viral RNA** and **viral proteins**. Then, whole viruses are assembled and released by host **cell lysis**, or by **exocytosis**.

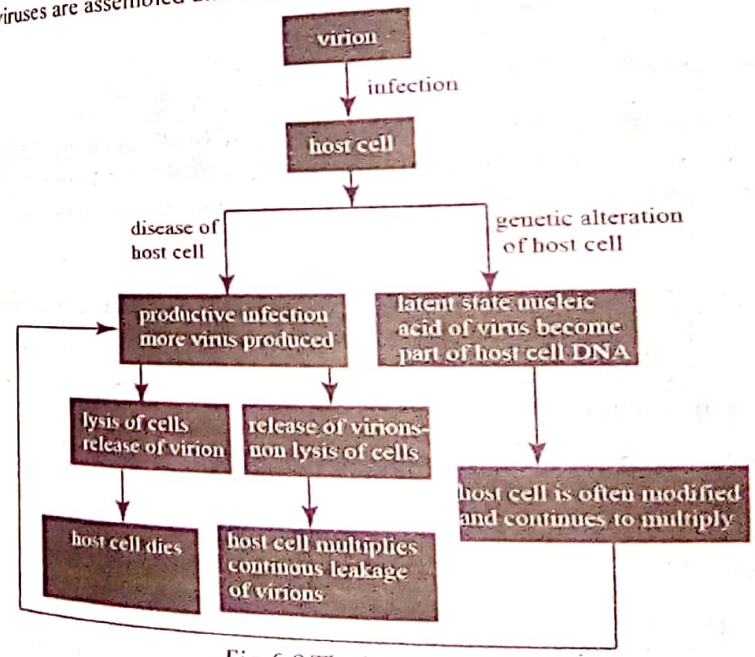

Fig. 6.8 The lytic cycle

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HIV and AIDS

HIV - Human Immune Deficiency Virus.

HIV is a **retrovirus** that attacks cells of the immune system.

Example: CD_4 T-lymphocytes (T-helper cells) which coordinate the immune system. As a result, their infection is serious. It is analogous to an attack on a "General" of troops. After the loss of their "General", the troops may try to attack their enemy in a disorganized way, and fails finally!

CD_4 - is a protein that **protrudes** from different human cells.

Example: T-lymphocytes, macrophages, natural killer cells, etc.

HIV has **spikes** on its surface. The spike consists of a **head** (gp - 120), and an extension (tail, gp - 41).

gp = glycoprotein

During infection, gp-120 of HIV fits into CD_4 of T-cells as an egg fits into its case. In the core of HIV, there are two molecules of **RNA**, and two molecules of **reverse transcriptase** enzyme. HIV's RNA is injected into host cell and the process continues as follows:

Viral RNA $\xrightarrow[\text{(By reverse transcriptase)}]{\text{Reverse transcription}}$ Viral DNA $\longrightarrow$ Incorporation into host cell's DNA $\xrightarrow{\text{Transcription}}$ Viral RNA $\xrightarrow{\text{Translation}}$ Viral proteins (Example: enzymes)

Once viral RNA and Viral proteins are produced, virus particles can be assembled and released by exocytosis. These infect other T-cells.

The immune system recognizes infected cells and destroys them. This cycle of **infection-destruction** and **replacement** continues so long as the body can replace the T-cells. When the body fails to do so, number of viruses increases and other tissues such as brain cell can be infected now. Then, opportunistic diseases attack the body.

Example: TB, Pneumonia, fungal infections, etc.

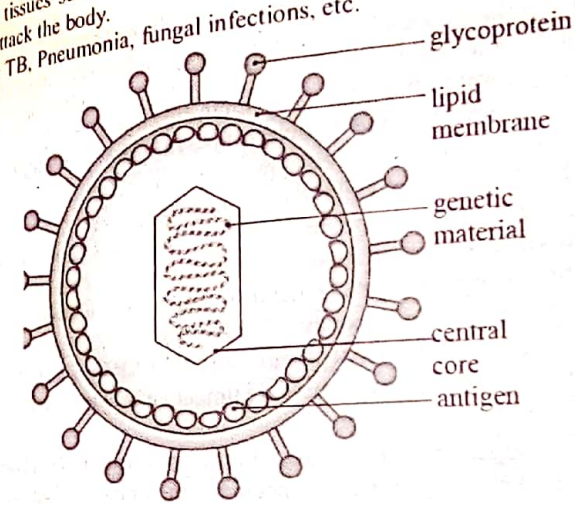

Fig. 6.10 Structure of herpes simplex

Stages of HIV infection

There are 3 (three) main stages of HIV infection.

a) **Acute stage** - soon after infection, white blood cells start producing **antibodies** against HIV. The antibodies are used in the diagnosis of HIV.

b) **Latency period** - is the asymptomatic stage when the body replaces T-helper cells as fast as they are destroyed.

c) **Disease period** - AIDS-Related-Complex (**ARC**) appears. Level of T-cells and antibodies declines.

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Virus level rises. Opportunistic diseases attack the patient, and death may follow. AIDS has no cure or vaccine, but some antiretroviral drugs are used in slowing down the progression of the infection to AIDS. These drugs are:-

1) **Entry inhibitors** - These are drugs that attach to CD_4 of T-cells, or gp-120 of HIV, and prevent their fitting together. Thereby, they minimize further infection of other cells.
   Example: Enfuvirtide
2) **Nucleotide Reverse Transcriptase Inhibitors (NRTI)** - Are incorrect nucleotides that mislead the enzyme and prevent reverse transcription.
   Example: Zidovudine (AZT), Didanosine, Zalcitabine, Stavudine, Lamivudine, Abacavir, Emtricitabine, Tenolavir.
3) **Non-Nucleotide Reverse Transcriptase Inhibitors (NNRTI)** - Attach to the enzyme and prevent reverse transcription.
   Example: Nevirapine, Delavivdine, Elavirenz.
4) **Protease-inhibitors** - prevent HIV's protease from facilitating the assembly of new virus particles.
   Example: Amprenavir, Atazanavir, Indinavir, Fosamprenavir, Nelfinavir, Lopinavir, Rotonavir, Saquinavir.

High Activity Anti - Retroviral Treatment (HAART) - is using the above drugs in combination. It is an effective treatment against the virus, but has many side effects.

HIV mutates (changes) into different strains from time to time. Therefore, it has been difficult to develop preventive vaccines and curative drugs against it. HIV/AIDS has social and economic impacts on an individual, a family and a nation as a whole. Its spread can be minimized by responsible sexual behaviours such as limiting sexual partner and using condom. Male circumcision can also help.

Illustrative Examples

46. A genetic material (DNA/RNA) embedded in a protein coat is
(A) Protozoa
(B) Fungus
(C) Virus
(D) Protista

Explanation:
- Viruses are simple acellular structures in which DNA/RNA is embedded in a protein coat.

Answer: C

47. The protein coat of virus is known as:
(A) Protease
(B) Capsid
(C) HAART
(D) Virion

Explanation:
- Viral protein coat is called Capsid.

Answer: B

48. Which of the following is not true of viruses?
(A) Viruses are true cells.
(B) Some viruses are free-living.
(C) Some viruses are as large as some bacteria.
(D) All of the above are incorrect.

Explanation:
- All viruses are acellular, parasitic and smaller than the smallest bacteria.

Answer: D

49. CD_4 is:
(A) A glycoprotein on HIV surface.
(B) A protein on T-cell's surface.
(C) A glycoprotein on T-cell's surface.
(D) A protein on HIV surface.

Explanation:
- CD_4 is a protein on the surface of some human cells such as T-lymphocytes.

Answer: B

50. Which of the following is the major killer of AIDS patients? **(UEE, 2003 E.C)**
(A) Pneumonia
(B) Anaemia
(C) Malnutrition
(D) Opportunistic infections

Explanation:
- Opportunistic infections are the major killers of AIDS patients because body's immunity is impaired.

Answer: D

51. Why are T-lymphocytes more vulnerable to HIV infection? Because they possess: **(UEE, 2003 E.C)**
(A) Thin cell membrane
(B) gp-120 on their surface
(C) HIV receptor protein
(D) Large pores in their membrane

Explanation:
- T-lymphocytes have HIV-receptor proteins called CD_4 on their surfaces.

Answer: C

52. Which of the following statements is **true** about bacteria?
(A) Bacteria found in the human body are all parasitic.
(B) All bacteria are harmful because they spoil foods.
(C) All bacteria cause diseases to plants, animals and humans in some ways.

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(D) Some bacteria living in the human body make vitamins needed by the body.

Explanation:

- Some bacteria living in human intestine can make vitamins that can be absorbed and used by the body. Example: Vitamin K (needed in blood clotting)

Answer: D

53. Which of the following is a DNA virus?
(A) Tobacco mosaic virus
(B) Influenza virus
(C) Adenovirus
(D) H_1N_1

Explanation:
Adenovirus.

- Adenovirus is a DNA virus. Others are RNA viruses

Answer: C

54. What is special about retroviruses?

Explanation:

- Retroviruses have RNA as their genetic material. But this RNA is reverse transcribed into DNA. Enzyme reverse transcriptase is involved in the conversion, and the conversion is made during infection.

55. Explain briefly how **anti-retroviral** drugs act at different stages of HIV AIDS progression.

Explanation

- First, entry inhibitors bind to CD_4 (on T-cells) or gp-120 (on HIV) and prevent their binding together. Once this is effectively done, the viruses can no longer enter into the cell. Secondly, after the virus injects its RNA into host cell, nucleotide and non-nucleotide inhibitors mimic normal DNA-nucleotides and stop viral DNA formation by reverse transcriptase. Finally, if the virus manages to manufacture all its spare-parts, protase inhibitors stop virus assembly by inhibiting enzymes that cut viral proteins into suitable pieces.

56. Which of the following is **not** eukaryotic cell?
(A) *Penicillium*
(B) Amoeba
(C) *Streptococcus*
(D) *Plasmodium*

Explanation:

- *Streptococcus* is a bacterium (Prokaryotic). *Penicillium* is a fungus. Amoeba and *plasmodium* are protozoans (protists, eukaryotic).

Answer: C

57. Which of the following is **acellular**?

(A) HIV
(B) *Lactobacillus*
(C) Mucor
(D) *Candida*

Explanation:

- HIV is a virus, and hence **acellular**. **Lactobacillus** is a bacterium, mucor and *candida* are fungi.

Answer: A

58. Which of the following is **incorrect** about viruses?
(A) They are made from nucleic acid and protein coat.
(B) They are all dependant on other cells for their reproduction.
(C) They live on dead organic matter for their nutrition.
(D) They attack plant, animal, bacterial and human cells.

Explanation:

- All viruses are considered living only when they are in living cells. They don't live on dead organic matter.

Answer: C

59. Which of the following makes viruses different from other micro-organisms
(A) They have no cellular organells.
(B) They can not carry out any metabolic processes on their own.
(C) They must enter other cells to reproduce.
(D) All of the above are correct.

Explanation:

- All of the above are correct

Answer: D

60. What are the abbreviations HIV and AIDS?

Explanation:

- HIV = Human Immune Deficiency virus.
- AIDS = Acquired Immune Deficiency Syndrome.

61. The period when the body replaces T-helper cells as fast as they are destroyed is
(A) Window period
(B) Latency phase
(C) Disease phase
(D) Acute phase

Explanation:

- **Latency phase** is the stage at which the body is able to replace what it lost (i.e. T-cells). The person is **carrier** at this stage, but can transfer to others.

Answer: B

62. The best treatment of HIV/AIDS in which several **Anti-Retroviral** drugs are combined to target different stages of the virus's infection process is known as:
(A) HAART
(B) Latency
(C) Accuracy
(D) Assembly

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Explanation:
HAART.

- High Activity Anti-Retroviral Therapy (HAART) is a combination of different drugs in the treatment.

Answer: A

63. Which of the following can limit the spread of HIV/AIDS?
(A) Responsible sexual behaviour.
(B) The use of condoms.
(C) Restricting the number of sexual partners.
(D) All of the above are correct.

Explanation:

- Male circumcision, using condoms, restricting sexual partners and developing responsible sexual behaviour are limiting ways of HIV spread.

Answer: D

64. Viral life cycle where the host cell continues reproducing, for a period, without producing more viruses, but copying the viral DNA is:
(A) Chronic release
(B) Lysogeny
(C) Lytic
(D) A & C

Explanation:

- In Lysogeny, the viral DNA is incorporated in host cell's DNA, and duplicated as the host reproduced.

Answer: B

65. Which of the following is **incorrect** about HIV/AIDS?
(A) AIDS affects the productive age of humans.
(B) AIDS has preventive vaccine and curative drugs.
(C) The causative agent of AIDS mutates regularly.
(D) HIV attacks the defence force of the body.

Explanation:

- HIV/AIDS has no cure or no vaccine.

Answer: B

Review Questions On Unit Six

Choose the correct answer

1. Which of the following is **not true** about T-helper cells?
(A) They have gp-120 on their surface.
(B) They possess CD_4 on their surface.
(C) They are easily attacked by HIV.
(D) They are part of the immune system.

2. In Nitrogen cycle, Nitrogen gas is taken back to the atmosphere by:
(A) Nitrifying bacteria
(B) Decomposers
(C) Denitrifying bacteria
(D) Ammonifying bacteria

Explanation: In lysogenic cycle, the viral DNA is incorporated in to host cells DNA, and duplicated as the host cell reproduces.

Answer: B

65. Which of the following is incorrect about HIV/AIDS?
A. AIDS affects the productive age of humans.
B. AIDS has preventive vaccine and curative drugs.
C. The causative agent of AIDS mutates regularly.
D. HIV attacks the defence fore of the body.

Answer: B

66. Which groups of bacteria consists of rod shaped cells?
A. Gram-positive
B. Bacilli
C. Cocci
D. spirilla

Explanation:

Bacteria	Shape
Bacilli	Rod
Cocci	Spherical
Sprina	Spiral

Answer: B

67. Which of the following do viruses share with living things?
A. Cell wall
B. Cytoplasm
C. Nucleic acid
D. Protoplasm

Explanation: Virus shave nucleic acid which is either RNA or DNA.

Answer: C

68. Which of the following is true about protozoa?
A. They are a group of bacteria
B. They are one-celled plants.
C. They represent both plants and animals.
D. They are unicellular organisms.

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Explanation: Protozoans are:

- ✓ Unicellular
- ✓ Heterotrophs (Freeliving or parasitic)
- ✓ Eukaryotes

Example: Amoeba, plasmodium, etc.

69. Choose organisms that belong to eukaryotes? Answer: D
    A. Bacteria
    B. Blue green algae
    C. Amoeba
    D. viruses

70. Choose the characteristic that viruses share with living things. Answer: C
    A. They are made up of many specialized cells.
    B. They contain genetic material.
    C. They reproduce by mitosis.
    D. They contain chlorophyll.

Explanation: Viruses share genetic material in nucleic acid such as DNA or RNA but not both.

71. Which stage in the life cycle of HIV is inhibited by antiretroviral drug known as integrase – inhibitor? Answer: B
    A. Conversion of viral RNA to DNA.
    B. Replication of the viral genome.
    C. Insertion of the viral genome into host genome.
    D. Release of the viral oprogeny from the cell

Explanation: Integrase inhibitor prevents the insertion of the viral genome in to host genome.

72. From which organism is the first antibiotic used in medicine produced? Answer: B
    A. Gram negative bacteria

    B. Members of fungi
    C. Genetically modified protozoa
    D. Higher plants.

Explanation: The first antibiotic is developed from a fungial species called ***Penicillium notatum*** in 1928 by English bacteriologist called Sir. Alexander Fleming.

Answer: B

73. What is the difference between HIV and AIDS?
    A. AIDS is the virus and HIV is the disease.
    B. Both AIDS and HIV refer to disease.
    C. AIDS is the disease and HIV is the virus.
    D. Both AIDS and HIV refer to the virus

Answer: C

Conceptual Summary Example

1. What characteristics does a virus share with a living cell? What characteristics of life are absent in virus?

Explanation: Viruses share the following characteristics in common with a living cell:

- ✓ Presence of genetic material (DNA or RNA).
- ✓ Reproduction (only inside the host cell).

- Viruses do not perform almost all the life activities of a living thing except, their ability to reproduce inside the host cell.

2. What are the steps in lytic cycle?

Explanation: Lytic cycle involves five major stages:

- ✓ Attachement to the host cell.
- ✓ Penetration of viral nucleic acid in to the host cell.
- ✓ Replication of the viral nucleic acid.
- ✓ Assembly of newly synthesized components in to new viruses.

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✓ Release from the host cell.

3. How does a lysogenic cycle differs from lytic cycle?

Explanation:

- Lytic viruses reproduce inside the host cell and are released by brusting open the host call, this results in killing of the host cell.
- Lysogenic viruses reproduce without killing the host cell.

4. Trace the ovolutionary origin of viruses according to current hypothesis.

Explanation: There are different views concerning the evolutionary hypothesis of viruses. They are:

i) Viruses may be bits of nucleic acid that originally "escaped" from animal, plant, or bacterial cells.

ii) Some Biologists hypothesize that viruses must have evolved before both prokaryotes and eukaryotes.

5. Characterize bacteriophages, and contrast a lytic cycle with a lysogenic cycle.

Explanation: Phages (bacteriophages) are viruses that infect bacteria For more details see conceptual example number 3.

6. Explain how HIV infects human cells?

Explanation:

- Viruses enter human cells by membrane fusion or by endocytosis. HIV enters to the T-lymphocyte by binding itself to the CD_4 proteins on the surface of the T-lymphocytes. Then after it facilitates its way into the T-lymphocyte cells.

7. Describe the reproductive cycle of a retrovirus such as human immuno deficiency virus (HIV)?

Explanation: Retroviruses, such as HIV, use reverse transcriptase to transcribe their RNA genome in to DNA intermediate that becomes integrated in to the host DNA, copies of viral RNA are then synthesized.

Post – test 2

1. Which one is correct about pathogens?
   A. They are agents that cause disease.
   B. They include most bacterias
   C. They include most of funguses.
   D. They include authotrops.
2. Which one is contained in the viral genome?
   A. DNA only
   B. RNA only
   C. DNA and RNA
   D. DNA or RNA
3. The caspid of virus consists of:
   A. Protein
   B. Neucleic acid
   C. RNA and lipid
   D. carbohydrate
4. Viruses that kill the host cells are:
   A. Lytic virtuses
   B. Lysogenic viruses
   C. phrophages
   D. Temperate
5. Which of the following is the correct sequential order in viral reproduction?
   A. Attachment→penetration → Assembly→ Replication→Release
   B. Penetration→Absorption→Assembly→Replication→Release
   C. Attachment→penetration→Replication→Assembly→Release
   D. Attachment→penetration→Replication→Release→Assembly
6. In lysogenic conversion:
   A. Bacterial cell may exhibit new properties.
   B. The host cell dies
   C. Reverse transcriptase transcribes DNA in to RNA
   D. None

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7. Peptidoglycan is a chemical which is found in the cell walls of:
   A. Most viruses
   B. All prokaryotes
   C. Most eubacterias
   D. All
8. Retrovirus
   A. are DNA viruses
   B. are RNA viruses
   C. Use reverse transcriptase to transcribe DNA in to RNA.
   D. Use DNA viruses to cause AIDS.
9. This question is based on the stages of HIV multiplication below.

1. Assembly	4. Budding
2. Transcription	5. Entry
3. Maturation	6. Reverse transcription
	7. Integration

The correct sequential order is:

A. 2→ 4→ 6→ 7→ 1→ 3 →5
B. 5→ 6→ 7→ 2→ 1→ 4→ 3
C. 6→ 5→ 2→ 1→ 4 → 3 → 7
D. 7 → 6 → 4 → 2 → 5 → 1 →3

Critical thinking

1. How might life be different if all the microorganisms were killed?
2. Imagine that you discover a new microorganism. After careful study you determined that it should be classified as protozoa. What characteristics lead you to this classification?
3. Why do you think is the reason that diary products such as yoghurt and cheese do not be casily contaminated by pathogenic bacterias?
4. Why do you think is it difficult to make vaccination for HIV?

3. Which of the following is **incorrectly** matched?
   (A) Yeast-Fungus (C) Bacteriophage-Bacteria
   (B) Plasmodium-Protozoa (D) Candida-Fungus
4. Which of the following is common structure in all bacteria?
   (A) Cell wall (C) Flagellum
   (B) Pillus (D) Plasmid
5. Which of the following bacterial structures are usually used as vectors in genetic engineering?
   (A) Capsule (C) DNA
   (B) Ribosome (D) Plasmid
6. Which of the following diseases is caused by a **virus**?
   (A) Pneumonia. (C) Influenza.
   (B) TB (D) "Farmers lung"
7. Which of the following is not an infectious disease?
   (A) Cholera (C) Anaemia
   (B) Common cold (D) HIV/AIDS
8. Which of the following disorders is categorized as social disease?
   (A) HIV/AIDS (C) Goitre
   (B) Arthritis (D) Alcoholism
9. Which of the following is the role of *Agrobacterium* in agriculture?
   (A) It converts Nitrogen into Nitrate.
   (B) It transfers gene of interest to plant cells.
   (C) It converts Nitrates into Nitrogen.
   (D) It decomposes dead organic matter.
10. The bacteria that is involved in **denitrification** is:
    (A) *Rhizobium* (C) *Pseudomona*
    (B) *Klebsiella* (D) *Azotobacter*
11. The bacteria that play important role in **Nitrogen fixation** is:
    (A) *Azotobacter* (C) *Klebsiella*
    (B) *Rhizobium* (D) All are involved
12. *Acetobacter* is industrially involved i the production of:
    (A) Vinegar (C) Penicillin
    (B) Ethanol (D) Insulin
13. A virus is composed of:
    (A) Genetic material and protein coat.
    (B) Nucleus and protein coat.

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(C) Ribosomes and cell membrane.
(D) Cytoplasm and cell membrane.

14. The type of virus that attacks bacteria is:
(A) Herpes simplex
(B) H_1N_1
(C) Bacteriophage
(D) Tobacco mosaic virus

15. Which of the following is transmitted by a vector?
(A) HIV/AIDS
(B) Pneumonia
(C) Malaria
(D) Syphilis

16. What is the major transmission method of HIV/AIDS?
(A) Blood transfusion
(B) Infected needles
(C) Mother to foetus
(D) Sexual intercourse

17. _____ is a state of complete physical, mental and social well-being
(A) Infection
(B) Health
(C) Disease
(D) Efficiency

18. A condition in which part or all of a body is made less efficient is
(A) Health
(B) Malfunctioning
(C) Disease
(D) Infection.

19. Infectious (communicable) diseases are caused by:
(A) Pathogenes
(B) Mutation
(C) Aging
(D) Malfunctioning

20. **Deficiency diseases** are caused by lack of:
(A) Drugs.
(B) Nutrients
(C) Alcohol
(D) Cosmetics

21. A disease that may be caused by more than one factor is known as:
(A) Functional
(B) Multifactorial
(C) Social
(D) Genetic

22. The level of _____ increases in blood when an HIV carrier is changed into AIDS patient.
(A) White blood cells
(B) Antibodies
(C) T-lymphocytes
(D) Viruses

23. **Protease inhibitors** prevent virus _____ in HIV/AIDS treatment.
(A) Assembly
(B) Binding
(C) Entry
(D) Release

24. The level of HIV-virus, HIV-antibodies and T-lymphocytes relatively stays constant during:

(A) Acute stage
(B) Latency period
(C) Disease stage
(D) Window period

25. If AIDS patients are given **anti-retroviral** drugs that are targeted against the protease enzyme, which one of the following stages of the HIV life cycle would be inhibited?
(A) Entry in the host cell.
(B) Reverse transcription.
(C) Integration of viral DNA into host DNA.
(D) Assembly of viral parts into a whole virus.

26. Choose the disease that is caused by what is known as the "**droplet-infection**".
(A) Flu
(B) AIDS
(C) Cholera
(D) Malaria

27. To which of the following groups does HIV belong?
(A) Plasmids
(B) Retroviruses
(C) DNA viruses
(D) Bacteriophages

28. What are the individual strands of a fungal **mycellium** called?
(A) Fungus
(B) Spores
(C) Hyphae
(D) Hydra

29. Which of the following is **true** about the anti-retroviral drugs currently used to treat AIDS patients?
(A) They cure AIDS.
(B) They stop HIV transmission.
(C) They serve as anti-HIV vaccines.
(D) They slow down HIV multiplication.

30. What is the main mode of transmission of diseases such as cholera and typhoid fever?
(A) Sexual intercourse
(B) Bite of animal vectors
(C) Blood-to-blood contact
(D) Drinking contaminated water

31. Which of the following cells of the immune system are the major target of HIV?
(A) B cells
(B) Phagocytes
(C) Killer T-cells
(D) Helper T-cells

32. Cyanobacteria (blue-green algae) are:
(A) Photoheterotrophs
(B) Photoautotrophs
(C) Chemoautotrophs
(D) Chemoheterotrophs

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33. When bacteria are rod-shaped, they are called:
(A) Cocci
(B) Spirilla
(C) Vibrios
(D) Bacilli

34. A condition in which ammonium is converted into Nitrate in the Nitrogen cycle is:
(A) Nitrification
(B) Ammonification
(C) Decomposition
(D) Denitrification

35. The gene for insulin can be transferred from human DNA to a bacterium. The enzyme used to cut out the gene from human DNA is
(A) Ligase
(B) Helicase
(C) Restriction endonuclease
(D) Polymerase

36. The cells of which group of micro-organisms can described as prokaryotic? **(UEE, 2003, E.C)**
(A) Viruses
(B) Protozoa
(C) Algae
(D) Bacteria

37. Which of the following are rod-shaped bacteria?
(A) Cocci
(B) Spirochetes
(C) Spirilla
(D) Bacilli

38. What are bacteriophage? **(UEE, 2003, E.C)**
(A) Bacteria
(B) Viruses
(C) Protozoa
(D) Fungi

39. To which major area of relevance and application of biology is the production of transgenic organisms related? **(UEE, 2003, E.C)**
(A) Agriculture
(B) Environment
(C) Biotechnology
(D) Medicine

40. Which component of soil fertility is improved when farmers grow legumes in crop rotation? **(EUEE, 2005, E.C)**
(A) Phosphorus
(B) Nitrogen
(C) Sulphur
(D) Carbon

41. What are the most frequent causative agents of food poisoning? **(EUEE, 2005, E.C)**
(A) Bacteria
(B) Protozoa
(C) Viruses
(D) Worms

42. Which of the following practices does **not** normally transmit HIV? **(EUEE, 2005, E.C)**
(A) Sexual intercourse through anus.

(B) Sharing injection needles.
(C) Blood transfusion.
(D) Shaking hands.

43. Which of the following human diseases can be prevented by taking proper diets? **(EUEE, 2005, E.C)**
(A) Degenerative diseases
(B) Genetic diseases
(C) Social diseases
(D) Deficiency disease

44. To which of the following groups do those bacteriophage that integrate their DNA into the chromosome of their bacterial host belong? **(EUEE, 2005, E.C)**
(A) Virulent viruses
(B) Lytic viruses
(C) Lysogenic viruses
(D) Non-parasitic viruses

45. Which group of micro-organisms causes the disease known as athlete's foot? **(EUEE, 2005, E.C)**
(A) Bacteria
(B) Fungi
(C) Protozoa
(D) Viruses

46. Choose the one that is different from all the others. **(EUEE, 2005, E.C)**
(A) Genetically modified organisms.
(B) Genetically engineered organisms.
(C) Pathogenic organisms.
(D) Transgenic organisms.

47. What is the advantage of using HAART (Highly Active Anti-Retroviral Therapy) for the treatment of HIV? **(EUEE, 2005, E.C)**
(A) It gives a lasting immunity to HIV.
(B) It prevents mutation of HIV.
(C) It prevents re-infection by HIV.
(D) It helps to break the life cycle of HIV.

48. On which one of the following principles are most of the anti-HIV drugs currently in use working? **(EUEE, 2005, E.C)**
(A) Inhibition of enzyme action.
(B) Degradation of viral RNA.
(C) Digesting of viral particles.
(D) Phagocytosis of the virus.

49. If a new anti-HIV drug is to be developed to prevent the virus from entering the host cell, which one of the following processes should the drug target? **(EUEE, 2005, E.C)**
(A) Reverse transcription.

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(B) Binding of Gp-120 and CD_4.
(C) Integration of viral DNA into host DNA.
(D) Assembly of viral parts into a whole virus.

50. Which one of the following is a human genetic disease?
(A) Kwashiorkor
(B) Haemophilia
(C) Arthritis
(D) Cholera

51. Which of the following bacteria convert nitrates in soil into atmospheric nitrogen during nitrogen cycle?
(A) Ammonifying bacteria
(B) Nitrifying bacteria
(C) Nitrogen-fixing bacteria
(D) Denitrifying bacteria

52. In the process of genetic engineering, which enzyme is used to cut the genes of the donor and recipient organism?
(A) Ligase
(B) Restriction endonuclease
(C) Lyase
(D) Plasmid

53. Which of the following viral reproductive life cycle involves the incorporation of the DNA of the virus into the DNA of the host cell?
(A) Lysogenic cycle
(B) Lytic cycle
(C) Chronic release cycle
(D) Transduction cycle

54. Which one of the following viruses uses bacteria to replicate its genetic information?
(A) HIV
(B) Herpes
(C) H_1N_1
(D) Bacteriophage

55. What is the importance of gram staining technique?
(A) To identify bacteria as gram positive or negative.
(B) To help bacteria to resist antibiotics.
(C) To produce vaccines from bacteria.
(D) To isolate endotoxins from bacteria.

56. The structures on T-helper cells which bind to the gp-120 of HIV during infection are known as
(A) Gp-41
(B) CD_4 receptors
(C) Glycolipids
(D) Macrophages

57. Which of the following diseases is caused by a **protozoa**?
(A) Athlete's foot
(B) Sleeping sickness
(C) Cholera
(D) Candidiasis

58. Which of the following is the true about viruses?
(A) They are prokaryotes.

(B) They have nucleus.
(C) They are obligate parasites.
(D) They are larger than bacteria.

59. To which of the following groups does HIV belong?
(A) Retroviruses
(B) Bacteriophage
(C) DNA viruses
(D) Plasmids

60. What makes animal cells different from bacterial cells? Animal cells:
(A) Have mitochondria
(B) Are smaller
(C) Are prokaryotic
(D) Have ribosomes

61. What is the role of **reverse transcriptase** in the process of HIV replication?
(A) Production of protein
(B) Conversion of **RNA** into **DNA**
(C) Conversion of **DNA** in **RNA**
(D) Destruction of host cell **DNA**

62. Plants obtain nitrogen for protein synthesis from the soil in the form of
(A) Nitrogen gas
(B) Ammonia
(C) Nitrate
(D) Nitrite

63. What is the role of anti-retroviral drugs in the fight against HIV/AIDS?
(A) Interfere with the reproduction of the virus.
(B) Get rid of the virus from the body.
(C) Give immunity against the virus.
(D) Prevent transmission of the virus.

64. Generally, prokaryotic cells have:
(A) Membrane-bound nucleus
(B) Membrane-bound organelles
(C) Circular DNA
(D) Linear DNA

65. Which of the following structures are found in **all bacteria**?
(A) Nucleus and cell wall.
(B) Plamid & pilli.
(C) Cell wall and flagellum.
(D) Cell membrane and genetic material.

66. Which of the following is **correctly matched** (paired)?
(A) Protozoa-Spirochaete.
(B) Bacteria-Chlamydomonas.
(C) Algae-Plasmodium.
(D) Fungi-Candida

67. Which of the following is the **last step** in **Gram staining**?
(A) Counter staining
(B) Iodine treatment
(C) Decolorization
(D) Fixation

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68. Which of the following diseases is caused b a **fungus**?
(A) Phenomena
(B) 'Farmer's lung'
(C) Sleeping sickness
(D) Measles

69. Which of the following is an **infectious** disease?
(A) Arthritis
(B) Botulism
(C) Clinical depression
(D) Kwashiorkor

70. Which statement is correct about *agrobacterium*?
(A) It converts nitrogen into nitrates.
(B) It causes denitrification.
(C) It can act as a **vector** to carry genes.
(D) It causes common cold.

71. Which one of the following is a nitrogen fixing bacteria?
(A) Nitrobacter
(B) Pseudomonas
(C) Nitrosomonas
(D) Rhizobium

72. A virus contains:
(A) Both DNA and RNA.
(B) Protein coat and ribosomes.
(C) Cell wall and cell membrane.
(D) Genetic material and protein coat.

73. After a retrovirus infects its host, it can change its
(A) DNA to DNA
(B) RNA to DNA
(C) RNA to DNA
(D) Protein to DNA

74. Which of the following statements is **Incorrect**?
(A) It is a retrovirus.
(B) It has glycoprotein gp-120.
(C) It has CD_4 protein receptor.
(D) It attacks T-helper cells.

75. In the nitrogen cycle, nitrogen gas is returned to the atmosphere by:
(A) Nitrogen-fixing bacteria
(B) Decomposers
(C) Nitrifying bacteria
(D) Denitriying bacteria

Match the groups under column "B" with their examples under column "A"

"A"	"B"
76. Amoeba, Plasmodium	A) Algae
77. Yeast, Candida, Mucor, Penicillium	B) Viruses
78. Chlamydomonas, Phytoplankton	C) Protozoa
79. Streptococcus, Lactobacillus, Spirochaetes	D) Fungi
80. HIV, Common cold and Flu causative agents	E) Bacteria

Match the bacteria listed under column "B" with their role in Nitrogen cycle given under column "A"

"A"	"B"
81. Nitrifying bacteria	A) *Decomposers*
82. Symbiotic Nitrogen-fixing bacteria	B) *Nitrosomonas & Nitrobacter*
83. Denitrifying bacteria	C) *Klebsiella* and *Azotobacter*
84. Ammonifying bacteria	D) *Rhizobium*
85. Soil free-living Nitrogen-fixing bacteria	E) *Pseudomonas*

Fill in the blank spaces with the appropriate word or phrase

86. The counter stain used in Gram staining, added towards the end of the process, is _____.

87. Hereditary diseases that pass from generation to generation are called _____.

88. _____ is the bacterium used in the oxidation of ethanol into acetic (ethanoic) acid (i.e. vinegar).

89. Viruses that infect bacteria are known as _____.

90. _____ lives in root nodules of legumes, and fixes free elemental Nitrogen into ammonium.

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Answer For Review Questions

1. A
2. C
3. C
4. A
5. D
6. C
7. C
8. D
9. B
10. C
11. D
12. A
13. A
14. C
15. C
16. D
17. B
18. C
19. A
20. B
21. B
22. D
23. A
24. B
25. D
26. A
27. B
28. C
29. D
30. D
31. D
32. B
33. D
34. A
35. C
36. D
37. D
38. B
39. A
40. B
41. A
42. D
43. D
44. C
45. B
46. C
47. D
48. A
49. B
50. B
51. D
52. B
53. A
54. D
55. A
56. B
57. B
58. C
59. A
60. A
61. B
62. C
63. A
64. C
65. D
66. D
67. A
68. B
69. B
70. C
71. D
72. D
73. B
74. C
75. D
76. C
77. D
78. A
79. E
80. B
81. B
82. D
83. E
84. A
85. C
86. Safranin
87. Genetic diseases
88. *Acetobacter*
89. Bacteriophages
90. *Rhizobium*

UNIT SEVEN

ECOLOGY

Ecology - is the study of the interactions of organisms with each other, and with their physical environment

7.1 Recycling of matter through ecosystems

Importance of material recycling

There is limited amount of nutrients in ecosystems. The same atoms must constantly be re-used again and again in ecosystems and globally. Otherwise, life can not be sustained. Materials pass from one organism to another, and to the environment, and vice versa. Waste product of one organism may be used by another. All the organisms are interdependent, and interact with their environment. The following processes help recycling of nutrients in ecosystem:

- Feeding
- Excretion
- Respiration and breathing
- Death and decomposition.

Atoms which where once in your body (Example: Carbon) may be released to the environment in the form of CO_2. This may be taken up by plants in photosynthesis, and be assimilated. The plants may be consumed by other people or animals. So, nutrients that where once part of the body of an organism may become part of another in the course of time. **Materials** are **recycled** in this way, but **energy never recycles** in ecosystem. It flows from the ultimate source (the sun) through ecosystems, and finally lost in the form of heat.

Materials recycle as molecules whose bonds store energy. As nutrients move, energy is also transferred, but eventually lost as **heat**, and replaced as light from the sun. The nutrients tied up in organisms are taken back to the ecosystem by the action of **decomposers** (bacteria and fungi). Decomposition follows death of the organism.

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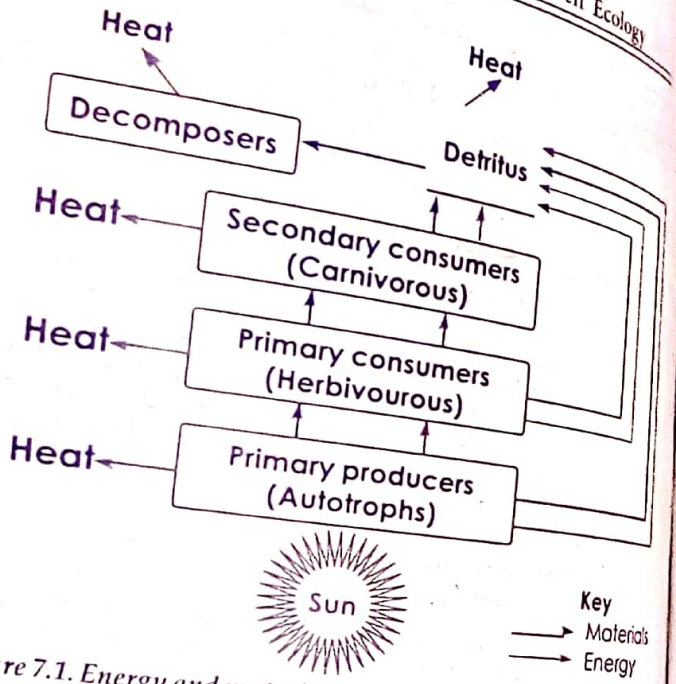

Figure 7.1. Energy and material movement in ecosystem

Decomposers

These are **macroscopic** or **microscopic** organisms that live on **dead organic matter**. They have **saprobiotic nutrition** in which they secrete **hydrolytic enzymes** on to the dead matter, and **digest** it outside their body (i.e. extracellular digestion). They, then **absorb** products of the digestion. They also produce different enzymes for various reactions. **Example:** Ammonifying enzymes.

Ammonfication: removal of amino group from amino acids (i.e. decomposition of proteins). Ammonification releases **ammonia**, and this is subsequently converted into **Nitrite** and **Nitrate**.

Amino acids $\xrightarrow{a}$ NH_3 $\xrightarrow{b}$ NO_2^- $\xrightarrow{c}$ NO_3^-

a = Enzymes in ammonifying bacteria.
b = Enzymes in *Nitrosomonas*.
c = Enzymes in *Nitrobacter*.

Nitrates (NO_3^-) are absorbed by plants, and assimilated. This is illustrated in the Nitrogen cycle presented in unit-1, but repeated here for more clarification.

The Nitrogen cycle

Nitrogen is part of **amino acid, proteins, DNA, RNA, ATP, ADP**, etc. So, it is essential for life on earth.

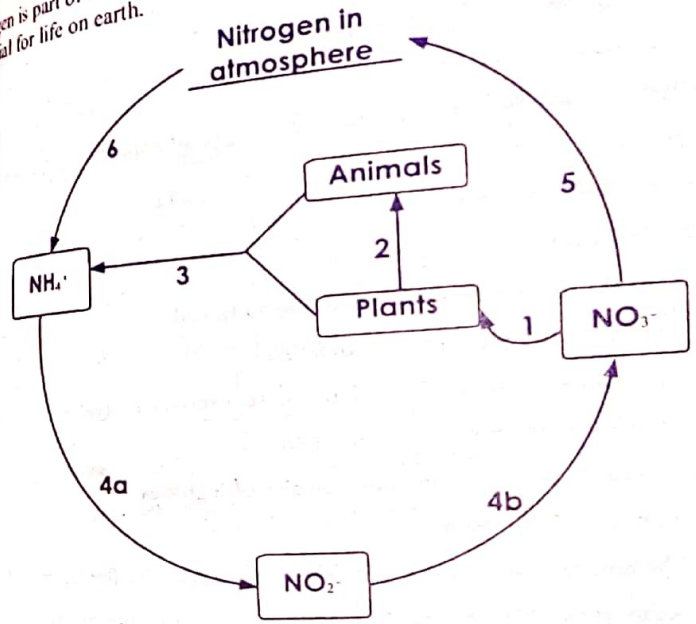

Figure 7.2. The Nitrogen cycle

Main processes

1. **Assimilation** - plants absorb **nitrate** (NO_3^-) from soil, and make **amino acids, proteins**, etc.
2. **Feeding and assimilation** - animals consume plants **directly** or **indirectly**, and make their own **proteins**, or others.
3. **Death and decomposition (ammonification)** - both plants and animals die and form **detritus** which is composed of nitrogen-containing organic compounds. **Detritus** and **excretory products** (Example: urea, ammonia) as well as **faeces** decay (decompose) into ammonium ions (NH_4^+). This is done by **aerobic and anaerobic** bacteria and **fungi.**
4. **Nitrification** - Ammonium ions (NH4)+ are oxidized into **nitrates** (NO_3^-) in two steps by nitrifying bacteria.

   a) $NH_4^+ \xrightarrow{\text{Nitrosomonas}} NO_2^-$ (Nitrites)
   b) $NO_2^- \xrightarrow{\text{Nitrobacter}} NO_3^-$ (Nitrates)

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5. **Denitrification** - reduction of nitrates in soil into elemental nitrogen by denitrifying bacteria. Example: *pseudomonas*.

$$NO_3^- \xrightarrow{\text{pseudomonas}} N_2$$

6. **Nitrogen - fixation** - conversion of free elemntal nitrogen into ammonium ions by nitrogen-fixing bacteria.

$$N_2 \longrightarrow NH_4^+$$

Generally, two types of bacteria involved.

a) **Free-living nitrogen-fixing bacteria in soil**

Example: ***Azotobacter*** and ***klebsiella***

The ammonium so formed by these bacteria is available to any plant growing in the area, or other organisms.

b) **Symbiotic bacteria in root nodules of legumes**

Example: ***Rhizobium***

The ammonium so formed, in this case, is used by the legumes (example beans, peas) themselves. It is made available to other plants when they die and decompose.

Research is being conducted to:

a) Isolate and transfer **nitrogen fixing genes** from bacteria to plants by genetic engineering, or

b) Persuade nitrogen-fixing bacteria to make symbiosis with non-legume plants.

This reduces the use of artificial fertilizers and minimizes pollution. It also solves the problem of food crisis as yield may increase.

Dangers of overuse of artificial fertilizers

Overuse → Leaching → Eutrophication

Leaching - the washing down of excess fertilizers into lakes and rivers.

Eutrophication - death of organisms in the water bodies. The process is as follows:

Algae overgrowth due to excess fertilizer --- light is blocked from reaching underneath the water --- plants die --- algae run out of nutrients and die (because of high metabolism) --- bacteria feed on the dead algae and plant and breed rapidly --- oxygen is depleted --- other organisms (example: fishes) migrate to other area or die.

The carbon cycle

The basic processes in carbon cycle are **photosynthesis** and **respiration**.

$$6CO_2 + 6H_2O \underset{\text{Respiration}}{\overset{\text{Photosynthesis}}{\rightleftharpoons}} C_6H_{12}O_6 + 6O_2$$

Organisms need **carbon-containing compounds** as **source of energy** and **raw materials** for body building.

Example: proteins, carbohydrates, fats, etc

Producers obtain these compounds from photosynthesis or by manufacturing them in different path ways using photosynthesis products and other nutrients as inputs. Proteins, for instance, are produced by combining nitrogen and carbohydrates. Consumers obtain the required nutrients directly by eating plants, or indirectly by eating other consumers or their products. Green plants, algae and some bacteria can photosynthesize. So, the forward reaction in the above equation is accomplished by them. The backward reaction is undertaken by all organisms. All producers and consumers respire. They break down carbon-containing compounds such as glucose, and release energy for their metabolic processes.

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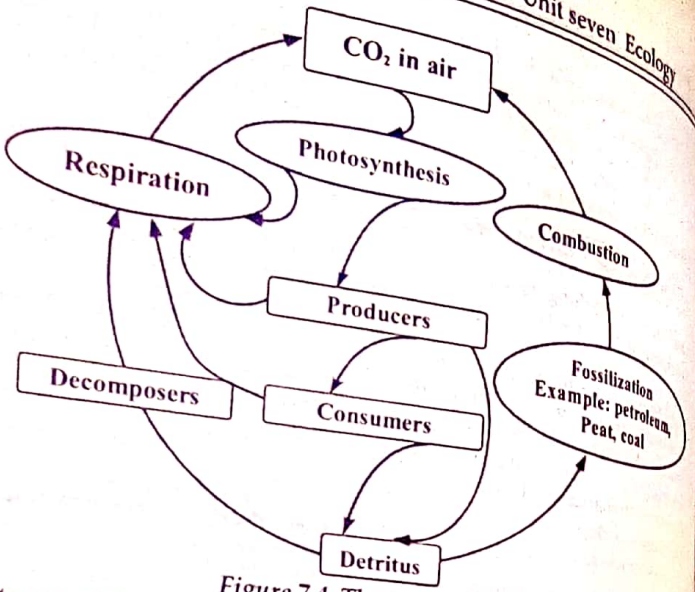

Figure 7.4. The carbon cycle

Main stages of the cycle

- **Photosynthesis** - fixes carbon from carbon dioxide (inorganic source) into organic compounds (**Example**: Glucose).
- **Feeding and assimilation** - passing carbon atoms from one trophic level to another using food chains. **Example**: From producers to consumers.
- **Respiration** - release of carbon dioxide from organic compounds.
- **Fossilization** - formation of fossils as dead organic matter fails to decompose.

  Example: Fossil fuels such as coal, petroleum, peat, etc.
- **Combustion** - burning of fossil fuels for different purposes. This releases carbon dioxide.

The earth has "carbon-based life forms" as all organic compounds found in organisms are based on carbon. So, the cycle is essential to the living world.

The phosphorus cycle

Phosphorus (P) is an important nutrient as part of **DNA, RNA, Phospholipids, ATP**, etc. It is absorbed from **soil** or **water** in the form of **phosphate** (PO_4) by plants. Then, it is taken back to soil and water as the producers and consumers die and decompose. Rock weathering and fertilizers can also be sources of phosphorus. Phosphate leaching leads to formation of phosphate-containing sedimentary rocks over millions of years.

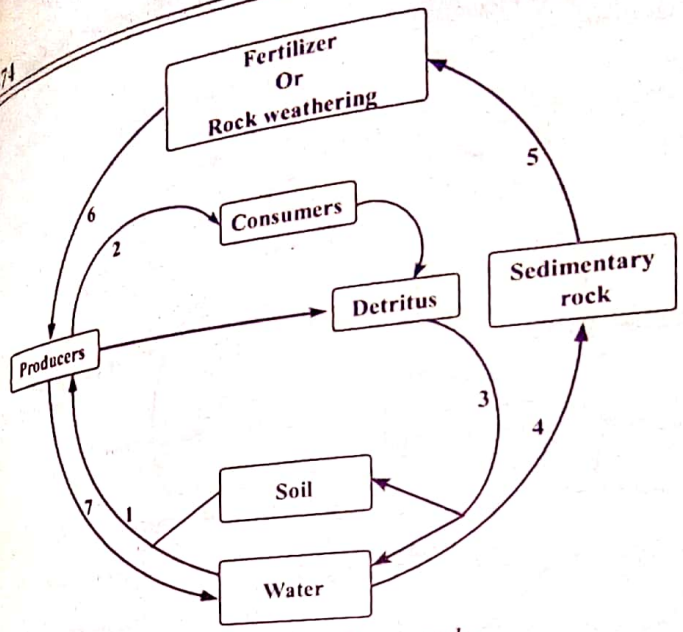

Figure 7.5. The phosphorus cycle

Key
1 = Assimilation by plants
2 = Feeding and assimilation by consumers
3 = Decomposition
4 = Sedimentation
5 = Weathering and/or fertilizer production
6 = Fertilizer application
7 = Leaching

- Refer to unit six for sulphur cycle.

The water cycle

Water is essential for all organisms in many ways. It makes up about **70%** of cells. It is **raw material** in **metabolic reactions** such as **photosynthesis**, and also acts as a **means** of **transport** and **excretion** in the body of organisms.

Additionally, water is also essential in:

- Washing clothes, utensiles, our body, etc.
- Flushing away waste.
- Making paper, steel, beer, etc.

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- Generation of electricity
- "**Hydroponics**" (plant growth in a soil-free medium).

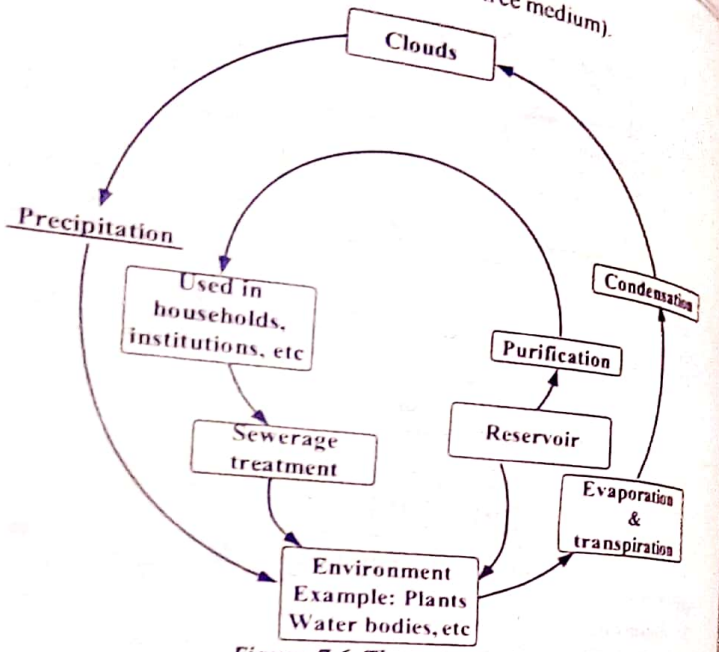

Figure 7.6. The water cycle

Explanations

- **Environment** - includes all terrestrial, water bodies, and the atmosphere.
- **Evaporation** - release of water into the atmosphere mainly from water bodies in the form of **steam**.
- **Transpiration** - release of water from plants mainly through stomata.
- **Condensation** - conversion of **gaseous water** (steam) into liquid form as it cools down.
- **Precipitation** - the coming back of water to earth in different forms. Example: rain, snow, etc.
- **Reservoir** - large dams where water is stored before treatment for purification.
- **Purification** - water treatment for the production of **pure water** used for different purposes. **Example**: drinking, washing, food preparation, etc

- **Sewage treatment** - the use of micro-organisms to breakdown organic matter found in **dirty water** (sewage). **Sewage** is then made **harmless** and released to the environment (river or field).
- The water that comes to earth in different forms may make **run-off** if vegetation cover is eradicated. But, if there is adequate such cover, the water **percolates** slowly into the ground, and forms large underground reservoirs called **Aquifers.**
- **Run off** - flooding of water over the surface.

Illustrative Examples

1. Why is it important to recycle materials in ecosystems?

Explanation:

- Nutrients should be recycled in ecosystems due to the fact that they are limited in amount. Organisms come into being, grow, reproduce, age, and finally die and decompose, there by returning the nutrients tied up in their body to the environment. Such restored nutrients can then be used up by new generations of organisms. Life is sustained in this way.

2. Which of the following life-processes are important in the recycling of material in ecosystems?

(A) Feeding and excretion
(B) Respiration and breathing
(C) Death and decomposition
(D) All are needed

Explanation:

- All the above processes are crucial in nutrient recycling between organisms and their environment. The processes are **bridges** that join different organisms together, as well as organisms and ecosystems.

Answer: D

3. Generally, our body is second - handed! Explain this statement.

Explanation:

- This is fairly true! Most of the atoms in our body, were in the bodies of other organisms before they became part our body. Nutrient-recycling made them part of our body. For example: the CO_2 breathed out by Bin Clinton some years ago might have ended up in a palm-tree. You might have consumed cooking oil produced from this plant. The carbon atoms that were once in Bin Clinton's body are now part of your body.

4. Which of the following recycles in ecosystems?

(A) Light
(C) Sound

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(D) Heat

(B) Water

Explanation:

- Light, sound and heat are different forms of energy, and energy simply flows, in ecosystems it is not recycled. As water is a material, it recycles in nature.

Answer: B

5. What is the ultimate source of energy for ecosystems?

(A) Universe (C) Producers

(B) The sun (D) Consumers

Explanation:

- The energy that flows in ecosystems originally comes from the sun. It first arrives in the form of **light** and **heat**. Light energy is converted into chemical energies by producers, and pass to others.

Answer: B

6. Which of the following is **incorrect**?

(A) Energy recycles between the sun and the earth.

(B) Nutrients recycle in ecosystems.

(C) Energy flows in ecosystems.

(D) Energy and materials move around in ecosystems.

Explanation:

Energy recycles between the sun and the earth.

- Energy simply flows from its ultimate source (the sun) through ecosystems, and finally lost to the universe.

Answer: A

7. Explain what would happen if there were no **decomposers** in ecosystems?

Explanation:

- All the nutrients in ecosystems would be **tied up** in the bodies of dead organisms if there were no organisms that decompose them. Life would end up in the course of time because of lack of nutrients.

8. Which of the following organisms directly obtain energy from the sun?

(A) Autotrophs (C) Carnivores

(B) Herbivores (D) Decomposers

Explanation:

- Autotrophs (producers) such as green-plants directly obtain solar energy from the sun, and convert it into **chemical energy** in **bonds** between atoms in molecules of **organic compounds** they form.

Answer: A

Example: Glucose.

9. Which one of the following has a unidirectional flow in an ecosystem? **(UEE, 2003 E.C)**

(A) Nitrogen (C) Phosphorus

(B) Carbon dioxide (D) Energy

Explanation:

- Energy flows in ecosystems; others are nutrients, they recycle in ecosystem.

Answer: D

10. In a given ecosystem, members of which one of the following feeding groups release nutrients that are locked up in organic molecules? **(UEE, 2003 E.C)**

(A) Herbivores (C) Carnivores

(B) Producers (D) Decomposers

Explanation:

- Decomposers are responsible for releasing nutrients tied up in dead organic matter and organic wastes.

Answer: D

11. How do animals obtain nitrogen? **(UEE, 2003 E.C)**

(A) By absorbing nitrates and ammonia from soil.

(B) From protein in the organisms they consume.

(C) By absorbing nitrogen gas from the atmosphere.

(D) Through a mutualistic relationship with nitrogen fixing bacteria.

Explanation:

- Animals have no capacity to absorb nitrogen from soil or atmosphere, they have no symbiotic bacteria that provide them with nitrogen. The only option they have is to obtain nitrogen from the organic compounds they consume. Example: Proteins, nucleic acids, etc.

Answer: B

12. Which of the following **is not** recycled between organisms and the environment in an ecosystem? **(UEE, 2004 E.C.)**

(A) Energy (C) Nitrogen

(B) Carbon (D) Phosphorous

Explanation:

- Energy flows in ecosystems, it is not recycled. Carbon, nitrogen and phosphorus are nutrients that are recycled in ecosystems both between organism, and between environment and organisms.

Answer: A

13. Which one of the following **is not** one of the roles that micro-organisms play in ecosystem? **(UEE, 2004 E.C.)**

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(A) Nutrient recycling
(B) Carbon fixation
(C) Nitrogen fixation
(D) Energy recycling

Explanation:

- Micro-organisms play important role in all the activities listed except energy recycling. Energy is not recycled, it flows in ecosystems.

Answer: D

14. Which of the following organisms have **saprobiotic** nutrition?
(A) Producers
(B) Decomposers
(C) Consumers
(D) All

Explanation:

- Decomposers are saprobionts because they live on dead organic matter.

Answer: B

15. Which of the following has external digestion?
(A) Amoeba
(B) Paramecium
(C) Fungi
(D) Giardia

Explanation:

- Amoeba, Paramecium and Giardia are protozoans that digest their foods intracellularly. Fungi secrete enzymes and digest their foods outside their body. They are decomposers. They absorb the digested food.

Answer: C

16. What are the **enzymes** produced by **decomposers** in addition to their digestive enzymes?
(A) Ammonifying enzymes
(B) Nitrifying enzymes
(C) Nitrogen fixing enzymes
(D) All

Explanation:

- In addition to digestive enzymes, different enzymes such as the above are produced by decomposers.

Answer: D

17. Which of the following is a **nitrogen fixing bacteria?**
(A) *Pseudomonas*
(B) *Rhizobium*
(C) *Nitrosomonas*
(D) *Nitrobacter*

Explanation:

- *Rhizobium* is a nitrogen fixing **symbiotic** bacteria, *Pseudomonas* are **denitrifying bacteria**, while ***Nitrosomanas*** and *Nitrobacter* are nitrifying bacteria.

Answer: B

18. In the nitrogen cycle, nitrogen gas is returned to atmosphere by
(A) Nitrogen-fixing bacteria
(B) Decomposers
(C) Nitrifying bacteria
(D) Denitrifying bacteria

Explanation:

- Denitrifying bacteria convert fixed nitrogen into free elemetnal nitrogen which is released to the atmosphere.

Answer: D

19. Decomposers feed by
(A) Parasitic nutrition.
(B) Intracellular digestion.
(C) Autotrophic nutrition.
(D) Saprobiotic nutrition.

Explanation:

- Decomposers are saprobionts

Answer: D

20. _____ is the movement of water along the surface of the earth.

Explanation:

- **Run-off** is the movement of water along the earth's surface. This is exaggerated when there is no vegetation cover. It washes away top soil which is relatively fertile.

21. _____ is the process of growing plants in a soil free medium using water.

Explanation:

- **Hydroponics:** the nutrients required are added, and the plants are supported by using different techniques instead of using soil as an anchoring material.

22. The following equation is part of the nitrogen cycle. Give the micro-organisms denoted by numbers in the equation.

$$\text{Amino acid} \xrightarrow{1} NH_4^+ \xrightarrow{2} NO_2^- \xrightarrow{3} NO_3^-$$

Explanation:

1. Ammonifying bacteria (decomposers)
2. *Nitrosomonas* Nitrifying bacteria
3. *Nitrobacter*

23. What is the form of nitrogen absorbed from the soil by plants?
(A) Nitrite.
(B) Nitrate.
(C) Ammonia.
(D) Protein.

Explanation:

- The usable form of nitrogen by plants is **Nitrate**. So, for plants to absorb and assimilate nitrogen, nitrites, ammonia, protein and other nitrogen containing compounds should be converted into **Nitrate**. This is accomplished by micro-organisms freely living in the soil, and symbiotic bacteria living in root nodules.

Answer: B

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24. What are the two genera of bacteria (free-living) involved in nitrogen fixation?

Explanation:

- ***Azotobacter*** and ***Klebsiella*** are free-living in soil, and they convert free elemental nitrogen into **ammonium** which is finally converted into **Nitrate** and absorbed by plants.

25. The death of organisms in water bodies due to the overuse of fertilizers is known as_____.

Explanation:

- **Eutrophication**, which has a chain of processes

26. What are the two areas of research in **genetic engineering** to reduce the use of **artificial fertilizers** and minimize **pollution**?

Explanation:

(a) **Isolation** and **transfer** of **nitrogen fixing genes** from bacteria to plants by genetic engineering.

(b) **Persuasion** of nitrogen-fixing bacteria to make **symbiosis** with non-legume plants.

27. Which of the following processes ties carbon in the bodies of living organisms.

(A) Fossilization
(B) Combustion
(C) Respiration
(D) Photosynthesis

Explanation:

- **Photosynthesis** fixes carbon into organic compounds such as carbohydrates which are part of living organisms. **Respiration** and **combustion** release carbon to atmosphere. **Fossilization** ties carbon in dead organic matter for some time, which is then converted into **fossil fuels** such as **petroleum**, **peal** and **coal**.

Answer: D

28. Phosphorus is absorbed from soil by plants in the form of_

(A) Phosphoric acid
(B) Phosphate
(C) Phosphite
(D) Phosphorus acid

Explanation:

- The usable form of phosphorous by plants is phosphate.

Answer: B

29. Which of the following is the use of water?

(A) Acting as raw material in metabolic reactions.
(B) Acting as a medium in hydroponics.

(C) Flushing away waste and electricity generation.
(D) All are uses of water.

Explanation:

- Water is a multi-purpose material. Sometimes it is said that "water is life", and that is truth as there is almost no activity that excludes water.

Answer: D

29. Which of the following processes involved in water cycle is the conversion of **gaseous water** into **liquid water**?

(A) Evaporation.
(B) Transpiration.
(C) Condensation.
(D) Precipitation.

Explanation:

- **Condensation** converts **gaseous water** into **liquid water**, and facilitates the returning of water to the earth's surface.

Answer: C

7.2 Ecological succession

Ecological succession is the process of sequential Community development over time, in which one species is replaced by another. Mature community develops gradually overtime. It does not happen instantly.

Community: - all the populations living in a given area.

Example: All the insects in a pond, algae population, higher plant population in the pond, the bacteria and nematod populations in the pond, etc form the **pond community**.

Population: group of organisms of the **same species** occupying the **same area** at the **same time**.

Example: A group of zebra living together, a group of insects under decaying log, individuals of maize plants growing on a plot of land, group of students in a class, etc.

Ecosystem: Communities of organisms in an area along with their habitat. In other words, an ecosystem consists of the **biotic communities** and the **abiotic components** in an area.

Biosphere: is part of the earth where life exists. It is made up of all the ecosystems on earth.

Different communities are established in biosphere by the process of community (ecological) succession which is the gradual change with the replacement of one biotic community by another. The stages in succession are called **Seres**. The process starts on bare land or in a newly formed ponds. The first community that colonizes the area is known as **pioneer community**. Example: lichens,

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algae, mosses, small dicots, etc. In **succession**, **ecosystems** change from **simple** to **complex**. Plants with symbiotic associations play important role in early succession.

Example:
- Lichens (algae + fungi)
- Legumes (with symbiotic bacteria in their root nodules)
- Mycorrhizal plants (with symbiotic fungi associated to their roots).

Such plants are **resistant** (i.e. **tolerant**) to harsh conditions such as **dessication** and other **extremes**.

There are two types of succession: **Primary** and **secondary**.

Primary succession - occurs on bare lifeless substrate such as rocks or open water of newly formed ponds, as well as new land (mountain) formed by volcanic eruption. The following organisms may occupy a bare rock during the early stages of succession.

Cyanobacteria
+
Algae
+
Lichen
} ---Mosses---Ferns---Small dicots---Mycorrhizal plants (trees)

Secondary succession - starts on areas that have been disturbed after having been occupied by living things earlier. Example: Deforested area, an area cleared by fire, etc.

Climax community - is the final most complex community at the end of succession. Example: Forest, grassland, etc.

Generally, succession follows the following trend.

Pioneers (primary colonizers) eg. Mosses → Secondary colonizers → Annual herbs eg. Weeds → Perennial herbs → Scrubs (shrubs) → Climax community eg. Forest

As **succession proceeds**, the **amount** and **depth** of **soil increases**, **ecological niches** and **species diversity** of the area also **increase**. Conditions also improve.

Niche is the role of a particular species in an area (habitat). The climax **community** of an area is controlled by **climate**. As climate changes gradually, **stagnation** can not be expected from **climax communities**. They may show changes. Successions that may start on bare rock are called **Xeroseres**, and those initiated on water bodies, **hydroseres**. In addition to the major factor affecting climax community (i.e. climate), **grazing** and **soil type** may also contribute their share. **Primary succession** may take **thousands** of **years**, or **longer**. **Secondary succession** is **quicker** as it occurs on an already **established environment** with **soil** and **seed bank**.

Illustrative Examples

31. What does an ecosystem possess that a community does not?

Explanation:
- An ecosystem encompasses the **abiotic component** in addition to the **biotic component** of the environment a community is comprised of.

32. Communities at early successional stages have _____ relative to the more advanced communities of later stages.
(A) Higher biomass
(B) Lower productivity
(C) More species
(D) More stability

Explanation:
- Communities at early stages of succession have **lower productivity**, **lower biomass**, **less species diversity** and **less stability**.

Answer: B

33. Which one of the following terms is most appropriate to collectively refer to all the plants, animals and micro-organisms found in YAYO forest in Ethiopia? **(UEE, 2003 E.C).**
(A) Ecology
(B) Ecosystem
(C) Population
(D) Community

Explanation:
- All the populations of organisms in an ecosystem are referred to as community.

Answer: D

34. What is the term that refers to all parts of the earth where living things are found? **(UEE, 2004 E.C).**
(A) Population
(B) Ecosystem
(C) Biosphere
(D) Environment

Explanation:
- Part of the earth where life exists is **biosphere**.

Answer: C

35. Which of the following **is not** correct?
(A) Succession is an **event**.
(B) Mature communities develop gradually overtime.
(C) Succession is a **process**.
(D) Ecosystem includes both **biotic** and **abiotic** components of the environment.

Explanation:
- Succession is not an event; it is a **process** that takes thousands of years.

Answer: A

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36. Which of the following refers to a group of organisms of the species at a time in a given area?
(A) Community
(B) Ecosystem
(C) Population
(D) Biosphere

Explanation:
- **Population** refers to a groups of organisms that can **interbreed**, and live together in a given area.

Answer: C

37. Which of the following groups of organisms **do not** fulfil the criteria for population?
(A) Bacterial colony growing on agar plate.
(B) A field of eucalyptus tree plantation.
(C) A group of human families living in a town.
(D) None of the above.

Explanation:
- None of the list can be an answer because all fulfil the definition of population.

Answer: D

38. The type of **succession** that starts on **bare land** or in a **newly formed pond** where life-had never existed is known as:
(A) Early succession
(B) Primary succession
(C) Late succession
(D Secondary succession

Explanation:
- **Primary succession** is the **colonization** by organisms of newly emerging **land forms** and **water bodies.**

Answer: B

39. During which phase of succession is the **total biomass** of an area is **greater**?
(A) Pioneer community
(B) Early succession
(C) Tolerant communities
(D Later succession

Explanation:
- During later stages of succession, species diversity, community stability and biomass increase.

Answer: D

40. What factors lead to the initiation of secondary succession in a given area?

Explanation:
- Community devastation by **drought, forest clearing, fire accident** etc. that eradicate vegetation of an area, agriculture, disturbance of an area by construction activities and the like.

41. Which of the following plants would likely colonize bare land first?
(A) Annual herbs
(B) Mosses
(C) Lichens
(D) Perennial herbs

Explanation:
- **Lichens** are **symbiotic** combinations between **algae and fungi**, and they are the most tolerant organisms that withstand harsh conditions in new environments.

Answer: C

42. Which of the following plant types would most probably occupy a newly emerging land last? The land may be a mountain that has been created by volcanic eruption
(A) Forest
(B) Shrubs
(C) Perennial herbs
(D) Annual herbs

Explanation:
- In most cases, depending on climate of the area, **forests** are the last comers to colonize a bare land.

Answer: A

43. Which of the following factors affects the type of climax community of an area?
(A) Grazing
(B) Soil type
(C) Climate
(D) All of the above

Explanation:
- Climate is the major factor affecting the **climax community** of an area. Factors such **grazing**, **soil type** and **soil depth** have their own share.

Answer: D

44. Which of the following statements is **incorrect**?
(A) Once succession reaches climax, communities will not change any more.
(B) The different stages of succession are called **seres.**
(C) Succession that starts on a **water body** is called **hydrosere.**
(D) Ecosystems are not **stagnant**, they show different degrees of change.

Explanation:
- **Ecosystems** and their **communities** are **always** in **change**. Even **climax** communities slightly change because there is **climate change**, at least. Human interference is another disturbance.

45. Succession that starts on **bare land** is called _____.

Answer: A

- Xerosere refers to succession that starts on bare land, while one that starts in water is **hydrosere.**

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7.3 Biomes

Biomes are large **geographical regions** with their specific **climate, soil type** and **organisms** adapted in similar ways to the area. They are large **global ecological units** distinguished by their characteristic **flora** and **fauna**, typical **plants** and **animal** species, respectively. **Similar biomes** are created in different parts of the world by **similar environmental conditions**. Though there is no definite number of biomes, they are **categorized** for the sake of describing the world's **fauna** and **flora**. They are generally characterized by their **vegetation**, but **animals** and **micro-organisms** are also included. **Biomes** are mainly defined using **climatic zones**, but they can also be defined by **rainfall patterns, soil types**, and **hydrological conditions**. **Temperature** and **precipitation** depend on **geographic location**, and are the main **determinant factors** of a **biome**.

There are two types of biomes: **Terrestrial** and **aquatic**.

Terrestrial biomes

The **land biomes** are defined by **temperature, rainfall, soil type, fauna** and **flora** of the area.

Example: desserts, tropical rain forest, tundra, etc.

1. **Deserts** - are dry areas in **temperate, sub-tropical** or **tropical** regions.
   Cold deserts - are deserts in **temperate** regions. Temperature is cold, almost no rain, and **soil** is **poor**. Inhabited by some **micro-organisms, lichens, polar bears** and **seals**.
   Warm/hot deserts - are found in **tropics** and **sub-tropics**. It has almost **no rain, temperature** is **hot** and the **soil** is **poor**. **Fauna** and **flora** include **insects, arachnids, reptiles, birds, succulents** (Example: cactus), etc.
   Dessert atmosphere has very low water vapour, leading to daily temperature changes from **very hot** to **very cold**. Rainfall is usually less than 250 mm per year.
   Example: - **Namib** and **Kalahari** deserts in **Southern Africa**.
   - The **Sahara** in **Northern Africa**.
   - The **Arabian desert**.
   - The **Gobi desert** to the north of the **Himalayas**.
   - The **Sonora** and **Mojave deserts** of the **USA**.
   - The **Coastal Atacama desert** of **Peru** and **Chile**, etc.

   Soil mineral content is **high**, and sometimes **poisonous** to the sparse **plants**. Organism of dessert biomes have **structural, behavioural** and **physiological** adaptations to **conserve water**.

- **Ephemerals** complete their whole life cycle of **germination, growth, flowing** and **seeding** with in very short period of time (i.e. a few weeks).
- **Perennials** pass the long **dry periods** as **adults** rather than **seeds**.
- **Geophytes** ("ground plants") have underground **bulbs** or **corms**. These live through long periods of **drought** and produce their above ground parts as soon as it rains.
- **Succulents** such as **cacti** and *Euphorbia* have no or **reduced leaves**. In some, leaves are reduced to **spines** and photosynthesis takes place in the **stem**. The sharp spines also protect them from herbivores. **Example**: **Cacti**. In others, **stomata** open only **at night**.
- **Desert vertebrates** such as **rats** live in **deep, cool** and sometimes **moist burrows** and come out **at night**.
- **Camels** can lose up to **30%** of their ideal water content. When it runs short of water, its body temperature can decrease by upto **10° C**. This **conserves** water that would have been lost by **evaporation** to keep it cool. Camel's **hamp** stores **fat** that releases water during drought periods through chemical break down. Camels also drink much water when they get it.

2. **Thorn forest (scrub)** - has **dry hot summer** and **rainy cool winter**. They occur around the **Mediterranean**, in **California, western Australia**, parts of **Chile** and parts of **South Africa**. It is called the '**Chaparral**' in southern California. The soil is very thin and poor in nutrients. Drought and fire resistant plants and animals inhabit it. Example: Oak, shrubs, pine, lizards, wood rats, different types of birds. During **summer**, the plants are **dormant**, and **green** during **winter**.
3. **Tundra** - is an open, often **boggy** grassland with **long harsh winters** and extremely **short summers**. It is located farthest **north** in **Eurasia** , **North America** and their islands, between permanent ice and taiga. Rainfall is low (<250 mm), trees are small and mostly confined to lake and stream margins. Rain water is **unavailable** because it is **frozen**. The **soil** is permanently frozen within a **meter** from the surface (i.e. **permafrost**). Tundra covers about **one-fifth** of the terrestrial environment. This limits growth of roots of woody trees. The **species diversity** and **primary productivity** are **low**. Dominant plants are **mosses, lichens** and some **grasses**. Animals such as lemmings, voles, weasels, Arctic foxes, grey wolves, snowshoe hares, snowy owls and musk-oxen are dominant. Hiking, oil explorations and military activities have damaged tundra, and it is a slowly regenerating

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biome. Growing season is very short (about 50 days) with long days. Soils are poor in humus.

4. **Boreal forest (Taiga)** - is the **evergreen forest** of the **northern Eurasia** and North America dominated by coniferous trees such as spruce, hemlock, pine and fir. Winters are long and cold. Most of the rains are in summer. Days are **longer in summer** and **shorter in winter**. Marshes, lakes and pond are common. **Animals** are adapted to the harsh condition in this biome. They **migrate** to warmer regions, hibernate, or pass the conditions as resting stages or eggs or pupae. The animals include elk, moose, deer, wolves, bear, lynx, birds, and wolverines. The biome is important for fur trapping and lumber production. **Siberia** is in this biome. It is the coldest in the northern hemisphere with temperatures ranging from -50°C to 40°C. The deeper layer of taiga's soils is frozen, snow remains on the surface most of the year. Crops and other plants grow rapidly during the short growing season.

5. **Temperate deciduous forests** - cover large areas in **eastern USA, Canada** and extensive region in **Eurasia**. There are **hot summers** and cold **winters** (i.e. continental type of climate). Rainfall ranges between 750 and 2500mm, and well distributed through out the year, but **unavailable** to organisms as it is **frozen** during **winter**. Top **soil** is **rich** in **organic matter** and dipper **clay layer** is rich in **minerals**. The **vegetation** is **broad-leaved** hard wood trees such as oak, hickory, maple and beech. The colour of leaves of these trees change before they fall (i.e. **autumn forests**), and look beautiful. They again form their leaves in spring. The animals of this region are deer, bears, beavers, birds, reptiles, amphibians, insects, and mammals and racoons. The area is affected by human interference as many animals and plants have been extincted. Example: puma, wolf, bison. The forests in North America were cleared for **timber production** and **agriculture** as well as **establishment** of **cities**. In some places, they are being **regenerated** to create a **semi-natural** state.

6. **Tropical montane forests (also known as tropical montane cloud forests)** - are ecosystems in the mountainous areas of the tropics. They show high altitudinal variation ranging from 500m to 3500m above sea level, but majorly occurring from 1200m to 2500m. Rainfall is between 500 and 10,000mm/year. Frequently, covered in **cloud** or **mist** at vegetation zone leading to **reduced evapo-transpiration** and **enhanced net precipitation**. There is also **horizontal precipitation** of **cloud water**. The soils are water logged, and rich in organic content. The trees are of reduced height and compact crowns with small thick and hard leaves. It has fewer plant

species and more open under story than tropical rain forest which is found at a lower elevations below the mountain. The biome shows relative diversity in **form, composition** and **adaptation**. The **flora** show some endemism. There is **high biomass** of **epiphytes** (bryophytes, lichens and filmy ferns), but lower proportion of **woody climbers**.

Tropical montane forests cover **11%** of all tropical forests in **tropical Africa, central and Latin America** and **South eastern Asia**. They regulate water supply of lowlanders and increase **ground water** level and stream flow. They also **prevent erosion**: They are threatened by human encroachment for **agriculture**.

7. **Tropical rain forests** - are the **richest** biomes in **species diversity** (i.e. more than **two million** species, probably half of the species of terrestrial organisms). They are found in **South** and **Central America** (around Amazon basin), **west** and **equatorial Africa, Southeast Asia, Indonesia** and **North-east Australia**. The biome is located **near** the **equator**, and its climate is **hot** and **wet**. Rainfall is on average from 2000–4500mm/year. It rains almost everyday and ensures continuous growth of plants. The soils are poor in organic matter, but moist. Many species of organisms are found in a unit area, but each species is represented by very few individuals. The ways of tropical organisms are often **specialized** and highly **unusual**. Example: a very **sluggish** animal called **sloth** (3-toed) moves in trees by hanging under a branch, a leaf-like insect (katydid), very huge trees, etc. Tropical rain forests have **high net productivity**, with nutrients being held with in the plants themselves. The nutrients are recycled when the plants die or when parts such as leaves fall and decompose. The soils get **much mineral** when the forests are **cleared** and **burnt**. They support **crops** for **few years**. But, as the minerals are depleted, farmers should move on clearing more forest to get agricultural land. Human **population burst** and the increase in transnational **loggers** are additional threat to this biome, unless appropriate conservation measures are taken, the biome with its rich biodiversity will soon be lost.

Fully developed rain forest has **three** or **four levels** of vegetation (canopy). The topmost is made from crowns of very **huge trees**, and exposed to direct sun light. The **middle layer** gets **less light**. The **lower layer** is consisted of plants adapted to **shade**. Many of the trees have **epiphytes** and **lianas**. Ninety percent of the animals of this biome live in the upper canopy. These include various insects, reptiles, amphibians, birds, mammals (Example: Sloths, monkeys), etc There are also large land-dwelling animals such as elephants.

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8. **The grasslands (temperate grasslands/prairies)** - were widespread in the interior of **North America, Eurasia** and **South America**, and are highly **productive** when converted into **agriculture**. Much of the rich agricultural lands in USA and Canada were originally **prairies**. The biome is referred to as **Prairie** in central North America, **steppe** in eastern Europe and Asia, and **pampas** in South America (Argentina). Rainfall is moderate (20-700mm), summers are hot, and winters cold. **Flora** of the biome is largely **perennial grasses** with **deep roots** and **rhizomes**. These die each year and make the soil fertile. There are two types of grasslands: **tall grass prairies** and **short grass prairies**, the first receiving more rain fall. s grow in tall grass prairies, especially, **near rivers** and **streams**.
The prairies in USA were once inhabited by **American Indians**, huge **herds** of **buffalo** and **pronghorns**, and predators such as wolves, bears, etc. These are now a history because the area is converted into the world's richest agricultural land.

9. **Savannas** - are **open grasslands** in the **tropics** where the precipitation is **reduced** with **prolonged** annual **dry seasons**. The **trees** and shrubs their leaves during the dry seasons and form **tropical** and sub-tropical **deciduous forests**. African huge herds of **grazing mammals** and their **predators** are **good examples of its fauna**. Savannas are transitions between **tropical rain forests** and **deserts**. Rain fall is between 500 and 1500mm, wider **temperature fluctuation** and seasonal droughts are common. The biome is characterized by scattered clumps of trees found mostly in **Africa**, but also found in **Australia, South America** and South **Asia**. Unlike temperate grasslands, seasons are regulated by rain fall than **temperature** which changes very little. But, there is greater temperature **fluctuation** than in tropical rain forests. The **trees** and grasses here are fire-**adapted**. They have **extensive underground root systems** that survive the fires from lightening or human beings. Some of the wild animals in savanna include **herbivores** such as Wilde beast, antelopes, giraffes, zebras and elephants, and **carnivores** like lions, leopards and hyenas. The biome is being converted into **agricultural** area, and the **wild animals** are being replaced by **domestic animals**. Crops are also taking the place of grasses and the sparce trees.

10. **Temperate rain forests** - are found on north west coast of North America, in South-eastern Australia and in southern South America. Rainfall is high and weather is cool due to dense fog. Precipitation ranges between 2000 and 3800mm, on average. They are situated near coasts, temperatures fluctuate slightly. Activity of decomposers is reduced by the cool

temperature and decay of fallen leaves and parts of coniferous trees is delayed. Consequently, soils in this biome are very poor. Its **flora** include large evergreen trees like western hemlock, douglas fir, western red cedar, and epiphytes such as mosses, lichens and ferns. The epiphytes grow and form a **carpet-like** mass on the floor. The **fauna** include wood-rats, squirrels, mule deer, elk, birds, reptiles and amphibians.
Temperate rain forests are one of the richest wood producers in the form of **lumber** ("tawula") and **pulp**. They have also high species diversity, relatively. Therefore, over harvesting should be avoided; they should be exploited in a sustainable way.

Aquatic biomes

The aquatic biomes are divided into **two** major types:-

a) Marine biomes, and
b) Fresh water biomes

Each of the two types is again divided into different biomes. Table 7.1 gives the main features of aquatic biomes.

Table 7.1. The main features aquatic biomes.

Biome	Salt content	Movement	Other features	Organisms
Marine 1. Oceanic, pelagic	High	Moving	Region of ocean where light penetrates	Many fishes, mammals, and plankton
2. Oceanic, abyssal	High	Less movement	Region where light does not reach	Angler fish, sulphur bacterial & other decomposers
3. Coral reef	High	Moving	Most diverse of all marine habitats, has many strata like rain forest	Corals, many fish, many sea weeds, etc.
4. Estuarine	Inter-mediate	Extreme movement	Unique habitat due to mixing of salty water and fresh water	Shore birds, fish, crabs, mangrooves, kelps, sea grass

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Fresh water 1. Pond and lakes	Fresh water	Standing	Stratified as top layer absorbs more heat and light	Large number of plants and animals in top layer
2. Streams and rivers	Fresh water	Moving	Water is highly Oxygenated.	Algae, plankton, plants and fish
3. Wetlands	Fresh water	Standing	Water is very nutrient rich	Many plants & animals (highest of all aquatic biomes)

Table 7.2. Net primary productivity of some biomes (dry organic matter/ m^2/ year in grammes)

Biome	Productivity	Biome	Productivity
Tropical rain forest	> 2000	Temperate grassland	600
Coral reef	1800	Lake and stream	250
Temperate forest	1250	Tundra	140
Tropical grassland	900	Open ocean	125
Cultivated land	650	Desert & semi dessert	40

Ecosystems of Ethiopia

Ethiopia has diverse **topography** and **land forms** ranging from the **Afar depression** which is **116m** below sea level to the mountain peaks such as **Ras Dejen** which rises to **4620m** above sea level. There are **plateaus, highlands, lowlands, valleys, gorges, steep slopes** and **flat lands**. These produced a wide range of different ecological habitats resulting in different ecosystems. There are **nine** known ecosystems in Ethiopia.

1. **Dessert & semi-dessert** - found in the **Afar** and **Somali** regions, Southern **Borena** of the **Oromia region**, and **South Omo** of the **Southern region**. It ranges up to 800m.a.s.l., and has **drought tolerant plants and animals**. Soils are vulnerable to wind erosion because there is little grass cover. The plants are shrubs, **Euphorbia** ('Qulqual'), **Aloe** ('Iret') and grass while the few animals include gazelles,Oryx, wild ass and zebra. Daily temperatures can be very hot, but at night, it may drop rapidly. The only true desert in Ethiopia is the **Danakil depression** in the Afar region.
2. **Acacia-commiphora (small-leaved) deciduous woodland ecosystem** - is distributed in the **eastern** and **southern lowlands**, in the rift valley and the

eastern slopes of the northern plateau between 900 to 1900m. It is very dry as rainfall may delay up to 10 months. **Deforestation** for agriculture and **grazing** is a threat, soils are fine textured and vulnerable to wind and water erosion. The vegetation of these areas is **drought tolerant** trees and shrubs with deciduous leaves, or leathery persistent leaves. Different species of **Acacia** ('yegrar zaf') and **Commiphora** ('yekerbe zaf') are common. Major mammals are oryx, kudu, zebra, gazelle and harte-beest.

3. **The lowland forest ecosystem** - a dry forest in the **Gambella** region between 450 and 600m.a.s.l. on well drained sandy soils. There is high temperature, but a good annual rainfall ranging between **1300** and **1800mm**. The vegetation consists of semi-deciduous trees, shrubs and grasses. Typical trees include *Baphia abyssinica* ('adedeguay'/Annuak), ***Alstonia boonei*** ('Jorga'/Annuak) and *Mimulopsis solmii* (Heraya'/ Oromo). Dominant mammals include elephant, giraffe, rhino, lion, white eared kob, and cheetah. The ecosystem is being threatened by Gambella town expansion and other settlements. The Abobo dam has also flooded a substantial portion of it.
4. **The Combretum-Terminalia (broad-leaved) Deciduous Woodland and Savanna ecosystem** - is distributed in south and south western Ethiopia as well as in the western parts. Example: in Gamo Gofa, Gambella, Gondar, Tigray, Wallaga and Kafa areas. It ranges between 500 and 1900m. It is characterized by small deciduous trees, shrubs, herbs and grasses. It is now diminishing due to agricultural expansion, and charcoal and fire wood supply for major towns. **Malaria** and **trypanosomiasis** limited its exploitation in western Ethiopia.
The vegetation in this area is dominated by different species of Combretum ('Zegdom'), Terminalia ('weiba') and Grewia ('lenquata/betremuse'). The grasses are dominated by Cymbopogon, hyparrhenia, sorgum and pennisetum. Dominant mammals are buffalo, zebra, water buck, gazelles, Nile lechwe, elephant, de Brazza's monkey, oryx and giraffe. Mago National park occurs in this ecosystem. The plants and animals of this area are adapted to **recurrent fires**.
5. **The Moist Evergreen Montane Forest ecosystem** - are distributed in Wallaga, Ilu-Aba-Bora, Bale and Southern kefa between 800 and 2500m along mountain ranges.
The average temprature is from 18-20°C, and the rain fall is between 1500 and 2400mm. Example: The Harena forest on the southern slopes of the Bale massif. The biome is in the zone where coffee is grown. The ecosystem is severely threatened for this, and timber production and

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supply of fire wood. Dominant trees are *Podocarpus gracilior* ('zigba'), *Albizia gummifera* ('sessa'), *Milletia feruginea* ('birbira') and *Prunus africana* ('tikur inchet'). Many endemic herbs are also found here. Blue monkey, Bush buck, Mountain Nyala, Great forest hog and Wild pig are the dominant mammals in the area.

6. **The Evergreen Scrub ecosystem** - found on the slopes of the northern plateau at various altitudes. It has been replacing montane forests as they are cleared for different purposes. Scrub ecosystem is not economically useful for grazing, agriculture or even firewood. No large mammals are found in it. Few small mammals associated with it are olive baboon, duiker and klipspringer. Typical plants in the area are *Euclea schimperi* ('dedeho'), *Dadonaea angustifolia* ('kitkita'), *Carissa edulis* ('agam'), *Scolopia theifolia* ('kokofla'), etc.

7. **The Dry Evergreen Montane ecosystem** - is a complex region combining **forest** and **grassland** consisting of the southern and northern plateaus at altitudes ranging from 1500 to 3200m. It is associated with small to large trees and extensive grass lands rich in legumes. Typical trees are *Juniperus procera* ('Tid') *Olea europea* ('Weira'), *Acacia abyssinica* ('Grar') and others. Large mammals such as elephant, buffalo and lion were common in the past, but now, species that can tolerate close association with man are persistent. Example: Leopard, Gelada Baboon (endemic), Monkeys, etc. Overgrazing, population pressure and agricultural encroachment are the main threats to this area. It is an area where sedentary cereal-based agriculture has been practiced for thousands of years. Most of Ethiopia's population lives in this area, therefore, its **biodiversity** is badly affected. The economy of the southern and eastern parts of this area is based on mixed cereal agriculture mainly growing 'teff', finger millet, sorghum, maize, barley, wheat, pulses and oil crops. The northern area, especially Wallo, is highly populated. The economy of the western parts of the south and central Ethiopia is based on mixed root crop agriculture growing enset, yam, taro, Oromo dinch, sweet potato and cassava. The enset producing areas of this ecosystem are those with the highest population densities in the country. Land degradation has been very severe in north eastern Ethiopia lying in this ecosystem. Forests have disappeared; streams and springs stay dry during the dry seasons. So, the ecosystem requires urgent conservation measures.

8. **The Afro-alpine and Subafro-alpine ecosystem** - is the vegetation native to the highest African mountains. It occurs only on the slopes and tops of the highest mountains, Mount Abune Yosef, Guna massif, Mount Guge and

Menz at altitudes greater than 3200m. Their vegetation is dominated by *Kniphofia* ('shemetmet'), *Helichrysum* ('tuka'), *Alchemilla* ('deshuala')/ keficho), *Lobelia rhynchopetallum* ('jibira') and *Erica arborea* ('asta'). Endemic mammals such as **Walia Ibex**, the **Ethiopian Wolf, Mountain Nyala** and the **Gelada Baboon** are the main **fauna** of the ecosystem. Human population density is low although there is still encroachment for barley cultivation. There is slow growth rate of vegetation.

9. **The Swamp, Lake and Riparian ecosystems** - are found along banks of rivers and shore lines of inland lakes. They are **wetland** areas. Example: Abay (Nile) river system, Awash valley, Wabishebelle basin, Lower Omo valley, Baro river basin, Lake Tana and Rift valley lakes. As there are many rivers and lakes in Ethiopia, riparian vegetation and animal life is very significant. This ecosystem is a conservation area for wetland plants and animals such as hippopotamus, crocodiles, fishes and many bird species. Its degradation has been the cause for environmental deterioration and severity of floods in the lowlands.

Recommended conservation measures

The above ecosystems are threatened by erosion, overgrazing, drought, fire, deforestation, agricultural expansion and/or population growth. Unless immediate serious actions are taken, much of our ecosystems will be destroyed with catastrophic outcomes. Therefore, the following measures should be taken.

Reforestation - deforested area should be **reforested** to rehabilitate them, and meet fuel wood and industrial timber needs. Areas which had no forest should also be **afforested**. Communities of local people should be rewarded with the benefits of afforestation so that they can participate with energy and enthusiasm.

Conservation - is the **wise use** of **natural resources without depleting** them. Forests, for example, are cleared in such a way that they can **regenerate**, or **replaced** immediately. **Water bodies** should be utilized **without poisoning** them with **toxic substances** coming out from **industries**. **Soils** and **water resources** (surface and underground) can be conserved by increasing vegetation cover of catchment areas. So, resources should be conserved by employing **sustainable utilization** of them.

Gazettement of all protected areas - Gazettement of a park is its having full legal status as a **national park** so that complete law enforcement is possible. This solves management problems posed by human encroachment and other disturbances. Offenders can be prosecuted in this case. The

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re-establishment of the conservation status of protected area should also be accomplished.

Education - local communities **should not** be simply **told** what they **should** and **should not do**. They should be **taught practically** the value of conservation practices such as protecting an area. Then, they will be far more willing to participate in conservation measures. Intensive training and education programmes about the benefits of ecosystem conservation, and the dangers of their degradation, should be employed.

Conservation of genetic diversity - habitat degradation led to the loss of our biodiversity in the wild. The replacement of our indigenous crops by the improved ones has also eroded our crop genetic diversity. So, when national conservation measures are thought of, such important burning issues should also be considered.

Establishment of newly protected areas - Ethiopia has many ecosystems with **endemic plants** and **animals** as well as high **species richness**. Such areas should be placed under immediate and urgent protection. Almost, all the ecosystems of our country are under threat. There are areas with wild relatives of crops grown for food, for example. As many ecosystems with such unique features as possible should be legally protected in addition to the already existing ones.

Illustrative Examples

46. Which of the following is expected to have a positive effect on the biodiversity of Ethiopia? **(UEE, 2003 E.C.)**
(A) Diverse ecology (C) Overgrazing by livestock
(B) Extensive deforestation (D) Logging and fuel wood removal

Explanation:
- There are diverse ecosystems in Ethiopia because of the diverse topography ranging from the Danakil depression to mountain peaks reaching more than 4000 m.a.s.l. **Answer: A**

47. In which of the following biomes are epiphytes typically found as characteristic elements? **(UEE, 2003 E.C.)**
(A) Tundra (C) Cold desert
(B) Boreal forest (D) Tropical rain forest

Explanation:
- Epiphytes are plants that grow on trees, and they are common in the tropical rain forest. **Answer: D**

48. Large geographical regions with their specific **flora** and **fauna** are known as:
(A) Biosphere (C) Biomes
(B) Ecosystems (D) All

Explanation:
- The best term that describes large global ecological units distinguished by their characteristic flora and fauna, as well as climate and soil type, is biome. **Answer: C**

49. Biomes are mainly defined by their
(A) Climate (C) Hydrological conditions
(B) Soil type (D) Vegetation

Explanation:
- Climate, which includes precipitation, wind, temperature and the like, is the major factor used to define a biome. **Answer: A**

50. Which of the following factors is **not** used in the characterization of terrestrial (land) biomes?
(A) Rain fall (C) Salt content
(B) Temperature (D) Soil type

Explanation:
- Salt content is used as criterion for the characterization of aquatic biomes, not land biomes. **Answer: C**

51. What are the **differences** and **similarities** between **cold** and **hot** deserts?

Explanation:
- Both are similar because there is limited rain fall, and there is less or no water available for organisms. Differences include the location and temperature conditions of the two. Cold deserts are found in temperate regions, they are very cold, and water is frequently, frozen. Warm/hot deserts are found in the **tropics** and **sub-tropics**, they are very hot, and there is no water here.

52. Which of the following deserts is not found in Africa?
(A) The Sahara desert (C) Atacama desert
(B) Namib desert (D) Kalahari desert

Explanation:
- Atacama desert is found in Latin America **Answer: C**

53. Which of the following desert plants are adapted to their environment by having underground **bulbs** or **corms**?

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(A) Ephemerals
(B) Perennials
(C) Geophytes
(D) Succulents

Explanation:

- Geophytes pass the long periods of drought as **corms** or bulbs, and regenerate from them as conditions restore.

Answer: C

54. Desert plants which complete their life cycle with in very short time are:
(A) Perennials
(B) Annuals
(C) Succulents
(D) Ephemerals

Explanation:

- Ephemerals germinate, grow, flower and produce seeds within very short period of time.

Answer: D

55. What is the collective name for plants that are well adapted to live in very dry environments? **(UEE, 2003 E.C)**
(A) Epiphytes
(B) Xerophytes
(C) Mesophytes
(D) Hydrophytes

Explanation:

- **Xerophytes** are adapted to live in a very **dry** environment. Epiphyte live on trees, **Mesophytes** in **well-watered** areas, and **Hydrophyte** are water dwellers.

Answer: B

56. By how much degrees would camel's temperature is reduced when it run short of water?
(A) 30°C
(B) 10°C
(C) 15°C
(D) 50°C

Explanation:

- Camels can lose up to **30%** of their body water content. This time their body temperature drops by 10°C.

Answer: B

57. What is the advantage of this temperature drop in camels in question 56 above?

- When camel's body temperature is reduced, the water that would have been lost by evaporation to cool it down, is conserved.

58. Tundra covers about one-fifth of the terrestrial environment. What limits the growth of tree-roots in this biome?

Explanation:

- Permafrost, frozen soil within a meter from the surface, is an obstacle for plant root growth down into the ground.

59. Which of the following is more likely to ensure the conservation of all the rest? **(UEE, 2003 E.C)**

(A) Soil conservation
(B) Plant conservation
(C) Wild life conservation
(D) Watershed conservation

Explanation:

- If vegetation is conserved, wild life uses it as shelter, and soil is protected by their roots. Both underground and surface water is also conserved as the result of the vegetation's spongy character to absorb the water and help to seep down in to the ground slowly.

Answer: B

60. Which of the following biomes is dominated by evergreen coniferous vegetation?
(A) Boreal forest (Taiga)
(B) Tundra
(C) Temperate rain forest
(D) Tropical rain forest

Explanation:

- Taiga is characterized by coniferous trees.

Answer: A

61. Which of the following terrestrial biomes experiences hot days and cold nights?
(A) Tundra
(B) Tropical rain forest
(C) Desert
(D) Grasslands

Explanation:

- Deserts experience greater temperature fluctuations due to the absence of much water vapour in the atmosphere.

Answer: C

62. Much of the rich agricultural lands in USA and Canada were originally:
(A) Temperate grasslands.
(B) Tropical grasslands (Savannas).
(C) Temperate rain forests.
(D) Tropical monate forests.

Explanation:

- Most of the rich agricultural lands in USA and Canada were **temperate grasslands (prairie)**. But those in the tropics were **savannas**.

Answer: A

63. Which of the following biomes is the **richest** in species diversity?
(A) Tropical monate forests
(B) Temperate rain forests
(C) Tropical rain forest
(D) Temperate deciduous forest

Explanation:

- Tropical rain forest is the richest showing different species of plants, animals, and micro-organisms.

Answer: C

64. The soils of tropical rain forest are poor in **Inorganic nutrients**. This is due to the fact that:
(A) The area is depleted from nutrients naturally.

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(B) The nutrients are tied up in the trees of the area.
(C) Water has washed the nutrients away.
(D) The nutrients have evaporated away by high temperature.

Explanation:

- Most of the nutrients in the soil of tropical rain forest are held up in the huge trees. Unless these trees fell down and decay, the nutrients can not be recycled.

Answer: B

65. Which of the following areas is not the site of **equatorial rain forest**?
(A) South America
(B) West Africa
(C) Europe
(D) Indonesia

Explanation:

- Tropical rain forest is found in South and Central America, West and equatorial Africa, South-east Asia, Indonesia and Northern Australia.

Answer: C

66. Which of the following ecosystems has the **least net primary productivity** per unit area per time?
(A) Open ocean
(B) Tundra
(C) Coral reef
(D) Desert and semi-desert

Explanation:

- Deserts and semi-deserts have the least productivity.

Answer: D

67. Which of the following aquatic biomes has many strata like that of equatorial rain forest?
(A) Estuarine
(B) Coral reef
(C) Wetlands
(D) Pelagic zone

Explanation:

- Coral reef is the most diverse of all the marine biomes, and has many layers similar to the equatorial rain forest.

Answer: B

68. Which of the following statements is **Incorrect?**
(A) Ethiopia is located in the tropics.
(B) There is different landforms in Ethiopia.
(C) Areas below sea-level are not found in Ethiopia.
(D) The diverse topography gave Ethiopia different ecosystems.

Explanation:

- Ethiopia's topography ranges from the **Danakil depression** which is **116m** below sea level to tall mountain peaks such as Ras Dejen (4620m a.s.l).

Answer: C

69. Which of the following ecosystems of Ethiopia is not economically useful for grazing, **agriculture**, or even **fire wood**?
(A) Lowland forest ecosystem.
(B) Combretum-Terminalia and Savanna.
(C) The moist evergreen montane forest.
(D) The evergreen scrub ecosystem.

Explanation:

- Evergreen scrub is an ecosystem that has been replacing montaine forests as they are deforested.

Answer: D

70. Which of the following measures are important for the rehabilitation of the degraded Ethiopian ecosystems?
(A) Reforestation and conservation of existing resources.
(B) Gazettement and establishment of protected areas.
(C) Educating people and conservation of biodiversity.
(D) All are measures ought to be considered.

Explanation:

- All the listed approaches of ecosystem restoration should be encouraged.

Answer: D

13 Biodiversity

In its broader sense, **biodiversity** may refer to the **genetic diversity** of organisms of an area, species **richness** and **ecosystem variability**. It may also include cultural diversity which is manifested by **diversity** in **language, religion, art,** music, social structure, diet and **land management**.

Here, we focus on **biodiversity** as referring to successful **species richness** of an ecosystem or area; which is measured by **index of biodiversity**.

Example: Simpson's index of diversity (d)

$$d = \frac{N(N-1)}{\sum n(n-1)}$$

d = Index of diversity
N = Total number of organisms in the area.
n = Total number of organisms of each species.

The following table shows the distribution of six species of plants in three different areas. The calculation and conclusion of Simpsons's index of diversity follows the table.

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Table 7.3. Example of three ecosystems with six species each.

Species	Species Number of organisms of each species		
	Area 1	Area 2	Area 3
A	86	16	23
B	5	17	25
C	2	16	27
D	3	17	5
E	1	17	12
F	3	17	8

$$d = \frac{N(N-1)}{\sum n(n-1)}$$

$$d_1 = \frac{100(99)}{(86x85)+(5x4)+(2x1)+(3x2)+(1x0)+(3x2)} = 1.348$$

$$d_2 = \frac{100(99)}{(16x15)+(17x16)+(16x15)+(17x16)+(17x16)+(17x16)} = 6.314$$

$$d_3 = \frac{100(99)}{(23x22)+(25x24)+(27x26)+(5x4)+(12x11)+(8x7)} = 4.911$$

- **High 'd'** value indicates **high species diversity** and **stable ecosystem**, while **small 'd'** value imply **low species diversity** and **hostile environment**.

So, in the above example, area 2 with diversity index of $d = 6.314$, has high diversity. In all the three areas there are six types of species, but these are not equally successful in area 2. They are not equally succesful in the remaining areas. In area 1, only species A, and in area 3, species A, B and C only are successful. There are **more niches** in stable ecosystems with higher d-value, and such areas are less affected by environmental changes.

Human influence on biodiversity

- Humans have been affecting biodiversity in two main ways: by **deforestation** and **agriculture**.
- Forests are cleared to make the land ready for **mining**, **agriculture** or **construction**. They are also needed to obtain **timber** to make paper,

furniture, charcoal, or to use them as building material.

The deforestation of tropical rain forest, for example, is destroying 25% of the known species and exposing 7% of the earth's surface. In modern agriculture, usually only one type of crop species is produced year after year (i.e. Monoculture). Other plants and animals are considered as **pests** and controlled by **herbicides** and **pesticides**. All these acts destruct habitats and niches, and reduce biodiversity of the area. In some cases, local extincition of species may follow.

Biodiversity in Africa and Ethiopia

There is high floral and fauna divesity in africa, and these overlap as plants provide habitats and niches for the animals. Africa has:

- 1/4th of world's mammal species (i.e. ~ **1229** out of ~ **4700**)
  - 960 sub-saharan Africa(SSA).
  - 137 in Madagascar.
  - 79 antelope species in eastern and southern savannas.
- 1/5th of world's bird species (i.e. ~ **2000** out of ~ **10,000**).
  - 1600 endemic to SSA.
- 950 species of amphibians.

Ethiopia has better biodiversity than most African nations though it is inferior to some countries such as Kenya and Democratic Congo. The different altitude and climate range gave rise to different **biomes** and led to the **high biodiversity**. There are about **6500-6700** plant species out of which about **1150** (~ 10-12%) are endemic. The country is also one of the **12** centres of origin of some cultivated crops (i.e. Vavilov's centeres). Eleven crops have their center of genetic diversity in Ethiopia. These are:

1. *Coffea arabica* **(Coffee)**
2. *Eragrostis tef* **('Tef')**
3. *Ensete ventricosum* **('Enset')**
4. *Coccinia abyssinica* **('Anchote')**
5. *Guizotia abyssinica* **(Niger seed, 'Nug')**
6. *Brassica carinata* **(Ethiopian rape, 'Gomen zer')**
7. *Carthamus tinctures* **(Saff lower, 'Suf')**
8. **Sorgum species**
9. *Hordeum* **species (Barley)**
10. *Linum usitatissimum* **(Linseed, 'Telba')**
11. *Ricinus communis* **(Caster bean, 'Gulo')**

There are also diversified **forage crops** (plants used as animal feed) in Ethiopia. There are about 46 endemic species of legumes important as forge crops and

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increase **soil fertility**.
Example: *Trifolium* (clover), vigna (a type of bean) and lablab (all parts of which is edible).
Every group of vertebrate animal is represented in Ethiopia as the following table shows.

Table 7.4. Vertebrate diversity of Ethiopia

Vertebrate group	Species	Endemic to Ethiopia	Proportion of endemism
Mammals	277	22	0.08
Birds	861	27	0.03
Reptiles	78	3	0.04
Amphibians	63	17	0.27
Fish	101	4	0.04

From the above table, **27%** of the **Amphibians** in Ethiopia are **endemic**. The next higher endemism is that of **mammals** with **8%** endemism.
Recently, 69 important bird areas are identified and protected.

Direct and Indirect effects of human practices on biodiversity

Practices with direct effects

- ✓ Deforestation
- ✓ Fuel wood collection and illegal logging.
- ✓ Overgrazing by stock animals.
- ✓ Introduction of improved crop varieties (decreases genetic diversity of indigenous crops).
- ✓ Over-hunting (poaching).
- ✓ Introduction of alien invessive species (out-compete native ones, and may make them extinct).

Practices with indirect effects

- ✓ Undervaluation of biodiversity resources by government bodies.
- ✓ Promotion of unsustainable exploitation by legal and institutional systems.
- ✓ Disregard of traditional communal land management systems.
- ○ Ethiopia's highland cover scenario from estimation:
  - Before 1950 ~ 87%.
  - In 1950 ~ 40%.
  - In 1980 ~ 5.6%.
  - In recent days ~ 3%.

Biodiversity concern

According to the millennium assessment findings, biodiversity loss contributes to:

- Worsening health.
- Food insecurity.
- Increasing vulnerability.
- Low material wealth.
- Worsening social relations.
- Less freedom for choice and action.

Biodiversity increases flexibility in food choice, and spreads(reduces) risk of food insecurity. It gives physical protection in some cases.
Example: coral reefs and coastal forests prevent **flooding** and **storms**.
The poor and landless people can get balanced diet from wild. Many drugs are obtained from plants and animals.
Example: Aspirin (oldest painkiller) is from willow tree. Spiritual values are attached to the components of ecosystems (i.e hills, trees, rivers, etc). Their loss reduces bondage between societies, and cause resentment towards benefited groups.

Ecosystem services important for human well-being

- Supporting of nutrient recycling, soil formation and primary production.
- Provision of food, fresh water, wood, fiber and fuel.
- Regulation of climate, flood and diseases, and water purification.
- Cultural values (aesthetic, spiritual, educational and recreational).

We are losing *10%* of our biodiversity every *50* years! But, this rate of loss is estimated to increase by *10* times in the future!

So, action is needed at individual, local, regional, national and international levels. We have to conserve biodiversity. **Conservation** is the **sustainable** use of resources, including **preservation** and **restoration** of ecosystems.

Guiding ideas of conservation

a) **Research** (to know what to do).
b) **Minimum intervention.**
c) **Repair rather than replace** (better and easier).

Ecological principles of conservation

1. **Biodiversity** is supported by **protection** of any **species** and **varieties.**
2. **Habitat maintenance** is fundamental to species conservation.
3. **Large areas** contain **more species** than smaller ones with similar habitats.
4. **Habitat disturbances reshape** an ecosystem.
5. **Climate change** will **affect** all **ecosystems.**

We can start planting trees at an individual and community level to restore our degraded environment. These can act as shelter for birds and other animals. Complex food webs can develop in the new ecosystem. Coppicing (cutting trees

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for timber) will be possible with in 20 years.

Mycorrhizal (symbiosis between fungi and plant roots) can also develop in the soil. Mycorrhizal plants are tolerant to environmental stress because fungal hyphae grow around roots and increase **surface area** for **nutrient** absorption. The fungus obtains organic nutrients from the roots in exchange to inorganic **minerals**.

Illustrative Examples

71. Which of the following **does not** indicate biodiversity?
(A) Genetic diversity of organisms of a species.
(B) Number of species in a given ecosystem.
(C) Ecosystem variability and cultural diversity of an area.
(D) None of the above.

Explanation:
- Because all the listed diversities indicate biodiversity of an area.

Answer: D

72. Which term better captures the concept of variety of organisms present in a whole continent as well as all the genes found in them? **(UEE, 2003 E.C)**
(A) Niche
(B) Biodiversity
(C) Ecosystem
(D) Biome

Explanation:
- Biodiversity encompasses both species richness and genetic variability of an area.

Answer: B

73. Using the following table, calculate Simpson's index of diversity for both areas and determine the area with higher biodiversity.

Species	Number of organisms	
	Area 1	Area 2
A	20	10
B	17	8
C	13	32

Explanation:

$$d = \frac{N(N-1)}{\sum n(n-1)}$$

$$d_1 = \frac{50(49)}{(20x19)+(17x16)+(13x12)} = \frac{2450}{808} = \underline{3.032}$$

$$d_2 = \frac{50(49)}{10(9)+8(7)+32(31)} = \frac{2450}{1138} = \underline{2.153}$$

- Therefore, Area-1 has high Simpson's diversity index, and high biodiversity with successful species than Area-2.

74. Discuss how agriculture affects biodiversity.

Explanation:
- **Deforestation**-forests or grasses are cleared to get land for agriculture.
- **Monoculture**-only one type of crop is cultivated year after year in modern agriculture.
- Other plants, except the crop, are considered as pests and eradicated by **herbicides**.
- Insects and animals feeding on the crop are destroyed by **pesticides**.
- Soil organisms may be negatively affected by the **pesticides**, **herbicides** and **fertilizers** used.

75. Which of the following cultivated crops has **no** center of genetic diversity in Ethiopia?
(A) *Coffea arabica*
(B) *Zea mays*
(C) *Eragrostis tef*
(D) *Brassica carinata*

Explanation:
- *Zea mays* (maize) has no center of origin in Ethiopia

Answer: B

76. How is it possible for different species to coexist in an ecosystem? **(UEE, 2003 E.C)**
(A) Each species has a specific biome.
(B) Each species occupies a different niche within the ecosystem.
(C) There is no competition for resources among species in an ecosystem.
(D) All resources are equally shared among species in an ecosystem.

Explanation:
- Each species occupies a different niche and this reduces competition between species.

Answer: B

77. Which of the following vertebrates has high species diversity in Ethiopia?
(A) Fish
(B) Amphibians
(C) Reptiles
(D) Birds

Explanation:
- When species number is considered, Ethiopia has high bird diversity (~ 861).

Answer: D

78. Which of the following vertebrates has high proportion of endemic species in Ethiopia?
(A) Mammal
(B) Birds
(C) Amphibians
(D) Fish

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Explanation:
- About 27% of the Amphibian species found in Ethiopia are endemic.

Answer: C

79. What are the two uses of legume forage crops?

Explanation:
- Their direct use is to act as animal feed, and their second use is to fertilize the soil by fixing Nitrogen because they have symbiotic bacteria that fix nitrogen.

80. What proportion of the world's mammal species is estimate d to occur in Africa?
(A) 25%
(B) 20%
(C) 50%
(D) 15%

Explanation:
- One-forth of the world's mammals are believed to occur in Africa.

Answer: A

81. Which of the following practices has direct impact on biodiversity?
(A) Poaching
(B) Deforestation
(C) Overgrazing
(D) All

Explanation:
- Poaching, deforestation and overgrazing have direct effect on biodiversity.

Answer: D

82. Which of the following practices have indirect effect on biodiversity?
(A) Undervaluation of biodiversity.
(B) Unsustainable exploitation promotion.
(C) Population growth and disregard of traditional land management skills.
(D) All are correct.

Explanation:
- All the listed practices have indirect effects.

Answer: D

83. What is Ethiopia's forest cover now?
(A) ~ 5.6%
(B) ~3%
(C) 87%
(D) 40%

Explanation:
- Our forest cover has drastically diminished to about 3% only.

Answer: B

84. Biodiversity loss can contribute to:
(A) Worsening social relations.
(B) Increasing food insecurity.
(C) Reduced freedom of choice and action
(D) All.

Explanation:
- All the listed problems being encountered by human kind are directly or indirectly caused by loss of biodiversity.

Answer: D

85. Which of the following statements is **incorrect**?:
(A) We are losing 10% of our biodiversity every 50 years.
(B) The rate at which we lose biodiversity presently is increasing 10 fold in the future.
(C) Climate change can not affect stable ecosystems.
(D) Ecosystem services with cultural values are important for human well-being.

Explanation:
- Climate changes practically affect all ecosystems, but stable ecosystems are less affected than unstable ecosystems with less biodiversity.

Answer: C

7.5 Populations

A population - consists of all **individuals** of a particular **species** in a particular habitat at a specific **time**. Every population has characteristic features such as size (number of its individuals), **density** (how crowded it is), **dispersion** (how they are distributed /**evenly** or **unevenly** spaced), **sex ratio** (males to females) and **demography** (proportion of young, middle-aged or old).

A population is **dynamic**, not **static**. Many populations can live together in an area because each may exploit a different **habitat** and have a different **niche**.

Habitat - an area where a population lives and obtains the resources it needs within its geographical range.

Ecological niche - role of a population in its habitat.

Geographical range - part of earth where a particular species lives.

Example: pond as a habitat:
- ✓ Plankton exploit open water.
- ✓ Decomposers live on detritus at the bottom.
- ✓ Floating plants use sunlight, CO_2, water and minerals and photosynthesize.
- ✓ Tadpoles feed on insect larvae.

Factors affecting population size

a) **Natality** (birth rate).
b) **Mortality** (death rate).

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c) **Migration**-movement into (**immigration**) and out of (**emigration**) an area.

Normally, two species can not occupy the same niche. If they do so, one competes with the other and makes it extinct, or less successful. ∴ competitive **exclusion principle**.

Natality as **mortality** as well as **immigration** and **emigration** are linked in their effects.

- If **natality increases** and **mortality decreases, population** size **increases**, and **vice versa**. But, if both are balanced, constant population size will sustain.
- If **immigration decreases**, and **emigration increases, population** size **decreases**, and **vice versa**. Keeping other factors constant, if immigration and emigration are balanced, population size stabilizes.

Other factors controlling population size

a) **Abiotic factors** - are physical environmental factors such as light, temperature, carbon dioxide and oxygen concentrations, physical space, nutrients and others. Availability of light, oxygen, carbon dioxide, enough space and nutrients will increase population size. Extreme temperatures and PH, as well as toxic substances in the environment will check population size.

b) **Biotic factors** - refer to the effects that come on to a given population from other organisms of the same or different species. These are:

1. **Predation** - presence of **carnivore** (for animals) or a **herbivore** (for plants).
2. **Diseases** - presence or absence of **pathogenic** micro-organisms. These may reduce productivity and/or be fatal.
3. **Intra-specific competition** - takes place between organisms of the same species.

   Example: - Seeds germinating in large number at a point.
   - Many human families occupying a small plot of land.
   - Caterpillars hatched in large number on a tree leaves.

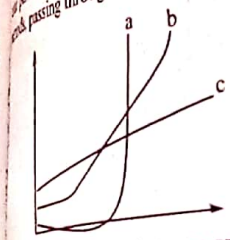
4. **Inter-specific competition** - occurs between organisms of different species. This is caused due to some overlap between niches of two species.

   Example: - Ethiopian wolf and domestic dogs hunting for rats.
   - Weeds and crops growing in the same field.
   - Lions and cheetah or leopard competing for zebra.

How do populations grow

There are two possible ways: Arithmetic and Exponential.

...metic rate - shows fixed growth rate and constant change per unit time.
...ple: 10 --15 --20 --25 --30 --35 --40 --45 --50 --- . . .
...ential rate - shows increasing growth rate and accelerating change per ...e. It may cause doubling in each time period.
...ple 10 --20 --40 --80 --160 --320 --640 --1280 --2560 --- . . .
...s illustrated above, two populations starting form the same number, ...with 50 individuals only in arithmetic growth, after 8 successive ...eriods, while the other ended up with 2560 individuals after similar time ...iod, passing through exponential growth rate.



a = Fast exponential growth
b = Slow exponential growth
c = Arithmetic growth

Figure 7.7. Comparison between arithmetic and exponential growth rates.

...a result of combined effect of **biotic** and **abiotic** factors, and growth potential ...ery population, **no population shows purely arithmetic or exponential** ...growth rate, but something between the two. All populations show growth ...pattern that has 4 phases (Figure 7.8)

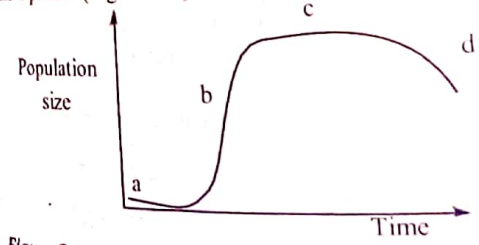


Figure 2.8. Main phases of population growth curve

a) **Lag phase** - organisms struggle to **adapt** and **establish** themselves to the environment. Most of them may die, but some survive and reproduce. Population size remains **low** and **static**, or increases **slowly**.

b) **Log phase** - all survivors are adapted and reproduce rapidly. Here population size increases rapidly.

c) **Phase** - here the caring capacity is reached, and death and reproduction rates are equal. Population size flactuates around a 'mean', and stays constant in general.

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Carrying capacity - the number of organisms of a species that an area or environment supports (carries) at a particular time. It may change from year to year due to environmental changes. Example: rain fall flactuations.

d) **Death/decline phase** - this is the stage at which nutrients are exhausted and new diseases emerge. Toxic excretory products also accumulate and cause death of the organisms. Consequently, population size declines rapidly.

Human populations

Human population was less than **5,000,000** ten thousand years ago, and was **6.8 billion** at **2009**. It was quickly increasing over the past 200 years because of increasing food **quality** and **quantity**, and improved **sanitation** and **medical care** which has **reduced pathogen impact**. But, high population size is obstacle to:

- Development (prevents or slows it down).
- Implementation of education and health programmes.
- Proper sanitation and others.

Although population size is increasing, its rate of increase is **decreasing**, and forecasted to level at **12 billion** by **2200**. The carrying capacity of earth is estimated to be about **10 billion**. Human population is subject to all factors affecting populations of other organisms. But, **agricultural development** and **industrial advancement** are affecting our **growth** and **death rates**, and life **expectancy**. As a result, every country passes through **demographic transition** depending on their stages of development.

Demographic transition - a model that tries to explain the transformation of countries from having **high birth** and **death rates** to **low birth** and **death rates**. Demographic transition has three major stages.

Stage-I: societies with **high birth and death rates**, population growth is very **slow**, or **stable**.
Example: Remote groups that are far from civilization.

Stage-2: societies in countries at this stage show **high birth rate** and low **death rate**, but **both rates** are **decreasing** (falling). population size is increasing.
Example: Egypt, Kenya, India, Brazil, etc.

Stage-3: Countries at this final stage show **low birth and death rates** with **stable population**. There may be **slow increase or decrease** in size, but on the average, population size is **constant**.
Example: USA, Japan, France, UK, Germany, etc.

Generally, population **size** is **increasing** in **developing countries**. It is stable in **developed** nations. During demographic transition, the relative number of

young and old people changes, and these are illustrated by **age pyramids**. (Figure 7.9).

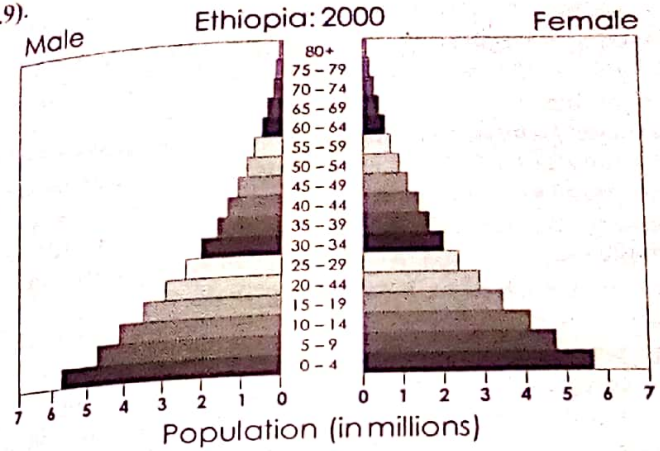
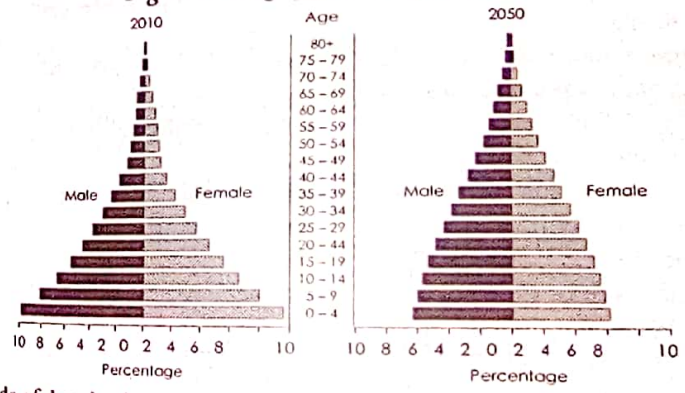


Figure 7.9. Age pyramid of Ethiopia



Pyramids of **developing countries** are **broad** at the base. This shows that there are more children. The **narrower tip** of pyramids of developing nations shows short life **expectancy** as there are few old people in the societies. The pyramids of developed countries is relatively **rectangular**. This is an indication that there is less gap between the different age groups. **Life expectancy** is **longer**, and population size is **stable**.

During demographic transition, median age and percentage of older population increases, overall shape of age pyramids becomes more rectangular, and population size stabilizes.

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Global population size depends on global **natality** and **mortality**. Migration has no effect on global population size as we go nowhere.

There are possible ways of reducing human population.

a) **Increasing mortality** - this is not something intentionally done, but war and conflicts are being observed over resources here and there in the world. Many people are losing their life because of such conflicts that may be given other pretexts though their root cause is competition over resources.

b) **Decreasing natality** - this **is the only option** we can **intentionally do**. People should be **educated**, and their **awareness** raised to **reduce** the number of children they will have in the future. Different birth control **methods** should be used. **Availability** and **accessibility** of these methods have to be **increased nationally** and **globally**.

At the moment, **Ethiopia** has one of the **highest birth rates** in Africa which is of course **decreasing**. Therefore, **population** size is **increasing** with **decreasing** birth **rate**. So, local communities have to get **access** to **general education**, **sex education** and **contraception**. The government has to introduce some feasible policies of population control.

Some countries introduced some kinds of population control policies. These are:

China - more than one child/family is discouraged, and unwanted pregnancy is punished with fine.

India - only people with two or fewer children may serve in local government.

Iran - compulsory contraception courses before offering marriage licence.

USA - free contraception services and sex education by giving priority to the poorest.

As the result of such policies in these countries, though population size is still increasing, its rate of increase is decreasing (Example: China, India, Iran). In developed countries such as USA, population size is stabilized, or increases slowly.

Illustrative Examples

86. A group of organisms of one species in the same place at the same time is
(A) Community (C) Population
(B) Seres (D) Ecosystem

Explanation:

- Populations are group of organisms of a species at a place at a time.

Answer: C

87. Part of the earth where a particular species is distributed is known as:
(A) Habitat (C) Ecological niche
(B) Geographical range (D) Biosphere

Explanation:

- Geographical range encompasses area where a species lives.

Answer: B

88. Characteristics of a population that shows whether members are evenly or unevenly distributed is:
(A) Demography (C) Population density
(B) Population size (D) Population dispersion

Explanation:

- Dispersion shows how members of a population are distributed in their habitat.

Answer: D

89. All other factors remaining the same, which of the following **decreases** the size of a population? **(UEE, 2003 E.C)**
(A) Birth rate that is grater than death rate.
(B) Emigration rate that is less than immigration rate.
(C) Immigration rate that is greater than death rate.
(D) Emigration rate that is greater than death rate.

Explanation:

- Emigration (moving out) rate that is greater than death rate decreases population size.

Answer: D

90. Which of the following factors have contradicting effects on population size?
(A) Natality & immigration (C) Natality & emigration
(B) Mortality & emigration (D) Emigration & death

Explanation:

- Natality (birth) has opposing effect to emigration (movement of individuals out of a population).

Answer: C

91. Which of the following population controlling factors is **abiotic**?
(A) Habitat (C) Pathogen
(B) Predator (D) Competitor

Explanation:

- Habitat is a place where an organism lives. The other alternatives are all **biotic** factors

Answer: A

92. When do populations of living things show exponential growth? **(UEE, 2004 E.C)**

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(A) When the resources are plentiful.
(B) Whenever they enter a new environment.
(C) When they face strong competition from other species.
(D) When the carrying capacity of the environment is reached.

Explanation:

- Populations grow exponentially so long as resources are available. As resources start being depleted, their growth is checked.

Answer: A

93. Which of the following is an **inter-specific** competition?
(A) Students running to biology laboratory to get front seats.
(B) Vulture and hyena fighting over dead animal.
(C) Dogs fighting over bone.
(D) Haile and Kenenisa running to win a gold medal.

Explanation:

- Vulture and hyena are different species. Inter-specific competition occurs between different species over a scarce resource.

Answer: B

94. As a result of combined effect of **biotic** and **abiotic** factors, the type of growth rate shown by populations is
(A) Purely arithmetic.
(B) Purely exponential.
(C) Purely geometrical.
(D) Something between arithmetic and exponential.

Explanation:

- Populations show a blending of **arithmetic** and **exponential** growth rates.

Answer: D

95. At which phase of the population growth curve is the fastest growth shown?
(A) Lag phase
(B) Log phase
(C) Stationary phase
(D) Death phase

Explanation:

- At the log phase, the organisms have already been adapted to the environment, and start fast reproduction. Resource is also available at this stage.

Answer: B

96. Which of the following factors have contributed to the faster growth of human population over the last 200 years?
(A) Improved food quality and quantity.
(B) Improved medical care and reduced impact of diseases.

(C) Improved sanitation and overall technological advancement.
(D) All of the above.

Explanation:

- All the listed factors have contributed much to human population growth.

Answer: D

97. What is the estimated carrying capacity of the earth for human population?
(A) 12 billion
(B) 22 billion
(C) 10 billion
(D) 200 billion

Explanation:

- The earth's carrying capacity is about 10 billion.

Answer: C

98. At which stage of the demographic transition is Ethiopia probably found?
(A) At the first stage where both birth and death rates are high.
(B) At the second stage where birth rate is high, and death rate is low.
(C) At the third stage where both birth and death rates are low.
(D) At the final stage where population size is stabilized.

Explanation:

- Ethiopia is probably at the second stage where birth rate is high, and death rate is low.

Answer: B

99. From the following countries, whose age pyramid is most probably broad at the base?
(A) Finland
(B) Norway
(C) Egypt
(D) Canada

Explanation:

- Egypt is a third world country (i.e. developing country). Its population is increasing. There are many young generations as compared to older ages.

Answer: C

100. As the demographic transition progresses:
(A) Percentage of older people increases.
(B) Median age decreases.
(C) Age pyramid becomes wider at the base, narrower at the top.
(D) Population size becomes less stable.

Explanation:

- Percentage of older people increases, median age increases, age pyramids become relatively rectangular, and population size stabilizes.

Answer: A

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UNIT SEVEN ECOLOGY

Conceptual Summary Example

1. Define population density and dispersion, and describe the main types of population dispersion

Explanation:

- **Population density** - is the number of individuals of a species in a unit of area or volume at a given time.
- **Population dispersion (spacing)** - a clumped dispersion.

2. Explain the factors immigration and emigration that produce changes in population size, and solve sample problems involving these changes.

Explanation:

- **Immigration** - is movement of people into a country or a given ecosystem. It results in increasing the population number of that area.
- **Emmigration** - is the movement of people out of an area which results in decreasing the population number of that area.

3. Define predation, and describe the effects of natural selection on predator-prey relationships.

Explanation:

- **Predation** - is a symbiotic relationship between two organisms in which one organism the **Predator** hunts and eats another organism is called **Prey**.
- **The effect of natural selection on the Prey** - Predator relationship that an organism that nature favors is less likely to be attacked by a Predator. The typical example for this is an experiment of the effect of industrial milanism on peppered moths.

4. Distinguish among mutualism, Commensalism, and Parasitism and give example for each.

Explanation:

There are different forms of mutualistic associations between organisms of different species, they include

- **Mutualism** - is an obligatory relationship between two organisms in which both are beneficial.

Example: Lichen (associations between algae and fungi).

- **Commensalism** - is a symbiotic relationship in which one organism is beneficial and another organism is neither benefited nor harmed.

Example: Epiphytes growing on a tree.

- **Parasitism** - the relationship in which one organism (parasite) harms the other organism (host).

5. Define ecological succession, and distinguish between Primary and Secondary succession.

Explanation:

- **Succession** - is the orderly replacement of one community by another.
- **Primary succession** - occurs in an area that has not previously been inhabited.
- **Secondary succession** - begins in an area where there was Pre-existing community and well-informed soil.

6. Contrast the influence of density - dependent and density - independent factors on population size, and give example of each.

Explanation:

- **Density - dependent factors** - regulates population growth by affecting a larger proportion of the population as population density rises. Examples are Predation,
- **Density - independent - factors** - limit population growth but are not influenced by changes in population density. Examples are hurricenes and blizzards.

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7. Distinguish between people overpopulation and consumption overpopulation.

Explanation:

- Developing countries tend to have people overpopulation in which population increases degrades the environment. Even though each individual uses few resources.
- Highly developed countries tend to have consumption overpopulation, in which each individual in a slow growing or stationary population consumes a large share of resources, resulting in environmental degration.

8. Describe the needs of the following cycles in an ecosystem:
   - Carbon cycle
   - Nitrogen cycle
   - Water cycle

Explanation:

Recycling in nature is one of the most important activities in an ecosystem. Thus, the following cycles and their importances will be discussed as:

Cycles	Importance
Carbon cycle	Plays an important role in the process of photosynthesis and respiration.
Nitrogen cycle	Important in preparation of useful proteins such as DNA, RNA, Enzymes, etc.
Water cycle	Useful in determining the rain fall (humidity) of an ecosystem.

9. What is Biodiversity? Describe its significance.

Explanation:

Biodiversity is the wealth of species in an ecosystem. Conservation of biodiversity is important in order to keep the balance of nature.

10. List out some of the major threats of Ethiopian biodiversity.

Explanation:

In history Ethiopia is one of the wealthy nations by their biodiversity. However, the wealth of species in Ethiopia is declining at faster rates, even some endemic animals are on the brick of extinction.

The threaths of biodiversity in Ethiopia are:

- Deforestation and burning of vegetation.
- Commercial logging.
- Subsistence farming.

11. What is conservation? What appropriate measures should taken to conserve Ethiopian biodiversity?

Explanation:

Conservation is the wise use of natural resources on a sustainable basis.

Appropriate measures to be taken to conserve Ethiopian biodiversity are:

- Avoiding deforestation and Planting trees.
- Creating awareness to the society on the ways of conserving their natural environment.
- Establishing legal ground on the ways of protecting the natural habitat from destruction.

12. Compare the following with brief discussions.
   - Intraspecific and interspecific competition.
   - Exponentrial and Arthimetic growth curves.

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Explanation:

Intraspecific Competition	Interspecific Competition
It is a completion between members of the some species on resources	It is a completion between organisms of different species

- **Arthimetic increase** - is where a population number increases by a set of amount in each time period.
- **Exponentrial growth** - is when the population number increases by an increasing amount in each time period.

13. How can natality and mortality influence the population size? Explain the interpretation of population growth curve.

Explanation:

- The number of individuals leaving the population is called Mortality. It results in decreasing human populatation.
- **Natality** - refers to the number of individuals entering the population by birth per thousand individuals in the population. It cases increase in population number.

14. How can the rapid population growth affect the development of a country? What appropriate measures should be taken to control the rapid populations growth?

Explanation:

Population growth affects the development of a country because resources, infrastructures and other supplies are scarce. This creates social, political and economic problems in the country.

The appropriate measures to be taken to regulate the rapid populations growth are:

- Effective use of contraceptives.
- Family planning
- Educating and creating awareness in the society, etc.

POST TEST

1. A rodent population leaves an ecosystem due to decrease temperatures and increased rain. Which **best** explains why the rodents left?
   A. Increase in predation.
   B. Decrease in parasitism.
   C. Change in biotic factors.
   D. Change in abiotic factors.
2. How are UV radiation and the ozone layer related?
   A. UV radiation is produced by ozone.
   B. UV radiation absorbed by ozone.
   C. UV radiation is destroyed by ozone.
   D. UV radiation shields the Earth from ozone.
3. Which has the greatest impact on determining the characteristics of different biomes?
   A. Climate
   B. Longitude
   C. Number of lakes
   D. Diversity of plants
4. Which completely describes the components of an ecosystem?
   A. All the living things on Earth.
   B. A group of population living in one area.
   C. A community of organisms and their physical environment.
   D. All the organisms of one species that live together and interbreed.

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5. Which list shows the **correct** order of organization from simple to complex?
   A. Cell, community, organism.
   B. Cell, population, community.
   C. Organism, community, population.
   D. Community, population, organism.
6. Which is the broadest level in the study of ecology?
   A. Biosphere
   B. Ecosystem
   C. Population
   D. Community
7. Which is the **greatest** threat to biodiversity?
   A. Hunting and fishing without a license
   B. Introducing non-native species into a habitat
   C. Growing flowers and vegetables in a home garden
   D. Increasing the amount of land set aside for conservation
8. Which is a biotic in an Arkansas forest ecosystem?
   A. Rainfall
   B. Bacteria
   C. Minerals
   D. Nitrogen
9. Which biome has low temperatures, small amounts of precipitation, and permafrost?
   A. Taiga
   B. Forest
   C. Tundra
   D. Grassland
10. Which human activity is a major part of the carbon cycle?
   A. Building dams.
   B. Burning fossils fuels.
   C. Using inorganic fertilizers.
   D. Removing species from their habitat.

11. A deer tick on the blood of a host and can transmit bacteria that can cause disease. The deer tick and the host are an example of what type of symbiotic relationship?
   A. Predation
   B. Parasitism
   C. Mutualism
   D. Commensalism
12. Which is a contributor to a decrease in biodiversity?
   A. Wetlands restoration.
   B. Use of alternative fuels.
   C. Human population growth.
   D. Formation of nature preserves.
13. The total variety of all living things on Earth is described as
   A. An ecosystem
   B. Biodiversity
   C. A population
   D. Competition
14. One of the roles of decomposers in an ecosystem is to
   A. Convert protein and waste into ammonia and nitrates.
   B. Produce animal and plant protein from nitrates.
   C. Convert nitrogen from the air into ammonia and nitrates.
   D. Release nitrogen into the air from nitrates.
15. What is the function of autotrophs in the carbon cycle?
   A. To use oxygen to produce glucose.
   B. To take in excess water.
   C. To use carbon dioxide to produce glucose.
   D. To feed on herbivores.

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16. Two different populations of birds live in the same area and eat the same types of food. Which **most likely** describes the relationship between these two populations of birds?

A. Competition
B. Mutualism
C. Parasitism
D. Predator-prey

17. What will **most likely** happen if the human population continues to grow at current rates?

A. There will be fewer natural resources available for future generations.
B. There will be an increase in nitrogen levels in the atmosphere.
C. There will be a decrease in water pollution.
D. There will be an increase in the number of strong hurricanes.

18. The yucca plant releases a strong scent at night when the yucca moth is active. The yucca moth, attracted by the scent, gathers pollen from the yucca flower. The yucca moth then deposits her eggs and the pollen on another plant. This ensures that the plant will be cross-pollinated and that the yucca moth larvae have a steady food supply. Which relationship does this best illustrate?

A. Commensalism
B. Mutualism
C. Parasitism
D. Predation

19. The construction of a new coal-burning power plant would have the **greatest** impact on which environmental issue?

A. Depletion of the ozone.
B. Production of acid rain.
C. Release of radiation.
D. Increase in deforestation.

20. A large population of cockroaches was sprayed with an insecticide. A few of the cockroaches survived and produced a population of cockroaches that was resistant to this spray. What can **best** be inferred from this example?

A. Species will adapt no matter what the environment.

B. The environment has no effect on the survival of an organism.
C. Insecticides cause mutations that are passed on to the next generation.
D. Individuals with favorable variation survive and reproduce.

21. A field ecologist wants to determine the interactions of various populations of organism living in a large grassland field. Which method is **best** for conducting this study?

A. Consulting with local farmers about the diversity of organisms in the field
B. Dividing the field in half, walking over it, counting organisms, and then doubling the numbers.
C. Walking over the entire field, collecting the organisms, and then compiling a total.
D. Observing the behaviors of different types of organisms several times during the year.

22. If worldwide deforestation is not regulated, what could most likely result?

A. Acid rain could lower the PH of rivers and lakes to dangerous levels for aquatic life.
B. CO_2 levels in the atmosphere could increase and contribute to global warming problems.
C. Plants and animals could become better adapted to living in desert conditions.
D. Future generations of human could have an excess of wood and paper products.

23. In which populations does genetic drift most often occur?

A. In small populations.
B. In large populations.
C. In marine populations.
D. In terrestrial populations.

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24. For allele frequencies to remain constant in a population it is essential that

A. Mutations occur.

B. Genetic drift occurs.

C. Mating occurs at random in the population.

D. There is regular migration into the population.

ANWERS

1.	D	7.	B	13.	B	19.	D
2.	B	8.	B	14.	A	20.	D
3.	A	9.	C	15.	C	21.	D
4.	C	10.	B	16.	A	22.	B
5.	B	11.	B	17.	A	23.	A
6.	A	12.	C	18.	B	24.	B

CRITICAL THINKING

1. Describe the simplest stable ecosystem you can imagine.
2. What will happen to nitrogen cycle if all bacteria were absent? Explain your answer.
3. How do ocean currents affect climate on land?

REVIEW QUESTIONS ON UNIT SEVEN

1. A leaf adapted to a warm, dry climate is
(A) Large and thin (C) Large and thick
(B) Small and thin (D) Small and thick

2. Which group of vertebrates has the highest proportion of endemic species as compared to its total number of species found in Ethiopia?
(A) Birds (C) Amphibians
(B) Mammals (D) Reptiles

3. Which alternative contains only crops known to have been domesticated within Ethiopia? **(UEE, 2004 E.C)**
(A) *Guizotia abyssinica, Zea mays, Pisum sativum*
(B) *Vicia faba, Carica papaya, Musa paradisiaca*
(C) *Oriza sativa, Triticum aestivum, Solanum tuberosum*
(D) *Coffea arabica, Eragrostis tef, Ensete ventricosum*

4. What could be the main reason behind the currently observed slow or stable rate of population growth in the industrialized countries?
(A) Good family planning (C) Poor health conditions
(B) Increasing death rate (D) High rate of child death

5. What is the projected rate of loss of biodiversity every 50 years?
(A) 50% (C) 5%
(B) 20% (D) 10%

6. Which of the following statements is **true** about the nitrogen cycle? **(UEE, 2004 E.C)**
(A) Plants fix nitrates from atmospheric nitrogen.
(B) The nitrogen used by animals largely comes from plants.
(C) Nitrogen is consumed by bacteria and removed from the soil.
(D) Nitrogen-fixing bacteria reduce the total amount of available nitrogen.

7. Which stage in a primary ecological succession contains more biodiversity?
(A) The pioneer community (C) The second seral stage
(B) The third seral stage (D) The climax community

8. Which seral stage of ecological succession has the least biodiversity?
(A) The first seral stage (C) The third seral stage
(B) The second seral stage (D) The fourth seral stage

9. How do human beings increase biodiversity? **(UEE, 2004 E.C)**
(A) By reducing species richness.
(B) By increasing genetic variability.

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(C) By promoting habitat uniformity.
(D) By narrowing ecological variability.

10. Which of the following can be given as a good reason for finding large number of plant and animal species in Ethiopia? **(UEE, 2004 E.C)**
(A) Lack of ecological disturbance.
(B) Environment free from predators.
(C) Presence of many biomes and habitats.
(D) Good ecological and biodiversity management.

11. The tropical rain forest is best characterized by:
(A) High rainfall and low temperature.
(B) Low rainfall and high temperature.
(C) High rainfall and high temperature.
(D) Low rainfall and high temperature.

12. Which of the following are **major factors** to determine **climax community**?
(A) Temperature and precipitation.
(B) Grazing and browsing animals.
(C) Radiation and reflection.
(D) Predators and preys.

13. Which of the following terms mean stages in an ecological succession? **(UEE, 2004 E.C)**
(A) Pioneers (C) Seres
(B) Climaxes (D) Niches

14. What is the best term that refers to the maximum number of individuals that a given environment can support over a long period of time without degradation? **(UEE, 2003 E.C)**
(A) Density (C) Carrying capacity
(B) Climax community (D) Limiting factor

15. The process that fixes **carbon atoms** from an inorganic source (i.e. CO_2) into organic compound is:
(A) Respiration (C) Photosynthesis
(B) Combustion (D) Assimilation

16. In the nitrogen cycle, plants absorb nitrogen in the form of
(A) Nitrite (C) Nitrate
(B) Nitrogen dioxide (D) Ammonium

17. Which of the following factors **does not** affect the type of climax community that may develop in an area?

(A) Soil type (C) Precipitation
(B) Thunderstorm (D) Temperature

18. Which of the following marine biomes is unique habitat due to the mixing of salt water and fresh water?
(A) Oceanic, abyssal (C) Coral reef
(B) Oceanic, pellagic (D) Estuarine

19. Which of the following cultivated crops has **no** center of genetic diversity in Ethiopia?
(A) *Brassica carinata* (C) *Eragrostis tef*
(B) *Zea mays* (D) *Coccinia abyssinica*

20. We are concerned about **biodiversity loss** because it increases:
(A) Food insecurity and vulnerability.
(B) Health problems.
(C) Worsening social relations.
(D) All.

21. In Ethiopia, which of the following groups of the vertebrates is the most diverse in terms of species number? **(UEE, 2003 E.C)**
(A) Mammals (C) Reptiles
(B) Birds (D) Amphibians

22. Which of the following land biomes occupies a large part of Brazil, parts of equatorial west Africa and south-east Asia? **(UEE, 2003 E.C)**
(A) Acacia forest (C) Coniferous forest
(B) Deciduous temperate forest (D) Tropical rain forest

23. Which of the following nations has stable population?
(A) Ethiopia (C) Brazil
(B) Canada (D) China

24. At which stage of the population growth curve does the population size decline rapidly?
(A) Log phase (C) Lag phase
(B) Death phase (D) Stationary phase

25. The global human population is currently:
(A) Increasing at a steady rate.
(B) Increasing at an increasing rate.
(C) Increasing at a slowing rate.
(D) Decreasing at an alarming rate.

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26. Which of the following is a threat to the survival of small populations? **(UEE, 2003 E.C)**
(A) Habitat destruction
(B) Breeding in captivity
(C) Disease resistance
(D) Absence of competition

27. Soils in _____biome are poor because many of the minerals from the soil remain "locked up" in the huge trees.
(A) Hot desserts
(B) Tropical rain forest
(C) Cold deserts
(D) Tundra

28. The **global biosphere** is divided into:
(A) Niches
(B) Habitats
(C) Ecosystems
(D) Biomes

29. _____is the process of growing plants in a soil-free medium using water and the required nutrients.
(A) Bioengineering
(B) Biotechnology
(C) Hydroponics
(D) Hydrolysis

30. The process that releases inorganic **carbon dioxide** from organic compounds is:
(A) Respiration
(B) Photosynthesis
(C) Fossilization
(D) Assimilation

31. In what ways are the modes of nutrition of **animals** and **saprophyta** similar? **(UEE, 2003 E.C)**
(A) No enzymes are produced.
(B) Digestion is external to their cells.
(C) They build up proteins from light.
(D) They are autotrophic organisms.

32. Which one is a **biotic** factor that can influence a plant? **(UEE, 2003 E.C)**
(A) The PH of the soil
(B) A pollinating animal
(C) CO_2 concentration
(D) The amount of radiant energy

33. In the sulphur cycle, plants absorb sulphur in the form of:
(A) Hydrogen sulphide
(B) Sulphate
(C) Sulphur dioxide
(D) Protein

34. From the aquatic biomes, the one which is the **most diverse** of all marine habitats with many strata like the tropical rain forest is:
(A) Coral reef
(B) Estuarine
(C) Oceanic, abyssal
(D) Oceanic, pellagic

35. Which of the following cultivated crops has **no** center of genetic diversity in Ethiopia.
(A) *Guizotia abyssinica*
(B) *Coccinia abyssinica*
(C) *Ricinus communis*
(D) *Manihot esculentia*

36. At which phase of the population growth curve does the population number increase rapidly?
(A) Decline phase
(B) Lag phase
(C) Stationary phase
(D) Log phase

37. Which of the following nations has increasing population?
(A) Unites states (USA)
(B) Afghanistan
(C) United kingdom (UK)
(D) France

38. Which of the following practices has **indirect effect** on biodiversity?
(A) Deforestation
(B) Overgrazing
(C) Population growth
(D) Over-hunting

39. An area consists of five different species, and the number of each species is 7,2,5,4,3. What is the Simpson's index of diversity of the area?
(A) 4.12
(B) 6.12
(C) 3.12
(D) 5.12

40. Which of the following processes is not involved in recycling of carbon through ecosystems?
(A) Respiration
(B) Combustion
(C) Transpiration
(D) Photosynthesis

41. The bacteria breaking down the protein in dead organic matter resulting in the release of ammonium is:
(A) Ammonifying bacteria
(B) Nitrifying bacteria
(C) Denitrifying bacteria
(D) Nitrogen fixing bacteria

42. In the food chain:
Grass --- Nematodes ---Arthropods ---Birds,
Which group contains the highest biomass?
(A) Arthropods
(B) Birds
(C) Nematodes
(D) Grass

43. Which of the following is the characteristics of a **primary succession**?
(A) Presence of soil
(B) Presence of seed bank
(C) Speedyness of the process
(D) Starting from bare rock

44. Which of the following organisms are pioneer colonizers of bare rock?
(A) Shrubs
(B) Ferns
(C) Lichens
(D) Mosses

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45. Which of the following factors is the deriving energy in photosynthesis?
(A) Solar radiation
(B) Chlorophyll
(C) Water absorbed by roots
(D) Carbon dioxide from air

46. Which of the following crops has not been domesticated in Ethiopia?
(A) 'Teff'
(B) 'Anchote'
(C) Maize
(D) Coffee

47. Which of the following is **mismatched** between bacteria and their role in nitrogen cycle?
(A) **Nitro-somonas** - convert **ammonia** into **nitrite**.
(B) **Rhizobium** - convert **nitrogen gas** into **ammonia**
(C) **Nitrobacter** - convert **nitrite** into **nitrate**.
(D) **Decomposers** - convert dead organic body into proteins.

48. Which of the following processes decreases carbon dioxide concentration in the atmosphere?
(A) Transpiration
(B) Combustion
(C) Respiration
(D) Photosynthesis

49. Which of the following processes has a negative effect on the biodiversity of a given ecosystem?
(A) Reforestation
(B) Poaching
(C) Afforestation
(D) Habitat maintenance

50. Which of the following combinations means the same thing?
(A) Consumers - autotrophs
(B) Auttrophs - hetrotrophs
(C) Producers - authotrophs
(D) Hetrotrophs - producers

51. What is biodiversity?
(A) Variety of living things in an area.
(B) Plant populations in an area.
(C) Living and non-living things in an area.
(D) Portion of the earth that contains life.

52. In which biome of the world are epiphytes found as characteristic organisms?
(A) Tundra
(B) Tropical rain forest
(C) Taiga
(D) Desert

53. Which of the following is a **biotic** factor that affects size of populations in an ecosystem?
(A) Carbon dioxide
(B) Light intensity
(C) Temperature range
(D) Infection by pathogens

54. Which of the following is the **primary source** of energy for all the others.
(A) Decomposers
(B) Producers
(C) Carnivores
(D) Herbivores

55. Which of the following **is not** true about secondary succession?
(A) It may occur after communities are destroyed by mining.
(B) It occurs on bare land created by volcanic eruption.
(C) It is faster than primary succession.
(D) It may occur after communities are destroyed by fire.

Match the population growth checking (controlling) factors given under column "B" with their explanations under "A"

"A"	"B"
56. Death of organisms	(A) Disease
57. Movement of organisms out of an area	(B) Temperature and pH
58. Struggle for a scarce resource within a species	(C) Immigration
59. Birth/hatching of organisms	(D) Living space
60. Movement of organisms into an area	(E) Intra-specific competition
61. Ailment that may kill or reduce effeciency of an organism	(F) Emigration
62. Struggle for a scarce resource between species	(G) Predation
63. The process of being consumed by herbivores (if plants) or carnivores (if animals)	(H) Natality
64. Abiotic factor that induces territorial behaviour	(I) Mortality
65. Abiotic factors that may denature enzymes of an organism.	(J) Inter-specific competition

66. Which of the needs of living organisms are provided by the sun? **(CEE, General science, 1995 E.C)**
(A) Water & minerals
(B) Light & heat
(C) Carbon dioxide & minerals
(D) Oxygen & carbon dioxide

67. Which aspect of biodiversity is best covered in a study that assesses the abundance of species in a given area? **(CEE, General science, 1995 E.C)**

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(A) Species diversity
(B) Genetic diversity
(C) Ecosystem diversity
(D) Cultural diversity

68. Which problem of the global environment results from materials added to it by human activity (action)? **(CEE, General science, 1995 E.C)**
(A) Hunting in a protected area.
(B) Pesticide in a natural forest.
(C) Deforestation in a natural forest.
(D) Habitat loss.

69. Of the following biomes of the world, which one has the highest primary productivity? **(CEE, General science, 1995 E.C)**
(A) Tropical rain forest
(B) Temperate grassland
(C) Temperate forest
(D) Savannah

70. Detritivores always include:
(A) Bacteria and fungi
(B) Leaf litter
(C) Decaying long
(D) Animal dung

71. The first tropic level in a food web is composed of
(A) Primary producers
(B) Primary consumers
(C) Secondary consumers
(D) Tertiary consumers

72. Which of the following is important in the classification of the ecosystem of Ethiopia? **(CEE, General science, 1995 E.C)**
(A) Amount of rainfall
(B) Altitude
(C) Major vegetation types
(D) Major soil types

73. Choose the biome that shows the least fluctuation in temperature? **(CEE, General science, 1995 E.C)**
(A) Desert
(B) Temperate forest
(C) Temperate grassland
(D) Tropical rain forest

74. Terrestrial biomes are largely defined by:
(A) Wind & temperature
(B) Temperature & light
(C) Rainfall & wind
(D) Rainfall & temperature

75. Loss of biodiversity **would not** include:
(A) Worsening of health problems.
(B) Decreasing insecurity of food supply.
(C) Increased vulnerability.
(D) Increasing insecurity of food supply.

76. Which ecosystem in Ethiopia has forest coffee in it? **(CEE, General science, 1995 E.C)**

(A) Lowland dry forest ecosystem.
(B) Dry evergreen montane ecosystem.
(C) Afroalpine ecosystem.
(D) Moist evergreen montane ecosystem.

77. Which of the following terms best includes all the rest? **(CEE, General science, 1995 E.C)**
(A) Heterotrophs
(B) Omnivores
(C) Herbivores
(D) Carnivores

78. Which of the following plants are dominant in the **boreal forest** (taiga) biome?
(A) Lichenes & mosses
(B) Conifers
(C) Deciduous trees
(D) Shrubs

79. One of the following biomes in Africa is sorting large wild mammals such as elephants, giraffes and lions. **(EUEE, 2005 E.C)**
(A) The Congo rain forest.
(B) The rain forest of Western Ethiopia.
(C) The Savanna grassland.
(D) The Sahara desert.

80. Dominant plants of the tundra biome are:
(A) Long grasses
(B) Mosses, lichens, short grass
(C) Epiphytes
(D) Large trees

81. Identify the process that is expected to have a positive effect on the biodiversity of an area.
(A) Over-hunting
(B) Fuel wood collection
(C) Deforestation
(D) Planting trees

82. What is the important role played by micro-organisms such as bacteria and fungi in the ecosystem? **(EUEE, 2005 E.C)**
(A) Antibiotic production.
(B) Recycling nutrients.
(C) Forming organic substances.
(D) Supplying energy to ecosystem.

83. In which of the following aspects is the tropical rainforest biome poor? **(EUEE, 2005 E.C)**
(A) Species diversity
(B) Amount of sunlight
(C) Annual precipitation
(D) Soil fertility

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84. Which of the following is **NOT** usually true as an ecological succession progresses to advanced seral stages? **(EUEE, 2005 E.C)**
(A) More ecological niches are formed.
(B) Species become more diverse.
(C) The depth of the soil increases.
(D) Less populations are supported.

85. Which of the following pairs of terms means the same thing?
(A) Consumers-Heterotrophs.
(B) Producers-Consumers.
(C) Autotrophs-Heterotrophs.
(D) Autotrophs-Consumers.

86. Which of the following processes increases population size?
(A) Mortality
(B) Immigration
(C) Overgrazing
(D) Overgrazing

87. In which one of the four phases of population growth is the number of the population the highest? **(EUEE, 2005 E.C)**
(A) Lag phase
(B) Log phase
(C) Constant phase
(D) Decline phase

88. What is the type of community called when it has reached the final and most complex stage of a succession? **(EUEE, 2005 E.C)**
(A) Climax community
(B) Pioneer community
(C) Seral community
(D) Secondary community

89. Which of the following s an ecosystem? **(EUEE, 2005 E.C)**
(A) A tropical rain forest.
(B) The African continent.
(C) All the organisms in a given area.
(D) The non-living components of an environment.

90. What is the main reason for the high species richness of plants and mammals observed in Ethiopia? **(EUEE, 2005 E.C)**
(A) Efficient management of the ecological resources.
(B) Presence of several biomes within the country.
(C) Lack of predators.
(D) Lack of disturbance.

91. Which of the following is **not** true about **secondary succession**?
(A) It occurs after communities are destroyed by fire.
(B) It occurs after communities are destroyed by humans.
(C) It starts on bare land.

(D) It is quicker than primary succession.

92. Which of the following is the primary source of energy for all the others?
(A) Producers
(B) Decomposers
(C) Consumers
(D) The sun

93. Which of the following is a **biotic** factor that affects populations in an ecosystem?
(A) Competition
(B) Predation
(C) Parasitism
(D) All are biotic factors

94. Which of the following concepts contains all the others? **(EUEE, 2005 E.C)**
(A) Community
(B) Species
(C) Genus
(D) Population

95. If the age pyramid of a certain country is narrowing at the base, what does this tell about the trend of the population size of the country? The population size is: **(EUEE, 2005 E.C)**
(A) Growing fast
(B) Declining
(C) Increasing slowly
(D) Stablizing

96. What is **biodiversity**
(A) Variety of living things in an area.
(B) Plant populations in an area.
(C) Living and non-living things in an area.
(D) Portion of the earth that contains life.

97. Much of the rich agricultural lands in USA and Canada were originally:
(A) Savannas.
(B) Prairies (temperate grasslands).
(C) Temperate forests.
(D) Tropical forests.

98. Which of the following aquatic biomes are stagnant (standing)?
(A) Streams and rivers.
(B) Estuarine and lakes.
(C) Ponds, lakes and wetlands.
(D) Pelagic and abyssal zones of oceans.

99. Which of the following best describes conservation? Conservation is:
(A) The provision of ecosystem services without disturbing the fauna and flora.
(B) Sustainable use of natural resources without their depletion.

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(C) Converting forest areas into agricultural land gradually after many generations.

(D) Passing (inheriting) natural resources to the coming generation without touching them.

100. *Erica arborea* ('asta') and *Lobelia rhynchopetallum* ('jibira') are characteristic plants of_____biome.

(A) Riparian ecosystem.

(B) Lowland forest ecosystem.

(C) The desert and semi-desert ecosystem.

(D) The Afroalpine and sub-Afroalpine ecosystem.

Answer For Review Questions

1. D	21. B	41. A	61. A	81. D
2. C	22. D	42. D	62. J	82. B
3. D	23. B	43. D	63. G	83. D
4. A	24. B	44. C	64. D	84. D
5. D	25. C	45. A	65. B	85. A
6. B	26. A	46. C	66. B	86. B
7. D	27. B	47. D	67. A	87. C
8. A	28. D	48. D	68. B	88. A
9. B	29. C	49. B	69. A	89. A
10. C	30. A	50. C	70. A	90. B
11. C	31. B	51. A	71. A	91. C
12. A	32. B	52. B	72. C	92. D
13. C	33. B	53. D	73. D	93. D
14. C	34. A	54. B	74. D	94. A
15. C	35. D	55. B	75. B	95. B
16. C	36. D	56. I	76. D	96. A
17. B	37. B	57. F	77. A	97. B
18. D	38. C	58. E	78. B	98. C
19. B	39. D	59. H	79. C	99. B
20. D	40. C	60. C	80. B	100.D

Unit Eight

Genetics

Genetics - is the science of **inheritance** (heredity) that deals with the way in which **traits** of an organism are **transmitted** from **generation** to **generation**. It is also concerned with **units of inheritance (genes)**. The **structure, change** and **dynamics** of **genes** at **molecular, organismal** and **population** levels are also covered.

8.1 Genetic crosses

Chromosomes, genes and alleles

Chromosomes - are long strands of **DNA** that have been wrapped around **histone** (protein) molecules. Each chromosome is then coiled on itself. A chromosome in non-dividing cell is **loosely coiled, long** and **thin**, and is called **chromatin** ('colored material'). Chromosomes carry **genes**.

Gene - is a section of DNA (or chromosome) that determines a specific feature of an organism.

Example: Height, color, earlobe attachment, seed shape, flower position, etc.

Alleles - are **alternative forms** of a **gene**. They occupy the **same locus** on homologous chromosomes.

Example: The gene controlling plant height may have **alleles** of **shortness** and **tallness**, the gene controlling our blood group has **3 alleles** (i.e the "A" allele, "B" allele and "O" allele).

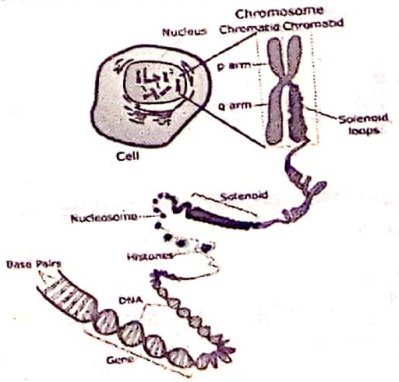

Figure 8.1 The structure of a chromosome. Chromosomes are located in the nucleus. A chromosome forms an exact copy of itself when the cell is ready for division, and hence two chromatids are observed.

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Sexually reproducing organisms receive **one set** of chromosomes from the **male** parent, and the **second set** from their **female** parent.

Example: In each cell of humans, there are 46 chromosomes (23 chromosomes from mother and 23 from father). Those coming from mother are called **maternal**, while those from father are **paternal**.

Paternal and maternal chromosomes come together during fertilization and form **homologous pairs**. Homologous chromosomes carry genes controlling the same trait/feature at the same position/locus on them.

Example:

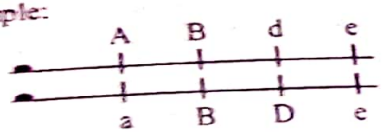


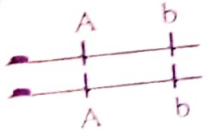
Figure 8.2 Homologous chromosomes with 4 loci (a gene may have similar or different alleles at a locus: genes at the first and third loci have dissimilar alleles; those at the second and last loci have similar alleles of their respective genes. Genotypes at the loci are Aa, BB, Dd and ee, respectively.

Homologous chromosomes - **structurally** and **functionally** similar chromosomes, one coming from each parent in eukaryotes. They are functionally **similar** because they carry **similar genes** at the **same locus**.

Locus (loci, pl.) - position of a particular gene on DNA (or chromosome).

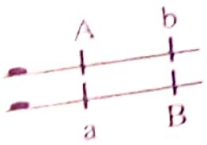
Homozygous organism - has **similar alleles** for a **particular gene** at a locus on homologous chromosomes.

Example:



Genotypes: AA, bb

Heterozygous organism - has **different alleles** for a **particular gene** at a locus on homologous pairs.



Genotypes: Aa, Bb

Genotype - **pair** of **alleles** for a **particular gene** in an organism (i.e. possessed by the organism). It is the genetic composition of an organism for a given trait.

Example: AA, Aa, aa, TT, Tt, tt, etc.

Phenotype - the observable characteristics of an organism resulting from the interaction between the **genotype** and **environment**.

Example: tall, dwarf, red, wrinkled, round, etc.

Monohybrid cross

A monohybrid cross (breeding) - is a process that is concerned with one trait/feature at a time. Two organisms are allowed to mate together (i.e. hybridized) naturally or artificially to see the effect of the process on one trait (Example: yield) of the organism. It may be conducted for research purposes, or to improve features of an organism.

Gregor Mendel (1822-1884)

Mendel was born to a **peasant parents** and **educated** in a **monastery**. He studied **science** and **mathematics** at the **university of Vienna, Austria**. Unfortunately, he **failed** an examination for a **teaching certificate**. Returning to the monastery, he spent the rest of his life as a **monk**, and eventually became an **abbot** (head of a monastery). Mendel conducted the first **quantitative** study of inheritance in his garden using garden pea (*Pisum Sativum*). He is considered the '**father of genetics**' as the results of his **experiments** have laid down **foundation** for most of the **rules** of heredity. Mendel chose garden pea because it showed 7 easily **distinguishable** contrasting **traits**, is **small, easy** to **grow**, produces large **number** of offspring and matures **quickly** relatively. Its **flowers** are also bisexual containing both the stamens and **pistil** within a flower where **self-fertilization** is **immediate** if undisturbed.

Table 8.1. The 7 contrasting traits of pea studied by Mendel and the F_2 results

No.	Traits	Variants		F_2 generation results		Ratio
		Dominant	Recessive	Dominant	Recessive	
1.	Plant height	Tall	Short	787	277	2.84:1
2.	Flower color	Purple	White	705	224	3.15:1
3.	Flower position	Axial	Terminal	651	207	3.14:1
4.	Pod color	Green	Yellow	428	152	2.82:1
5.	Pod shape	Smooth	Constricted	882	299	2.95:1
6.	Seed color	Yellow	Green	6022	2001	3.01:1
7.	Seed shape	Round	Wrinkled	5474	1850	2.96:1

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Mendel's techniques

1. First, Mendel allowed **self-pollination** and **fertilization** of different lines of garden pea. He took those **lines** which produced the **same trait** (phenotype) generation after generation during **self-fertilization**. He called these **'pure breeders'** or **'true breeders'**.
   Pure (true) breeders - are organisms that produce the **same feature** (phenotype) generation after generation during reproduction.
   Example: A **tall** plant self-pollinated and producing **all tall progeny**, a **black man** married to a **black woman** producing **all black children** generation after generation. The **plant** and the **couple** here are **pure-breeding** for the **genes** controlling plant-height and **skin color**, respectively.
2. He then crossed **pure-breeders** for the **7 contrasting characteristics**. Pure tall and pure short plants, for example, were crossed. For this, he removed **stamens** from flowers of the tall plants before maturity. This prevented self-pollination.
3. At flower **maturity**, he **transferred**, **pollen** from **flowers** of the short **plants** to the **stigma** of **tall plants**. He reciprocated his experiment by removing **stamens** from **short plants** and **dusting** them with pollen **grains** from **tall plants**.
4. When the plants matured, he collected the **pods** and extracted seeds. These are the **first filial generation** (F_1) resulting from the first cross.
5. Mendel grew again all the seeds from F_1 generation, and noted the plants at maturity. All of them were **tall**. Mendel called the trait expressed in F_1 as **dominant**, and the one masked, **recessive**. For each pair of contrasting traits Mendel studied, one proved to be **dominant**, and the other, recessive. In this case **tallness** is **dominant** and **shortness** is **recessive**.
6. The F_1 plants were allowed to **self-pollinate**, and produce seeds. These seeds were collected and planted to produce the **second filial generation** (F_2). The second filial generation exhibited nearly 3 **dominant** to 1 **recessive trait** in all Mendel's experiments (table 8.1.).

The first filial generation (F_1) - refers to the offspring from cross between different parents.
The second filial generation (F_2) - are the offspring of F_1.
Trait - is **feature** of an organism determined by its **gene(s)**.
Dominant allele - is the **allele (trait) expressed** in a **heterozygous** organism.
Example: **tallness** in garden pea.

Recessive allele - feature (allele) of an organism that is **masked** in a **heterozygous** state. It is expressed only when it exists in homozygous state.
Example: shortness in garden pea.
Conventionally, genetic traits are represented by a **letter** standing for the **recessive** or **dominant** allele. The **dominant** trait is designated by **upper case**, and the **recessive** by a **lower case** letter. In **heterozygous state**, the **dominant** allele is **written** first.
Punnett square - is a simple **table** that shows the **possible combinations** of **gametes** of **two parents**. It represents the **possible genotypes** of the **offspring** so produced, and the **ratio** in which they will occur. It was **named** after the **English geneticist Reginald Crundall Punnett**.

Examples of letter representations
Plant height: tall (**TT**), short (**tt**)
Flower color: purple (**PP**), white (**pp**)
Seed color: yellow (YY), green (yy)
Pod color green (GG), yellow (gg), etc.

Examples of Mendel's experiment

In this case P usually stands for parents, **F** for filial generations.

P_1: **Tall plants** (TT) X **Short plants** [both are pure breeders] (tt) [genotype]

Gametes: (T), (t)

F_1: **Tt** (all are tall, but heterozygous)

P_2: (selfing F_1) Tt X Tt

Gametes: (T), (t) (T), (t)

Here, **Punnett square** should be used to work out the **possibilities**. Gametes from one parent are placed in the **upper row**, and those from the **second parent** in the **left column**.

F_2:

♂ \ ♀	T	t
T	TT	Tt
t	Tt	tt

♀ = Female parent
♂ = Male parent
⇒ **Genotypic ratio**: 1TT : 2Tt : 1tt
Phenotypic ratio: 3 tall: 1short

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Mendel's explanations of his results

1. Every **trait** of an organism is controlled by a **pair of heritable factors**, now known as **genes**. The genes may take **two** or **more** forms called **alleles**.
2. If the **two alleles** of a gene in an organism are **different**, one is **dominant** and the other is **recessive**. The **trait** controlled by the **dominant** allele is **expressed** in the organism's **appearance** or **physiology**. The recessive allele is **masked**.
   Not all alleles of **genes** show **dominance-recessiveness**, of course. But, the traits Mendel studied exhibited the situation.
3. Genes pass from generation to generation by **gamete cells** which are the only **physical link** between generations.
4. The **pair** of **alleles** of a **gene** controlling any **trait** of an organism are **separated** during **gamete** (sex cell) **formation**. Each gamete contains only **a copy** of the **pair** of **alleles**. This is **Manedel's first law (the law of segregation)**. The gametes **fuse randomly** at **fertilization**.

Gametes are sex cells that pas characteristics of organisms from generation to generation (i.e. from parents to offspring). They are **specialized** in that they carry only **one set** of **chromosomes** (i.e. **haploids**).

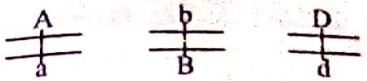
Example: Sperm and **egg** cells in **animals**, and **pollen grains** and ovules in **plants**.

After **fertilization**, ordinary body cells are formed by fusion of sperm and **egg cells**, or **pollen** and **ovules**. Body cells are **diploid** cells containing full set of chromosomes (i.e. one set from each parent).

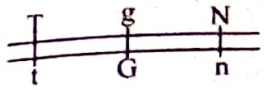
Haploid cells(n) - are cells that carry only **one set of chromosomes**.
Example: gamete cells contain 23 chromosomes only in man.
Diploid cells(2n) - are cells which contain **two sets of chromosomes**.
Example: human body cells contain **46** chromosomes.

Mendel's second law

Also known as **the law of independent assortment**, states that the inheritance of one trait does **not depend** or **affect** inheritance of the other provided that the two traits are controlled by **genes residing on different chromosomes**. That is, the alleles of one gene segregate independently of the alleles of another gene controlling a different feature. This is **not true** for **linked genes** that are found on the same chromosome. It was true for all the 7 traits studied by Mendel because they were **non-linked**.

Example: 1) 

Non-linked genes on different chromosomes

2) 

Linked genes on single chromosomes

In the above first example, 3 genes at 3 loci on different chromosomes are shown. They control 3 different features of an organism. Alleles of the 3 genes segregate independently during gamete formation. Gametes containing **AbD, aBd, Abd, aBD, abd, ABD, ABd** and **abD** can randomly be formed. In this case, the **dominant** allele from the **first gene** may combine with the **dominant** or **recessive** allele from the other genes, and vice versa. But, in the **second example** where the 3 genes are **linked** on the **same chromosome**, only **two types** of **gametes** (**TgN** and **tGn**) can be formed. The alleles do not assort independently, i.e. the inheritance of the dominant allele from the first gene means inheritance of the recessive allele from the second gene, and dominant allele from the third gene, and vice versa. Therefore, the inheritance of one allele, in this case, is somehow affected by inheritance of the other. So, **Mendel's 2nd** law does not apply for **linked genes**.

Illustrative Examples

Using punnett square work out the results of the following crosses. Make up your own symbols.

1. Across between heterozygous purple-flowered plants and white flowered plants.

Explanation:

- From table 8.1., **Purple** flower color is **dominant**, while **white** is **recessive**. So upper case letter is given to purple, and lower case letter to the white color. Let us represent purple with P, and white with p.

Pp X pp (Parents)

Ⓟ, Ⓟ Ⓟ, Ⓟ (Gamets)

♂ \ ♀	P	p
p	Pp	pp
p	Pp	pp

Result

Genotypic ratio: 2Pp:2pp 1:1 50%:50%

$\frac{2}{4}$Pp:$\frac{2}{4}$pp $\frac{1}{2}$:$\frac{1}{2}$ 0.5:0.5

Phenotypic ratio: 2 purple: 2 white

$\frac{2}{4}$:$\frac{2}{4}$ $\frac{1}{2}$:$\frac{1}{2}$ 0.5:0.5 1:1 50%:50%

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The purple flowered offspring in the above cross are **heterozygous**, not pure-breeders; Both ratios are similar here.

2. A cross between white-flowered plants.

Explanation:

- Here both parents are **pure breeders** for the **recessive trait** (white flower color).

} pp X pp (parents)

(p) (p) (only one type of gametes from each parent)

pp (offspring)

Whether we use Punnet square, or not, the outcome is **white-flowered** plants. So, the **phenotypic** and **genotypic ratios** are the **same**, and **100%** **white** and **pp**, respectively.

3. Two **heterozygous** tall plants are crossed.

Explanation:

- Regarding plant hight, **T** and **t** represent **tallness** and **shortness**, respectively because tallness is dominant and shortness is recessive.

} Tt X Tt (parents)

(T), (t) (T), (t) (gametes)

♂ \ ♀	T	t
T	TT	Tt
t	Tt	tt

Genotypic ratio of offspring

1TT:2Tt:1tt } 1/4 : 2/4 : 1/4 } 1:2:1 } 25%:50%:25%

} 0.25 : 0.5 : 0.25

Phenotypic ratio of offspring

3tall:1short } 3/4 : 1/4 } 3:1 } 75%: 25% } 0.75:0.25 Out of the **75%** tall plants, **25%** are **pure-breeders**, while **50%** are **heterozygous tall.**

4. A cross between a **heterozygous green** poded plant and a **yellow-poded** plant.

Explanation:

- Green pod is dominant over yellow pod (table 8.1). We represent green pod by 'G', and yellow by 'g':

Gg X gg (parents)

}

(G), (g) (g), (g) (gametes)

♂ \ ♀	G	g
g	Gg	gg
g	Gg	gg

Genotypic ratio of offspring

2/4 Gg: 2/4 gg } 1:1 } 50%:50%

Phenotypic ratio of offspring

50% offspring with green pods and 50% with yellow pods.

Remember!

Parents showing **similar phenotypes** may produce **offspring** with a **different phenotype**! This is possible if both parents are **heterozygous** for the trait under question

In the illustrative example 3 above, both parents are **tall phenotypically**, but **heterozygous genotypically**. So, they have **25%** probability of producing short plants.

Example: Ear lobe attachment - there are two alleles of the gene controlling this feature: alleles for **attached** and **free ear lobes**. Free ear lobe is **dominant** over **attached**. Lets represent free ear lobe with 'F', and attached ear lobe with 'f'.

Alleles: Free ear lob (F)

Attached ear lobe (f)

Genotypes: Free (FF)

Attached (ff)

If crossing starts from pure-breeders, the process proceeds as follows.

Free ear lobe (FF) X Attached ear lobe (ff) (parents)

(F) (f) (gametes)

Ff (all offspring with free ear lobe)

If these offspring are cross-bred, they will produce **25%** offspring with **attached** ear lobe which is not their own feature.

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Ff X Ff (parents)

Ⓕ, ⓕ Ⓕ, ⓕ (gametes)

♂ \ ♀	F	f
F	FF	Ff
f	Ff	ff

Genotypic ratio

1FF:2Ff :1ff } 25%:50%:25%

Phenotypic ratio

$^3/_4$ Free ear lobe: $^1/_4$ attached ear lobe } 3:1 } 75%:25%

This ratio occurs in all organisms when two heterozygous parents are cross-bred.

Genetic ratios shown by **Punnett square** (genetic diagrams) are **probabilities** that **predict** the **possible outcomes** of a cross between two parents. They may **not be realized** with small number of offspring as the **laws of chance affect** them. They are **highly likely** to be **realized** with **large number** of offspring

Test cross (Back cross)

Test cross is breeding experiment that helps us to **determine** the genotype of an organism showing **dominant phenotype** because the organism may be **true-breeding** or **heterozygous** for a trait. For this, it is crossed with a known **homozygous recessive** organism.

Example: Unknown tall garden pea plant (TT? or Tt?).

Crossed with known short plat (tt).

Tall plant (?) X tt

Possibilities

TT X tt Tt X tt

♂ \ ♀	T	T
t	Tt	Tt
t	Tt	Tt

Offspring: **100%** tall

♂ \ ♀	T	t
t	Tt	tt
t	Tt	tt

Offspring: **50%** tall and **50%** short

Conclusion: if the **outcome** of a test cross is **100% dominant phenotype**, it means that the **unknown** was **homozygous** for the **dominant allele** (TT). If the back cross results in **50% dominant** and **50% recessive phenotypes**, the **unknown** must be **heterozygous** (Tt) for the trait in focus.

To tell whether an allele is **dominant or recessive**,

1. We have to look at the **offspring** resulting from a **specific pair of parents**.
2. The information gained from **pedgree** may be used, especially for **humans**, because experimental **crossings** with humans are **immoral**.

Pedgree - is a **chart** that shows the **traits** of **humans** over several **generations**.

In a **pedgree**, different **symbols** are **used**.

Example: ◻ = male, ◯ = female, shaded symbols usually represent organisms showing the trait, **horizontal line** between a **male** and **female** shows marriage, and **vertical lines** connect children to their parents.

Sometimes, it is possible to work out **genotypes** from a **pedgree**. The 2 parents in the 2nd generation of figure 8.3, for example, have Widow's peak, but they produced 2 daughters (one with Widow's peak, and the other without). The only way this can happen is, if the parents are **heterozygous** for Widows peak. If we are sure that they are **heterozygous** from this evidence, and if they show the trait, it means that the **trait** (Widow's peak) is **dominant**.

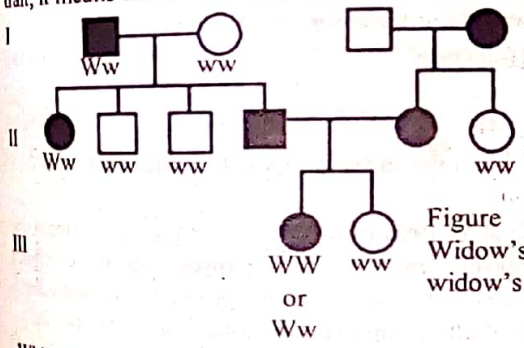

Figure 8.3 A genetic pedgree for Widow's peak (shaded = show widow's peak, unshaded = normal)

Widow's peak - is a situation in which the **hair** of an individual **grows** slightly forward on to the **center of forehead**, and forms a peak on the background.

Example: Tadele Gemechu (an Oromo singer).

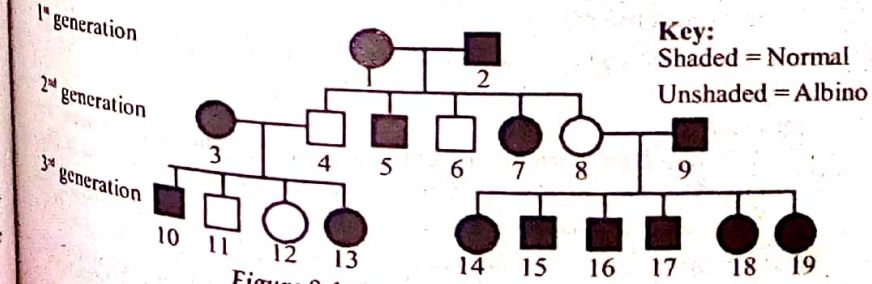

Figure 8.4. A pedgree for albinism

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Albinism - is a genetic **disorder** in which a man or an animal is born without the ability to produce the natural **coloring pigment (melanin)** in the **skin**, **hair** and the **eyes**. The skin and hair of such an individual (an **albino**) becomes white while its **eyes** become **pink**.

Illustrative Examples

Using figure 8.4, answer the following questions

5. What does the relationship between individuals 1, 2, & 6 prove?

Explanation:

- The fact that an **albino** (individual 6) is born to normally pigmented parents (individuals 1 & 2) indicates that the parents are **heterozygous**. A **heterozygous** is normal, in this case, if and only if the normal allele is **dominant**. So if we assign '**A**' for the normal allele, '**a**' will represent the **recessive** allele for albinism.

Individuals	Genotypes
1 & 2	**Aa**
6	**aa**

6. Why must the **genotypes** of all the **individuals** with **albinism** be '**aa**'?

Explanation:

- The allele for albinism is already proved to be recessive (from the pedgree). For a recessive trait to show (express) itself, its alleles should be found in pairs (i.e. be **homozygous**) in an individual. Therefore, **genotype** of all organisms with **albinism** has to be '**aa**'.

7. From where have the people with albinism inherited each of the '**a**' allele?

Explanation:

- Albino individuals inherit each of their alleles from both parents (one from each parent).

8. What allele will each of their children inherit from the individuals with albinism?

Explanation:

- Any **children** born to **albino** will always **inherit** the allele for **albinism**. Ofcourse, a child born to an **albino** may be **normal** if his/her other parent is **normal**.

9. What are the genotypes of individuals-3 and 17?

Explanation:

- Individual 3 is normal, but she has produced an **albino** (example individual-11) with her albino husband (4). She must have contributed

an allele of **albinism** to her albino child. For her, to pass on the recessive allele of albinism and be normal at the same time, she has to be **heterozygous** for this trait. Insividual-17 is also normal, but has an albino mother. He has to inherit an allele of albinism from his mother and a normal allele from his father. Therefore, he is a **heterozygote** for albinism. So, the genotypes of individuals-3 and 17 is '**Aa**'.

10. What is the genotype of individual-9?

Explanation:

- Individual-9 has produced normal offspring, and himself is also normal. We can be sure that he has one normal allele, but we can not be sure if he has a recessive allele or not because we do not see any albino offspring produced from him in the 3rd generation. So, he may be '**AA**' or '**Aa**'.

Dihybrid inheritance

Dihybrid inheritance - is the inheritance of **two pairs** of contrasted **characteristics (two genes)** considered at the same time.

Example: Seed color and seed shape, flower color and flower position, plant height and seed shape, or pod shape and pod color.

Let us take **plant height** and **seed shape**, and see what Mendel demonstrated.

P_1: Tall plants with round seeds X Short plants with wrinkled seeds

(TT RR) (tt rr)

Gametes: (TR) (tr)

0 A **gamete** is formed by taking **one allele** from **each gene**, and **only one** type of gamete is formed from **homozygous** organisms.

F_1: Tt Rr (all tall and round seeded)

P_2: Self-fertilization of F_1 generation

(Tt Rr) (Tt tr)

(TR), (Tr), (tR), (tr) (TR), (Tr), (tR), (tr)

Four different gametes can be formed from a heterozygote dihybrid since there is no linkage between the two genes in this case. An allele from the first gene can freely combine with any allele from the second gene to form a gamete.

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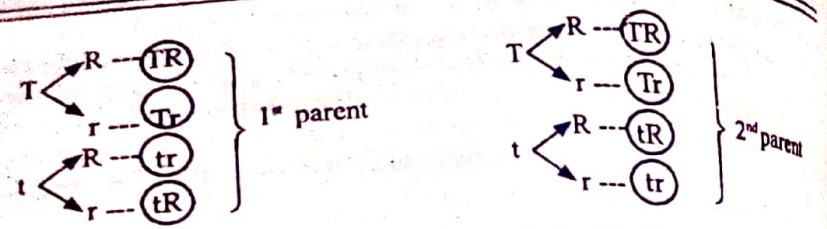


A Punnet square that accommodates 4 gametes **vertically and horizontally**, must be used to workout the outcome of the cross between the F_1 individuals.

F_2:

♂ \ ♀	TR	Tr	tR	tr
TR	TT RR 1	TT Rr 2	Tt RR 3	Tt Rr 4
Tr	TT Rr 5	TT rr 6	Tt Rr 7	Tt rr 8
tR	Tt RR 9	Tt Rr 10	tt RR 11	tt Rr 12
tr	Tt Rr 13	Tt rr 14	tt Rr 15	tt rr 16

Figure 8.5. A dihybrid cross for garden pea plant height and shape of its seeds.

There are **4 possible genotypes** in F_2 generation with the following genotypes and **probabilities**.

Phenotypes	Genotypes	Sites of genotypes in figure 8.5.	Proportion
Tall-round	T-R-	1,2,3,4,5,7,9,10,13	9/16
Tall-wrinkled	T-rr	6,8,14	3/16
Short-round	ttR-	11,12,15	3/16
Short-wrinkled	ttrr	16	1/16

Figure 8.6. Summery of a dihybrid cross

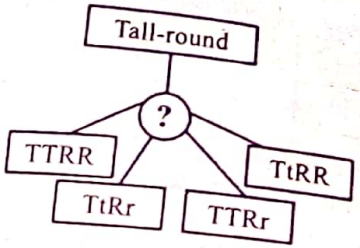
Here the **genotypic ratio** is not as such important but, the **phenotypic ratio** is **9:3:3:1**; i.e. **9 tall-round: 3tall-wrinkled: 3short-round:1 short-wrinkled**
This ratio is always found when ever a dihybrid cross is made between two **parents heterozygous** for **both traits** in focus.

Test cross for a dihybrid

The **genotype** of an organism showing **dominant phenotype** for **two genes** (traits) can be of **4 types**.

Example: Tall plant with round seeds may be:
- Y **Homozygous** for **both** features (**TTRR**), or
- Y **Heterozygous** for **both** features (**TtRr**), or
- Y **Homozygous** for the **first trait** and **heterozygous** for the **second trait** (**TTRr**), or
- Y **Heterozygous** for the **first gene** and **homozygous** for the **second gene** (*TtRR*).

So, how can we acertain ourselves about which **genotype** is really possessed by our plant? We need to cross our **unknown plant** with a **double recessive** (**ttrr**) plant (i.e short-wrinkled plant).



The test crosses

a) If the plant is **TTRR**, it produces (TR) gamete only.
TTRR X ttrr --- (TR) x (tr) --- TtRr
} All will show **dominant phenotype** for both traits.

b) If the plant is **TtRr**, it can produce 4 types of gametes:
(TR), (Tr), (tR) & (tr).
The test plant (ttrr) produces only one type of gamete, (tr).

♂ \ ♀	TR	Tr	tR	tr
tr	TtRr	Ttrr	ttRr	ttrr

Here, all the 4 possible phenotypes are produced in a 1:1:1:1 ratio.
i.e. Tall-round (TtRr) - 25%
Tall-wrinkled (Ttrr) - 25%
Short-round (ttRr) - 25%
Short-wrinkled (ttrr) - 25%

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c) If the plant is **TTRr**, it produces 2 types of gametes only. These are: (TR) & (TR).

♂ \ ♀	(TR)	(Tr)
(tr)	TtRr	Ttrr

Two phenotypes are produced in a 50%:50% or 1:1 ratio.
==: Tall-round (50%)
Tall-wrinkled (50%)

d) If the plant is **TtRR**, it can produce 2 types of gametes, (TR) & (tr).

♂ \ ♀	(TR)	(tR)
(tr)	Tt Rr	tt Rr

Two phenotypes in a 1:1 ratio.
==: Tall-round (50%)
Short-round (50%)

Conclusion: After a dihybrid test-cross, if the offspring show:

1) **Dominant phenotype** for both genes, the **unknown** is homozygous for both traits (TTRR).
2) All the **4 possible phenotypes**, in a 1:1:1:1 ratio the **unknown** is **heterozygous** for both features.
3) **Dominant phenotype** for the **first gene**, but some dominant and some **recessive phenotypes** for the **second gene**, the **unknown** is homozygous for the **first gene** and **heterozygous** for the **second gene**.
4) **Some dominant** and some recessive phenotypes for the **first gene**, but **dominant** phenotype for the **second gene**, the **unknown** is heterozygous for the **first gene** and **dominant homozygous** for the **second gene**.

Types of dominance

There are **three** types of **dominance**

1) **Complete dominance** - is a situation is which **one of the pair of alleles of a gene** in a **heterozygous** organism completely **masks** the other, and **expresses** itself.
Example: The allele for **tallness, roundness** and **purpleness** are completely dominant over **shortness, wrinkledness** and **whiteness**, respectively in garden pea height, seed shape and flower color.
2) **Incomplete dominance** - both alleles produce their **gene products** (**Example**: pigment) to some extent, but **not enough** for **dominance**, and an **intermediate phenotype** (Example: **color**) is produced.
Example: Snapdragon flower color: if **true-breeding red flowered plant** is crossed with **true-breeding white-flowered** plant, a **pink-flowered plant** is produced.

RR (Red) X rr (White) --- Rr (Pink)
Rr X Rr

Gametes: (R) (r) (R) (r)

♂ \ ♀	(R)	(r)
(R)	RR	Rr
(r)	Rr	rr

Genotypic ratio
1RR:2Rr:1rr
Phenotypic ratio
1 red:2 pink:1 white

3) ***Codominance*** - both alleles of a **gene** produce their **gene product** equally, and become **equally dominant**.
Example1: If **pure-breeding red cattle** is crossed with **pure breeding white cattle**, the **heterozygote** so produced will be **roan** ( ).
Let **R** represent the allele for **redness**, and **W white** allele.
RR (Red) X WW (White) --- RW (Roan)
Example2: '**A**' & '**B**' of the 3 alleles (A,B,O) of our **blood-group** system are also **codominant**.

A gene may have more than 2 alleles, and **multiple allele inheritance** is involved in this case.
Example: Our ABO blood system has 3 alleles; 'A', 'B' and 'O': 'I' represents the gene **locus** for blood-type, and stands for **isohaemagglutinogen**.

I^A, I^B Codominant

I^O - recessive to both 'A' & 'B'

Table 8.2 possible genotypes and phenotypes of ABO blood system

	Genotypes	Phenotypes (Blood-types)
1	I^AI^A, I^AI^O	A
2	I^BI^B, I^BI^O	B
3	I^AI^B	AB
4	I^OI^O	O

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Two parents with blood groups A & B can produce children with all the 4-blood types (A,B, AB & O) if they are heterozygous.

♂ \ ♀	A	O
B	AB	BO
O	AO	OO

The physical basis of inheritance

The physical basis of inheritance are **genes** which are **parts** of a **chromosome**. The chromosomes are found and transmitted from generation to generation in **gametes**. Gametes are produced when special cells in sex-organs (i.e. in ovary and testis, or anther) of an organism divide by **meiosis**.

There are 3 main outcomes of meiosis

1) **Four daughter cells** are produced.
2) Each daughter cell contains **half** the chromosome number found in the original cell (i.e. **haploid** cells with 'n' chromosome number produced)
3) The daughter cells show **genetic variation**.

Stages of meiosis

Meiosis is a **reduction** division in which the chromosome number is halved from the **diploid** number (2n) to the **haploid** number (n). Whenever a cell is ready for division, each chromosome **replicates** and **exists as a pair** of chromatids joined together by a **centromere**. At this stage, the cell is a **tetraploid** (4n), and the stage is called **inter-phase**. Here the chromosomes are loosely coiled threads called **chromatins**. The **centrioles** also replicate. Then, 2 meiotic cell divisions (meiosis-I and meiosis-II) will take place.

Meiosis-I

- **Homologous** chromosomes **separate**; 2 haploid cells form.
- Has **4** major stages: **Prophase-I**, **Metaphase-I**, **Anaphase-I** and **Telophase-I**.

Prophase-I

- Chromosomes shorten and become visible as single structures.
- **Homologous chromosomes** pair up (=**synapsis**), and form a **bivalent (a tetrad)**.

Homologous chromosomes:

- Are **paternal** and **maternal** pairs.
- Have the **same length**.
- Their **centromeres** are at the same position.
- Usually, they have the same number of genes arranged in the same order.

- **Chromatids** of **homologous chromosomes** join at several points along their length forming **chiasmata** (chiasma (pl.), a cross). This is called **crossing over**, and **different chromatids** from the **homologues exchange** some **piece**, and thereby, genetic material. This brings about **genetic recombination**.
- The **centrioles** (if present) move to **opposite poles**.
- The **nuclei** and **nuclear membrane** disappear.
- The **spindle fibers** form.

Metaphase-I

- **Bivalents** (**homologues**) line up on the **center** of the **cell** (**equator** of the **cell/metaphase plate**).
- Spindle fibers attach to each chromosome.
- **Spindle fibers** (**microtubules**) are **arranged** (organized) by **centrioles**.

Anaphase-I

- **Homologous chromosomes** are **pulled apart** to **opposite poles** by the **contraction** of the **spindle fibers**.

Telophase-I

- Two **nuclei** form, each with **haploid chromosome number**. But, each chromosome is still composed of **two chromatids**.
- If **crossing over** has occurred, these **chromatids** in the **same chromosome** are **not genetically identical**.
- **Chromotids** must be **separated** in **meiosis-II**.
- Spindle fibers disappear.
- Depending on the type of the cell, **cytokiesis** (cytoplasmic division/ actual cell division) may occur. In such cases, **interphase-II** may follow. In **interphase-II**, there is no further **DNA** replication. Chromatids uncoil.

Interphase-II

- Usually occurs in **animal cells** and **some plant cells**.
- **Cleavage** (in animal cell) or **cell wall formation** (in plant cells) takes place if **interphase-II** occurs. But, in many plants, there is no telophase-I, cell wall formation or intephase-II. In such cases, the dividing cell directly passes from **anaphase-I** to **prophase-II**.

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Chromatids of **homologous chromosomes** join at several points along their length forming **chiasmata** (chiasma (pl.), a cross). This is called **crossing over**, and **different chromatids** from the **homologues** **exchange** some **piece**, and thereby, genetic material. This brings about **genetic recombination**.

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Telophase-I

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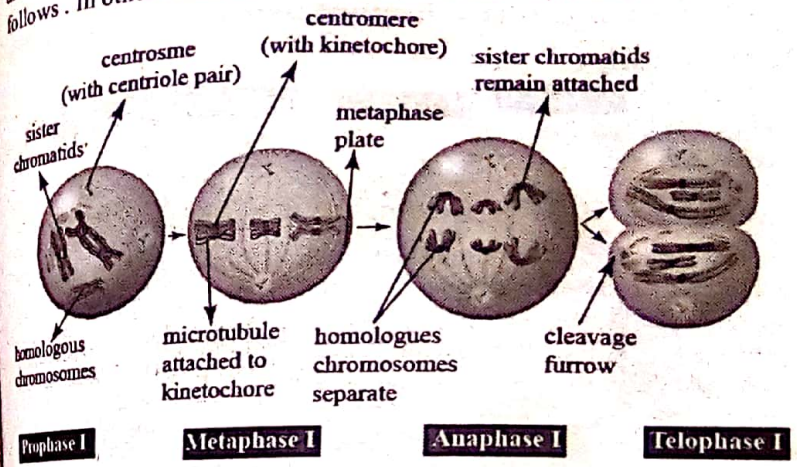
- **Chromotids** must be **separated** in **meiosis-II**.
- Spindle fibers disappear.
- Depending on the type of the cell, **cytokiesis** (cytoplasmic division/ actual cell division) may occur. In such cases, **interphase-II** may follow. In **interphase-II**, there is no further **DNA** replication. Chromatids uncoil.

Interphase-II

- Usually occurs in **animal cells** and **some plant cells**.
- **Cleavage** (in animal cell) or **cell wall formation** (in plant cells) takes place if **interphase-II** occurs. But, in many plants, there is no telophase-I, cell wall formation or intephase-II. In such cases, the dividing cell directly passes from **anaphase-I** to **prophase-II**.

Figure 8.7. Meiosis-I: (a) a cell not ready for division, (b) Interphase-I (a cell ready for division), (c) Prophase-I, (d) Metaphase-I, (e) Anaphase-I, (f) Telophase-I, and (g) Cytokinesis in animal cells and some plant cells, where interphase-II follows . In other cells, meiosisi-II continues directly from telophase-I.



Meiosis-II

- Sister **chromatids separate**, and finally **4 haploid gamete** cells form.

Prophase-II

- Nucleus and **nuclear envelope** disperse.
- **Chromatids shorten** and **thicken**
- Centrioles divide and **move** to **opposite poles**.
- **Spindle fibers** form and **arranged** at **right angle** to the **spindle of** meiosis-I by the **centrioles**.

Metaphase-II

- **Each chromosome**, with its **chromatids, lines** up **side by side** on the **equator** of the **cell**.
- Spindle fibers attach to each chromatid.

Anaphase-II

- Sister **chromatids separate** at the **centromere** by the **pulling** effect of **shortening** (contracting) **spindle fibers**.
- Each **chromatid** moves to **opposite pole**.

Telophase-II

- Nuclear envelope forms around daughter cells.

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- Spindle fibers disperse.

Cytokinesis-II

- Four haploid cells form.
- Chromosomes uncoil, lengthen and become indistinct.
- In plant cells, cell wall forms around each cell.

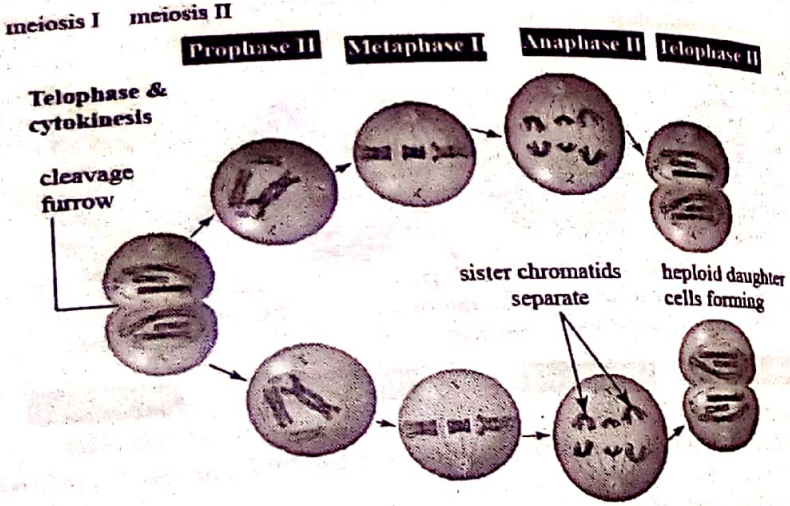


Figure 8.8 Meiosis-II: (a) last stage of meiosis-I, (b) Prophase-II, (c) Metaphase-II, (d) Anaphase-II, (e) Telophase-II, (f) Cytokinesis, (g) Cell after division entering interphase.

Crossing over (synapsis)

- Is the process that occurs in **prophase-I** of **meiosis-I**.
- **Non-sister chromatids** of **homologous chromosomes** exchange their **pieces**. This increases **genetic variation**.

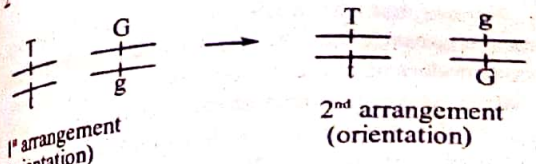
Example:

Figure 8.9 Illustration of the effect of **crossing-over** (synapsis) on a bivalent (homologous chromosomes).

At the end of the above process (synapsis), **4** different gamete cells would be produced. (i.e. gametes with chromosomes carrying AB, ab, aB & ab genes). If crossing-over did not take place, only two types of gamete-cells could have been produced (i.e. **50% AB & 50% ab**).

Due to the **independent assortment** of the **maternal** and **paternal** chromosomes in homologues, there are 2^{n-1} possible arrangements of **bivalent** at **metaphase-I** of **meiosis-I**. (n= number of bivalents). In addition to **crossing-over**, this is the second source of **genetic variation in sexually reproducing** organisms.

Example: with only 2 bivalents (i.e. a cell with only 4 chromosomes), there are $2^{2-1} = 2^1 = 2$ possible arrangements of bivalents.



This simple **orientation** of **bivalents** would **affect** the **number of gamete types** produced with in an organism. With 23 bivalents (i.e. 46 chromosomes per cell) in humans, for example, there are 2^{22} (=4,194, 304) **possible arrangements** of bivalents, and this would **substantially increase** the **genetic variation** among the offspring that would be produced within a **population** of **humans**.

Some terms

Sister **chromatids** - are **duplicates** of the **same chromosome** that have the same genes and **alleles** in the **same sequence** at the **same loci**.

Non-sister **chromatids** - are **duplicates** of **different chromosomes** of homologous **chromosomes**. They have the **same genes** at the **same loci**, but may have **different alleles** at the **same loci**.

Homologous **chromosomes** - occur in pairs in an organism (one **paternal**, the other **maternal**). They carry the **same genes** for the **same feature** at the **same** loci.

Haploid **cell** - is a cell (example: **gamete cell**) that has a **single set** of chromosomes (i.e. n)

Example: sperm cell, pollen grain, egg cell.

- Human sperm/egg cell has 23 chromosomes.

Diploid **cell** - is a cell that has **2 sets (full/complete set)** of **chromosomes** (i.e. 2n).

Example: Body cells of most organisms.

- **Human body** cells have **46 chromosomes**.

Synapsis - pairing of **homologous chromosomes** in **prophase-I** of **meiosis-I** that may lead to **crossing-over** and **exchange** of **pieces**.

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Bivalent - is a **pair of homologous chromosomes** on **metaphase plate** at **metaphase-I**. **Bivalent formation** is a **precondition** for the **separation** of **homologous chromosomes** in **gamete formation.**

Tetrad - refers to a **duplicated bivalent.**

Genetic linkage

Organisms have **thousands** of **genes**, but only a few chromosomes. In humans, there are only **46** chromosomes in **somatic** (body) cells, but about **20,000-25000** protein coding genes. To accommodate all these, each chromosome has to carry may genes. **Genes** that are **situated** on the **same chromosome** are termed as **linked genes**. Such genes pass to the **same gamete cell**, and inherited together. Bateson and Punnett (British geneticists) studied **flower color** (purple/red) and **seed shape** (long/round) of **sweet pea** which is different from **garden peas**. The **7 characteristics** of **garden pea** studied by **Mendel** were **non-linked** (i.e they are controlled by genes found on different chromosomes. In this regard Mendel was fortunate! He was experimenting with non linked-genes. Genes controlling the flower color and seed shape of sweet pea are linked.

Alleles

Flower colour — Purple (P) - dominant; Red (p) - recessive

Seed shape — Long (L) - dominant; Round (l) - recessive

In this case, the genotypes are written as **PL/PL** and **pl/pl** for the pure homozygotes **purple flowered-long seeded** and **red flowered-round seeded** plants, respectively rather than the **PPLL** and **ppll** used if the genes were non-linked.

P_1: Purple-long (P L / P L) = PL/PL X red-round (p l / p l) = pl/pl

Gametes: (PL) (pl)

PL/pl (all purple-long)

F_1: (P L / p l)

P_2: Self-fertilization of F_1 --- PL/p] x PL/p]

Gametes: (P L / p l) X (P L / p l) — (PL), (pl) and (PL), (pl)

F_2: have **2 possibilities** (**complete-linkage** or **linkage** with **crossing-over**)

Possibilities of dihybrid F_2 ratios of linked genes

If the 2 genes were **non-linked**, the **F_2 generation** would have shown the expected **9:3:3:1** ratio.

i) **Complete linkage (linkage without crossing-over)**

F_2:

♂ \ ♀	(PL)	(pl)
(PL)	PL/PL	PL/pl
(pl)	PL/pl	pl/pl

Genotypic ratio
1PL/PL : 2PL/pl : 1pl/pl
Phenotypic ratio
3 purple-long : 1 red-round

These ratios are expected under **complete linkage**, but **Bateson** and Punnett did not get this result. Therefore, complete-linkage is not expected between genes found on the same chromosome. There must be some process that causes **recombination** of **linked genes**. This process is **crossing-over**, but it does not restore **complete independent assortment** which could have led the outcomes to **Mendelian-ratios**.

ii) **Linkage with crossing-over**

If crossing-over occurs at **prophase-I** of **meiosis-I**, all the **4** predicted gamete types, and hence **phenotypes** could be obtained but not in the expected **9:3:3:1** ratio because the **process** (crossing-over) may not occur in all **progamete cells**.

Bateson and Punnett obtained **381** sweet pea plants at F_2 generation in their experiment.

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- 215 purple-long
- 71 purple-round
- 71 red-long
- 24 red-round

} Plants in the expected **9:3:3:1** ratio could have been obtained under the **absence of linkage**

Or

- **286** purple-long
- **95** red-round

} 3:1 ratio could have been obtained under **complete linkage without crossing-over.**

But according to Bateson and Punnett's experimental result, a new ratio different from the two but, closer to the **9:3:3:1** ratio was obtained as **recombinant gametes**, and hence **recombinant phenotypes** are formed due to **crossing-over**. The result was as follows:

- **284** purple-long
- **21** purple-round
- **21** red-long
- **55** red-round

Fruit fly (*Drosophila melanogaster*)

Fruit flies are **experimental animals** used in early **research** on crossing-over and **recombination** as they are convenient because:

a) They are **small** with **short life-cycle** (only 2 weeks)
b) It is **cheap** and **easy** to **breed** them in **large numbers.**
c) They have only **4 pairs** of **chromosomes** (i.e. **8** chromosomes per cell) which are **large** and **observable.**
d) They possess particular **genes** that can be **stained** and **observed as dark** bands.

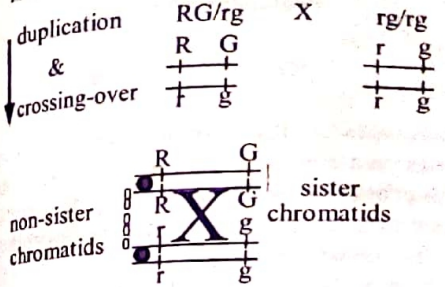
Example:

In *Drosophila*, **eye color** and **body color** genes are **linked** on the same chromosome.

Eye color: Red (R) - dominant; Purple (r) - recessive
Body color: Gray (G) - dominant; Black (g) - recessive

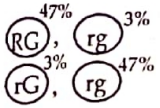
P: Heterozygous Red eyed-Gray bodied X Purple eyed-Black bodied

duplication & crossing-over: RG/rg X rg/rg



- duplication occurs but crossing-over does not result in change of gamete types produced.
- The same type of gametes formed whether crossing over takes place or not

Gametes:



rg -100%

So, the **heterozygous parent** produces **4 types** of **gametes**, while the **homozygous parent** forms only **one type** of **gametes**. The **heterozygote** could have produced only **2 types** of **gametes** (RG & rg) but **crossing-over** that occurred between **non-sister** chromatids brought about **2 additional types** of gametes (Rg & rG).

F_1:

♀ / ♂	RG	Rg	rG	rg
	47	3%	3%	47%
rg	RG/rg	Rg/rg	rG/rg	rg/rg

RG & rg — **Parental** gametes
Rg & rG — **Recombinant** gametes

The following result was obtained from the above experiment phenotypically.

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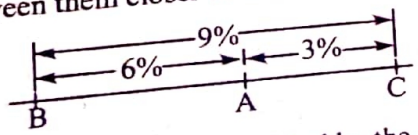
- Red eyed-gray bodied flies (47%) | **Parental** phenotypes
- Purple eyed-black bodied flies (47%) |
- Red eyed-black bodied (3%) | **Recombinant** phenotypes
- Purple eyed-gray bodied flies (3%) |

The **percentage** of the **recombinant** types is called **cross-over value**, and shows the **distance** between the **two genes** on a chromosome. **Low cross-over value** indicates **short distance**, and **high cross-over value** shows **long distance**. The **closer together two linked genes** on chromosome, the **lesser** the **possibility** of **crossing-over** occurring between them, and **vice versa**.

Example: Suppose 3 genes A, B, & C have loci on the same chromosome. Cross-over values were found to be 6%, 3% & 9% between A & B, A & C, and B & C, respectively. Arrange the 3 genes on the chromosome. Cross-over values are expressed in **centimorgan** or in percentage.

Solution: Distance between A & B = 6 centimorgan
Distance between A & C = 3 centimorgan
Distance between B & C = 9 centimorgan

To solve this problem, a line is drawn and the two furthest loci (i.e B & C are placed on it 9 centimorgan units apart. Then, the 3rd locus (i.e. A) is placed between them closer to C because A is further from C only 3 units.



Cross-over value is calculated by the following formula

$$\text{Cross-over value} = \frac{\text{Number of recombinant offspring}}{\text{Total number of offspring}} \times 100$$

Selective breeding

Is the **choosing** and **breeding together** of organisms for a better outcome. It involves **inbreeding** and **outbreeding** (cross-breeding). Selective **breeding** makes use of **genetics**.

Inbreeding - is the **breeding together** of **related varieties** of a species generation after generatation.

Example: interbreeding offspring produced by the same parents. Inbreeding **sustains** (allows the **continuity** of) **desirable characteristics**. But, prolonged inbreeding:

- Increases **homozygosity**, and this usually leads to:
- Reduced fertility

- Reduced disease resitance
- Overall reduction in productivity.

Cross-breeding - is the breeding together of **unrelated lines/breeds/varieties** of the same species. In some plants, it may be between closely related species. The offspring are called **hybrids**, and usually show **phenotypes superior** to either parent. This is **hybrid vigour (heterosis)**. Increased **hybrid-vigour** may result from **increased heterozygosity** or from specific **interactions** between particular alleles in the hetetozygote. Each parent that is crossed together may contain **some**, but not all **dominant alleles**. The heterozygote produced by cross-breeding may contain **all dominant allels** and becomes **vigrous** in the desired trait.

Example:

Parents: DDeeffGGhh X ddEEFFggHH

Gametes: (DefGh) (dEFgH)

F_1: Dd Ee Ff Gg Hh

- Cross-breeding:
  - Increases **heterozygosity** and thereby usually,
  - Increases **productivity** and **disease** resistance

The desired outcomes of cross-breeding can be achieved by:

a) Taking the **advantage** of **hybrid vigor** as illustrated above, or
b) Taking the **advantage** of **combined good qualities** of **two** or **more** varieties

Example:

- If **two pure-breeding** cattle are crossed, the resulting **hybrid** may give **more milk** than either parent. } **hybrid vigour**
- If **two crops** (one **high-yielding** but **less resistant** to desease, and the other, **less-yielding** but **more resistant**), are crossed together, **a crop** with **high yield** and **better resistance** may be obtained. } **Combined good qualities.**

Genes on sex chromosomes

Sex determination

Man has 46 **chromosomes** in his/her **somatic cells** (**diploid** body cells). Two of these are **sex chromosomes** and 44 are **autosomes** (non-sex chromosomes). Females have two x chromosomes and called **homogametic**, while **males** have one x and one y chromosome as sex-chromosomes, and termed as **heterogametic**.

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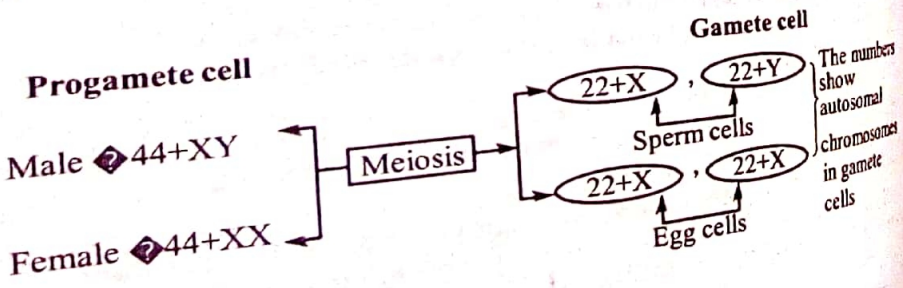
This is shown in human **karyotype** (Figure. 8.10). **Karyotype** is a **photograph** of **homologous chromosomes** of an organism arranged according to their **size** (larger to smaller).

Figure 8.10 The human **karyotype**. There are **23 homologous chromosomes** in man, and totally **46**. **Autosomes** (normal body chromosomes) are 44, and **sex chromosomes** are **2** (**XX** in **females, XY** in **males**). The **homologues** are numbered from **1** to **23** according to **size** except the **sex chromosomes**.

The **chance** of a **couple** gettting a **girl** or a **boy** is **1:1 (50% : 50%)** as shown below. But, this **deos not mean** that the **number** of **girls** and **boys** born to a **couple** (husband and wife) is **always equal**. There are many **families** in which all the children are **boys only** or **girls only**.

Example: a family having 6 girls or 5 boys only. The 1:1 chance is only a **possibility** that may or may not be realized practically.



To simplify the discussion, we leave out the **auto-somal chromosomes** and focus on the **sex-chromesomes** in **gamete cells** (**sperm** and **egg** cells), and use **Punnett square** to **workout** the **possibilities**.

♂ \ ♀	X	X
X	XX	XX
Y	XY	XY

So, a **marriage** between a **man** and a **woman** has the **possibility** of producing **boys** and **girls** with **equal chance (probability)**, i.e. 1:1 (50% : 50%).
} 2XX : 2XY } 1XX : 1XY

After **fertilization** in the early **development** of the **embryo**, the **urogenital** ridge develops into a **'bi-potential gonad'** with the **ability** to give rise to ovary or **testis**. If there is **Y-chromosome** in the **embryo**, and if the **SRY** (Sex Determining Region Y) gene on it is **activated**, **bi-potential** gonad develops into **testis**, and the embryo becomes **male**. If the **SRY gene** (i.e. Y-chromosome) is **absent** or **inactivated**, bi-potential gonald develops into **ovary** , and the embryo becomes a **female**.

Abnormality in sex chromosome number

These are caused by **faulty** (incorrect) **segregation** of the **sex chromosomes** during **meiosis**, (i.e. **non-disjunction**) and **lead** to some **abnormalities**.

a) **X** - Develops into a slightly **masculinized female** without **mental** retardation, but **sterile**. She would be **short wide chested** with under developed **breasts** and **narrow aorta**. } **Turner's syndrome.**

b) **XXY** - is a **sterile feminized male** without mental retardation } **Klinefelter's syndrome.**

a) **XYY** - is a **fertile male** taller than average.

a) **XXX** - is **sterile female** usually **normal** in other aspects. She may be with **mental abnormality**.

Birds - in birds, the **males** are **homogametic** with **76 autosomes** and **2Z sex chromosomes**, while **females** are **heterogametic** with **76 autosomes** and **Z & W sex chromosomes** per body cell.

} Birds → Male --- 76+ZZ (**Homogametic**)
Female --- 76+ZW (**Heterogametic**)

Grasshopper - male is with single X, and females with two X chromosomes. Both have 22 autosomes commonly.

♂ Male --- **22+X**
♀ Female --- **22+XX**

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Some insects - in insects such as **bees** and **ants**, **males** are **haploid** and **females** **diploid**. In such cases, **fertilized eggs** and **unfertilized eggs** develop into **females** and **males**, respectively.

Male --- (16) Chromosomes (n)
Female --- (32) Chromosomes (2n)

Chromosome **number** or **type may not** be the **only factor** that determines **sex**. **Environmental factors** or **features** of the **organism** may also play role. Some **reptiles** such as **alligators** have their **sex determined** by egg incubation **temperature**. **Some snails** start out as **males**, and later become females. In others, **dominant individuals** become **females** (Example: tropical clown fish), or **males** (Example: blue wrasse fish). Still in some other species, there is no sex **determination**. In such cases, an **individual** will **posses** both **male** and **female** **sex organs**, and called **hermaphrodite**. Example: some snails, earthworms, etc.

Sex linked genes

Genes found only on the **sex chromosomes** (**X & Y in man** and **mammals**) are **sex-linked genes**. The **X** and **Y** chromosomes are very **different** in **morphology** and in the **genes** they **carry**. However, they have **regions** that are similar (**homologous regions**), though these regions are very short.

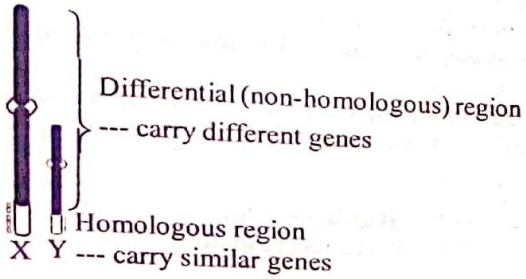

Figure 8.11. Morphology of the X and Y chromosomes.

As a result of difference in structure of **X & Y** chromosomes, there are genes found only on X or Y.

Y-only

- *Retinitis pigmentosa* - a gene controlling this **degenerative disease** where **eyesight** progressively becomes **weaker** and may end up with **blindness**. is found on **Y-chromosome** only.

- A form of **deafness** controlling gene.

These features can be **inherited only** by **males** because **males only** have **Y-chromosome**.

X-only

- A gene controlling **red-green color blindness**.
- A gene controlling **one form** of **haemophilia**.

Both **males** and **females** posses **X-chromosome** (at least one). So **features** on **X-chromosome** are **inherited** by **both**.

Genes found on the **non-homologous regions** of **X** or **Y** are **expressed** in an individual even if they are **recessive** because there is **no corresponding** allele on the other **sex chromosome** that may be **dominant** and **mask** them.

Characteristics of X-linked feature in the non-homologous region

a) **More common** in **males** as there is only one **X** in them, and even if a male carries a **recessive allele**, it would be **expressed** as there would be **no corresponding allele** that may be **dominant** and **mask** it.
b) Affected **males inherit** the allele from their **mother** because their only **X-chromosome always** comes from their **mother**.
c) Affected **females** inherit **one allele** from **each parent** since they **always** get **X-chromosome** from **each parent**.
b) Heterozygous females are called **carriers** of the feature. Heterozygous males are non existent as they **lack** a **second** corresponding **X-chromosome**.
e) They may '**skip**' a **generation** and appear in **males** only.

Example: For red-green color blindness,
B = allele for **normal vision**
b = allele for **red-green color blindness**.

X^BY - normal male
XBXB - normal female
XBXb - carrier female (non-color blind)
XbY - color blind male
XbXb - color blind female

Sex influenced & Sex-limited traits

These are **controlled** by **genes** carried on the **autosomal chromosomes**. They are **expressed differently** in **males** and **females**. **Sex-influenced traits** are expressed in either sex based on some **internal physiological differences**.

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Example: Pattern **baldness** depends on **testosterone** concentration in the individual. **High testosterone concentration** makes the **baldness** allele **dominant** in males.

Male	Female
BB --- Normal	BB --- Normal
Bb --- Bald	Bb --- Normal
bb --- Bald	bb --- Bald

In **heterozygous males** (Bb), **'b'** is made **dominant** by high **testosterone**. In **heterozygous females** the **normal** allele is **dominant**.

Therefore, **baldness** is more **frequent** in **males**.

Sex-limited traits are **expressed** in **one sex only**.

Example: Lactation gene is found both in **males** and **females**, but expressed only in **females**. It is **silenced** in **males**.

Illustrative Examples

11. If two parents of genotype Gg are crossed, what proportion of the offspring will differ from their parents phenotypically? The alleles do not show codominance.

Explanation:

P_1: Gg X Gg

G: Ⓖ, ⓖ Ⓖ, ⓖ

F_2:

♂ \ ♀	G	g
G	GG	Gg
g	Gg	gg

Genotypic ratio:
1GG : 2Gg: 1gg
Phenotypic ratio:
3 dominant phenotype : 1 recessive phenotype

So, the **parents** show **dominant phenotype** because they have at-least 1 **dominant allele** (G). The **proportion** of the **offspring** showing **recessive** phenotype is **25%** or 1/4. These are the **offspring carrying 'gg'** genotype. They differ phenotypically from their parents.

12. What proportion of the offspring will **look-like** their parents in the above question?

Explanation:

- **75%** or **3/4** (i.e those with **GG & Gg** genotypes)

13. A **woman** with **homozygous** blood group **A** and a **man** with **homozygous** blood group **B** could potentially have offspring with blood type _____
(A) A
(B) B
(C) AB
(D) O

Explanation:

- **Homozygous A** means **AA**, and **homozygous B** is **BB**. If these are crossed, what is obtained is **AB** only.

Answer: C

14. If a **man** with **heterozygous** blood type **A** marries a **woman** with **homozygous** blood type **B**, their offspring would potentially be:
(A) A
(B) AB
(C) B
(D) O
(E) B & C are correct

Explanation:

- Heterozygous **A = AO**; homozygous **B = BB**. So, phenotypic possibilities are blood types **B** & **AB**

♂ \ ♀	B	B
A	AB	AB
O	Bo	Bo

Genotypic ratio
50% AB: 50% BO
Phenotypic ratio
50% AB: 50% B

Answer: E

15. A **woman** which is **pure breeding** for **A** has married a **man** who is **heterozygous** for **B**. Workout **phenotypic possibilities** of their offspring.

Explanation:

- Pure-breeding **A = AA**; heterozygous **B = BO**. Their children may either be of blood type **A** or **AB**.

♂ \ ♀	A	A
B	AB	AB
O	Ao	Ao

Possible Phenotypes:
A & AB in a 1:1 ratio.

16. What is the **possible blood type** carried by **offspring** produced from a couple with blood type **A** that is **heterozygous**?

Explanation:

- Heterozygous **A = AO**. So, the offspring would be **A** or **O** type in a 3:1 ratio.

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♂ \ ♀	A	O
A	AA	AO
O	AO	OO

17. A **woman** who is **heterozygous** for A has married a **man** that is **heterozygous** for B. Potentially, they can have offspring with blood type
(A) O (C) B (E) All
(B) A (D) AB

Explanation:

- Heterozygous **A = AO**; heterozygous **B = BO**; when these are crossed, all types of blood groups can be produced.

Answer: E

♂ \ ♀	A	O
B	AB	BO
O	AO	OO

18. If an animal has genotype **GgRr without linkage** between the two genes, what gametes can the animal produce?

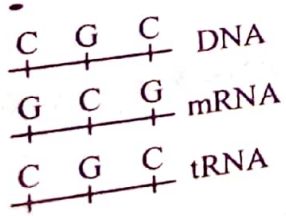
Explanation:

- **GgRr** } each allele from the 1st gene can combine with each allele of the 2nd gene, and form 4 different **types** of gametes.

19. The DNA triplet CGC would code for an **amino acid** carried by tRNA with the **anticodon**:
(A) GCG (C) CGC
(B) CGG (D) GGC

Explanation:

-



} The **same** as the **sequence** on DNA in this case, but it may differ from it if A & T have been considered (involved).

Answer: C

20. _____ is the science of **inheritance**.
(A) Genomics. (C) Genetics.
(B) Heredity (D) Cytology.

Answer: C

21. Which of the following **includes the others**?
(A) Chromosome. (C) Gene
(B) Allele (D) Chromatin.

Explanation:

- **Allele** is an **alternative version of a gene**, **gene** is **part of DNA** which together with **histones** form a **chromatin** (**'colored material'**). A **chromosomes** is **condensed chromatin**.

Answer: A

22. When two heterozygous individuals are crossed, the proportion of the **progeny** that are **true breeding** for **dominant** trait is _____
(A) 1/2. (C) 2/4
(B) 3/4 (D) 1/4.

Explanation:

- Example: Tt X Tt

♂ \ ♀	T	t
T	TT	Tt
t	Tt	tt

} **TT** is the **true-breed**, and its proportion is **one out of four (=1/4 or 25%)**

Answer: D

23. If **blue eye color** is **recessive** to **brown**, could **blue eyed parents** have a brown-eyed child?

Explanation:

- B = brown, b= blue, if this is so, a blue-eyed parent would be **'bb'** genotypically. Crossing individuals of these genotype would not result in a **brown eyed** child. All the outcomes of the cross would be **blue-eyed** children.

24. Can a **brown eyed parents** have a **blue-eyed child**?

Explanation:

- Brown eyed-parents can genotypically be **BB** or **Bb**. They can produce a child with blue eye if and only if both of them are **heterozygous** for the trait (Bb).

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♂ \ ♀	B	b
B	BB	Bb
b	Bb	bb

} Their probability of producing children with blue-eyes is **25%**.

25. Sexual reproduction involves **segregation**, **independent assortment** and **random union of gametes**. What evolutionary advantages these phenomena have over **asexual reproduction**?

Explanation:

- These cause **variation in a population**, and become **raw materials** of **evolution**.

26. Before making crosses, which part of the flower did Mendel remove to avoid self-pollination? **(EUEE, 2005, E.C)**

(A) Stigma. (C) Ovary.
(B) Ovule (D) Stamens.

Explanation:

- Mendel removed **stamens** (i.e. the male parts).

Answer: D

27. Which of the following is the correct F_2 phenotypic ratio of a mono-hybrid cross? **(EUEE, 2005, E.C)**

(A) 1:1. (C) 3:1
(B) 1:2 (D) 2:2

Explanation:

- 3/4 dominant phenotype and 1/4 recessive phenotype (i.e. 75% : 25%).

Answer: C

28. How many chromosomes do humans inherit from each of their parents? **(EUEE, 2005, E.C)**

(A) 23 chromosomes. (C) 46 chromosomes
(B) 23 pairs of chromosomes (D) 46 pairs of chromosomes.

Explanation:

- Any sexually reproducing organism (including man) inherits the **haploid chromosome number** from each parent, i.e. half the total chromosome number in the body cells of the organism. In man there are **46** chromosomes in his/her **body cells**, and **23** chromosomes in the **gamete cells**. When **sperm** and **egg** cells fuse, then the newly formed **human zygote** inherits **23 chromosomes** from each parent.

Answer: A

29. Which of the following is **not true** about the gene **SRY**? **(EUEE, 2005, E.C)**

(A) It is found on the Y-chromosome.
(B) It determines maleness.
(C) Testes develop in its presence.
(D) Females have two copies of this gene.

Explanation:

- SRY gene is found on Y-chromosome, and females lack this chromosome.

Answer: D

30. Suppose in a mono-hybrid cross, **80** F_2 plants were produced, what is the number of the plants that are expected to have the **dominant** and the **recessive** phenotypes? **(EUEE, 2005, E.C)**

(A) 70 dominant + 10 recessive. (C) 50 dominant + 30 recessive.
(B) 60 dominant + 20 recessive. (D) 40 dominant + 40 recessive.

Explanation:

- Phenotypic ratio of mono-hybrid cross at F_2 is 3/;1 (3 dominant to 1 recessive). By multiplying the total number of plants by these ratios, we can get the actual numbers.

$\Rightarrow \frac{3}{4} \times 80 = \underline{60}$ plants with dominant phenotype

$\Rightarrow \frac{1}{4} \times 80 = \underline{20}$ plants with receessive phenotype

8.2. Molecular Genetics

Eukaryotic chromosomes are composed of **DNA** and **histones** (globular proteins). The **DNA fiber** wraps itself around **8 histones molecules** at intervals forming **bead-like** structures called **nucleosomes**. The **nucleosomes** are packed together to form **chromatin**. The **chromatin** fibers **coil** and **condense**, and form a chromosome.

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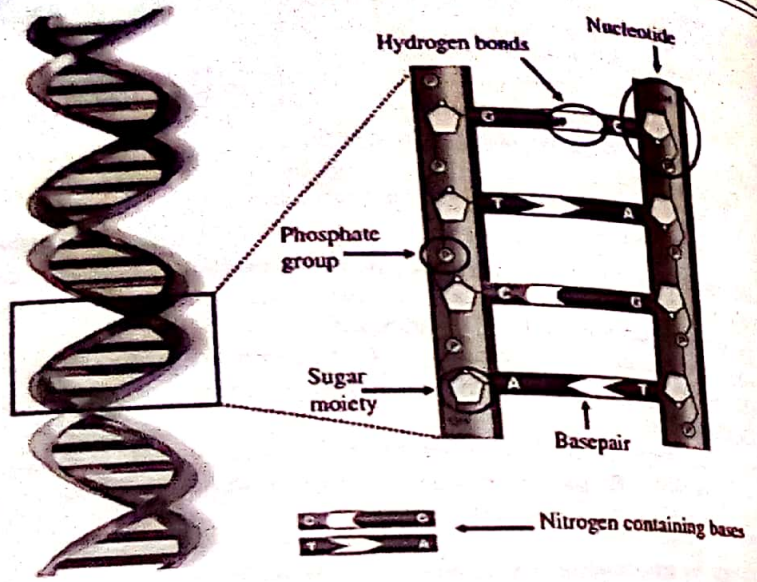

Figure 8.12 Detailed structure of the **genetic material**. **Nitrogenous base**, a **phosphate group** & a **pentose sugar** (**deoxyribose**) are organized into **nucleotides** which are put together into a **DNA strand**. **Two DNA strands** are joined via their **bases** and form **DNA double helix**. This is wrapped around **histone** molecules and forms a **chromatin**. The **chromatin** is **coiled** and packed to form a chromosome. And the chromosome is situated in the nucleus.

The **genes on DNA** are **active** when the **chromosome** is **loosely packed** and very **thin** to be seen under the microscope. The **genes** are **inactivated** when the cell **divides**. During this process, the **chromosomes thicken** and become **compact**. This time, they can **easily** be **observed** under the **microscope**.

Structure of DNA

DNA (deoxyribonucleic acid) and **RNA** (ribonucleic acid) are organic compounds grouped under **nucleic acids**, the building blocks of which are **nucleotides**. Nucleotides are made from 3 components:

a) A phosphate group (PO_4^{3-}).
b) A 5-carbon sugar **ribose** (in RNA) or **deoxyribose** (in DNA), and
c) A nitrogenous base **A, G, C** and **T** (in DNA) or A, G, C and U (in RNA).

A = adenine
G = guanine
C = cytosine
T = thymine
U = uracil

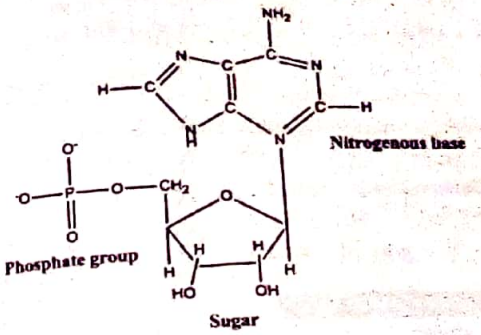

Figure 8.13. The structure of a nucleotide

There are **5** types of **nitrogenous bases**: **adenine (A), guanine (G), cytosine (C), thymine (T)** and **uracil (U)**. **A & G** are **double ringed** are called **purines**, while **C, T** and **U** are **single ringed** and called **pyrimidines**. There are **4** types of **nucleotides** in **DNA**. These are:

- **Adenine** containing nucleotides (a)
- **Thymine** containing nucleotides (b)
- **Guanine** containing nucleotides (c)
- **Cytosine** containing nucleotides (d)

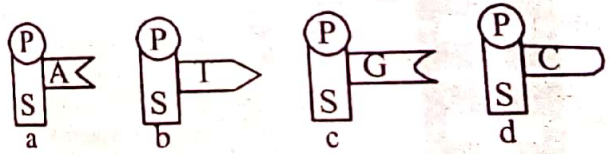

Figure 8.14. simplified structure of **nucleotides** (**P** = phosphate, **S** = sugar, A= adenine, **T** = thymine, **G** = guanine, **C** = cytosine)

DNA is made from **2 antiparellel polynucleotide** strands. The **backbone** of the **polynucleotide** is made from **sugar** and **phosphate**. The 2 strands are held together by their **bases**. **Adenine** always combines with **thymine**, and **cytosine** with **guanine**. The **base sequence** on each strand of of **DNA** are **complementary** to each other in their **base pairings** (i.e. A-T and G-C). Therefore, the **number**

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or **percentage** of **adenine** is equal to that of **thymine** in DNA molecule. The same holds true for **cytosine** and **guanine**. Consequently, the **number** or **percentage** of **all** the **bases** in a DNA molecule can be **worked out** if the **number** of **one base** is known.

Example: A DNA molecule contains **300** adenine bases. Workout the number of the remaining 3 bases if the molecule contains **1900** bases totally.

Explanation:

A - T } base combination in DNA double strands
G - C }

No. of A = No. of T = 300 } A+T = 600

<= if we substract 600 from 1900, the result = 1300

<= G+C = 1300 No. of G = No. of C = 1300/2 = 650

<= A = 300, T = 300, G = 650, C = 650

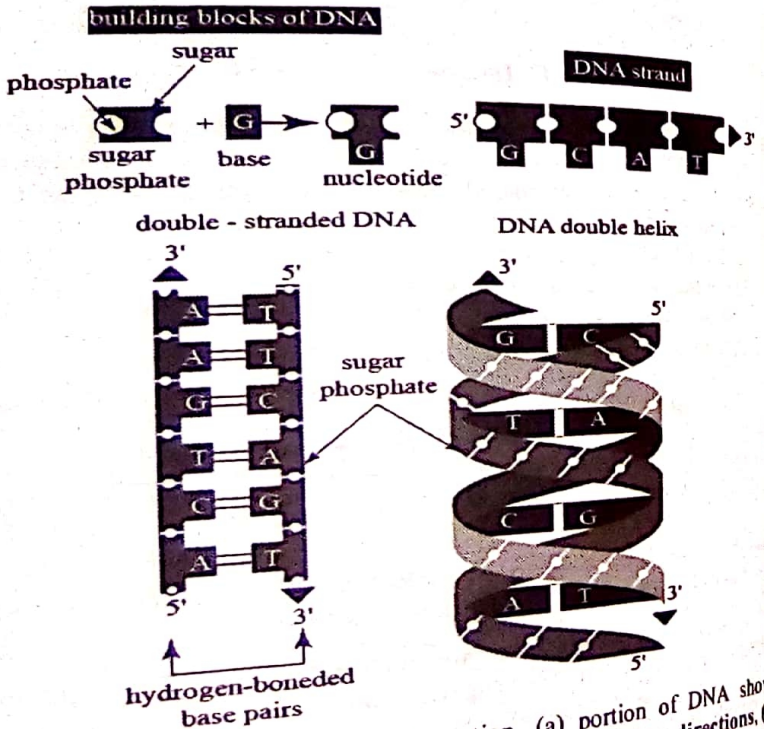

Figure 8.15. **DNA double helix** formation. (a) portion of DNA showing **4 nucleotides** in each strand, **each strand** running in **opposite directions**, (b) A **double-helix** as it exists in the cell.

DNA replication

There are **3 theories** which try to explain how **DNA** replicates: **conservative, semi-conservative** and **disruptive** replication methods.

a) **Conservative method** - according to this theory, the **old parental DNA** molecule is **conserved** when a **new DNA** molecule is **synthesized**.

b) **Semi-conservative method** - here the **newly formed DNA** molecules contain one strand from the **old** and a **second** from the **newly** synthesized strand. This is **more accepted** and **supported** by **experimental results**.

c) **Disruptive method** - the **newly** formed **daughter DNA** molecules contain **pieces** from **both** the **old** and **newly** synthesized strands. The **old** and the new strands are **cut** into **fragments** and put together to form the **new DNA** molecule. Each **fragment** is **equally distributed** in the **new** molecule.

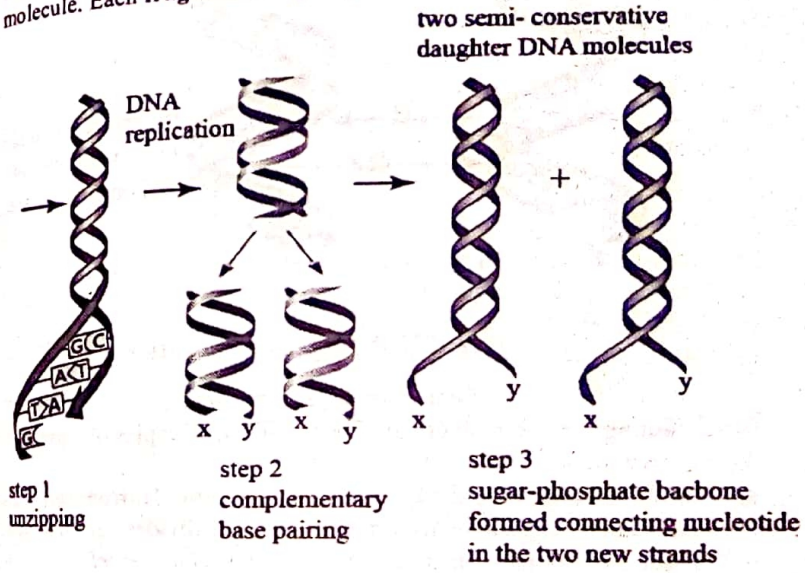

Figure 8.16 diagrams explaining 3 theories of DNA replication

Semi-conservative replication

Has 2 main steps

a) **Unzipping (opening)** - an enzyme called **DNA-helicase** breaks **hydrogen bonds** between the **bases** to open the **DNA-helix**, and forms **single-stranded** regions.

b) **Polymerization** - here **DNA-polymerase** assembles **new strand** from **nucleotides** in the cell using the **old strand** as a **template**. The **new strand** is **complementary** to the **old one**. Then, the **daughter DNA** molecules are twisted into a **helix**.

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